Team A1 — Complete Theory Compilation

Flexible Computing Loads, Grid Resilience, and Carbon Emissions

Every role, every week, every proof, every step

Team A1 (Roles 1–4)

Weeks 1–4, working compilation

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Part I

Role 1 — Optimal Contract and General Carbon Theory

Chapter 1

Week 1: Setup, unified FOC, and the uniform ground truth

[General Carbon Theory for Flexible Computing Loads] General Carbon Theory for Flexible Computing Loads:
The Capacity-Substitution Channel and a Distribution-Free
Net-Carbon-Reduction Condition

Summary. *We harden and generalize the carbon analysis of the flexible-computing-load (FCL) model beyond the uniform-distribution benchmark. Four results.* First*, we re-derive the uniform closed forms (paper eqs. 18–26) from the first-order conditions and verify them numerically against direct quadrature; along the way we establish the* knife-edge *structure for the entire power-law family and reconcile two transcription issues in the current draft.* Second*, we prove the central comparative-statics claim $K_R^* > K_R^\circ$—renewable capacity is strictly higher with the FCL program—for* any *residual-demand distribution satisfying two conditions: an economically transparent monotonicity of the* shift ratio $g(x) = F_{K_R}(x)/f(x)$*, plus interior regularity. The program raises renewable capacity whenever an extra megawatt of renewables relaxes residual demand more in deep-surplus states than in tight states. The power-law family satisfies the shift-ratio condition for every shape parameter, which is exactly why footnote 6's extension goes through.* Third*, we give the exact net-carbon inequality, decompose it into a distribution-free* utilization *effect and a sign-controlled* capacity-substitution *effect, and give a shape comparison to the uniform at matched cutoffs (which bounds the saving but—as Week 2 shows the uniform's interior value is negative—does not by itself sign interior carbon reduction).* Fourth*, we give a distribution-free reading of the effect:* at fixed equilibrium capacities *the carbon change is affine in two conditional means, is bracketed by closed-form* magnitude bounds *computable from cutoff data, and—among distributions sharing those capacities, within the natural shape class—is worst-cased by the* uniform*. Every analytical formula is verified against finite differences of the profit integral to 6–16 digits. Of the points left open by the uniform-only treatment, three are closed—the lattice (submodularity) obstruction, the endogenous-capacity feedback (at fixed capacities), and the renewable-cost threshold (general upper anchor; explicit window for the power-law family)—and the regime map is unified up to the small-FCL crossover; a companion memo keeps the same accounting.*

1.1 Setup and notation

We work with the model of Section 3 of the paper and adopt its notation. A regulated utility chooses renewable capacity K_R, dispatchable fossil backup K_F, and flexible-computing-load capacity K_M to maximize expected profit over one representative period. The state variable is *residual demand*

$$\tilde{x} := \tilde{D} - K_R \tilde{S} \in [-K_R, \bar{D}], \tag{1.1}$$

where $\tilde{D} \in [0, \bar{D}]$ is stochastic consumer demand and $\tilde{S} \in [0, 1]$ the renewable capacity factor. Let $F(x; K_R)$ be the CDF of $\tilde{x}$ with density $f(x; K_R)$, extended by $F \equiv 0$ for $x < -K_R$ and $F \equiv 1$ for $x > \bar{D}$. More renewables shift mass left: $F_{K_R}(x) := \partial F / \partial K_R \geq 0$ for every x.

Per-MWh quantities: FCL revenue R, fossil variable cost c, retail price p, blackout penalty β. Amortized per-MW capital costs: C_R, C_F, C_M. We maintain the paper's parameter regularity.

Assumption 1.1 (Regularity)**.** $R > c + C_M, \quad R < c + C_F + C_M, \quad \beta + p > c + C_F, \quad \beta + p > R.$

Under the optimal within-period dispatch rule $d(x) = \min\{\max\{K_F - x, 0\}, K_M\}$ (paper Lemma 1), plugging into expected profit and integrating by parts gives the unified profit

$$\Pi(K_R, K_F, K_M) = p\,\mathbb{E}[\tilde{D}] + R\int_{K_F - K_M}^{K_F} F\,dx - c\int_{-K_M}^{K_F - K_M} (1 - F)\,dx - (\beta + p)\int_{K_F}^{\bar{D}} (1 - F)\,dx - C_F K_F - C_R K_R - C_M K_M, \tag{1.2}$$

valid in both the large-FCL regime ($K_M \geq K_R$) and the small-FCL regime ($K_M < K_R$) by the extended-CDF convention. Setting $K_M = 0$ gives the no-FCL benchmark Π_0.

Lemma 1.2 (Unified first-order conditions)**.** *Differentiating* (1.2) *by Leibniz's rule,*

$$\frac{\partial \Pi}{\partial K_M} = (R - c)\,F(K_F - K_M; K_R) + c\,F(-K_M; K_R) - C_M, \tag{1.3}$$

$$\frac{\partial \Pi}{\partial K_F} = R\big[F(K_F) - F(K_F - K_M)\big] - c\big[1 - F(K_F - K_M)\big] + (\beta + p)\big[1 - F(K_F)\big] - C_F, \tag{1.4}$$

$$\frac{\partial \Pi}{\partial K_R} = R\int_{K_F - K_M}^{K_F} F_{K_R}\,dx + c\int_{-K_M}^{K_F - K_M} F_{K_R}\,dx + (\beta + p)\int_{K_F}^{\bar{D}} F_{K_R}\,dx - C_R. \tag{1.5}$$

In the large-FCL regime $F(-K_M; K_R) = 0$, and (1.3)–(1.5) *collapse to the paper's eqs. (12), (13), (15) respectively.*[1]

Lemma 1.3 (Distribution-free equilibrium probabilities)**.** *At any interior optimum, the no-FCL threshold satisfies $F(K_F^\circ; K_R^\circ) = \varphi^\circ := \frac{\beta + p - c - C_F}{\beta + p - c}$, independent of the shape of F. In the* large-FCL *regime ($F(-K_M^*; K_R^*) = 0$) the two FCL thresholds are likewise distribution-free:*

$$F(K_F^*; K_R^*) = \frac{\beta + p - c - C_F - C_M}{\beta + p - R}, \qquad F(K_F^* - K_M^*; K_R^*) = \frac{C_M}{R - c}. \tag{1.6}$$

In the small-FCL regime the term $cF(-K_M^*; K_R^*)$ in (1.3) *does not vanish, and these two values are not pinned by the FOCs alone.*

Proof. The first identity is the no-FCL FOC (1.4) with $K_M = 0$; the other two solve (1.3)–(1.4). Each fixes a *value* of F, not a capacity, so no distributional assumption enters. *(Verified on the non-uniform* Beta$(2, 2)$ *scale-mixture: both equal $\frac{5}{9}$ and $\frac{17}{18}$ to 10^{-9}.)* □

These give the adoption threshold

$$\bar{C}_M = (R - c)\,\varphi^\circ = \frac{(R - c)(\beta + p - c - C_F)}{\beta + p - c}, \tag{1.7}$$

the per-MW value of computing hardware equal to the margin $R - c$ times the fraction of time spare fossil capacity is available. Lemma 1.3 is the lever that makes the carbon comparison of §1.4 general: the program shifts the equilibrium *capacities*, but the *probabilities* at which fossil cuts in and FCL saturates are pinned.

[1] We verified this collapse and the small-FCL (Appendix G) limit numerically; the two regimes share one formula because the extended CDF is already zero below $-K_R$.

1.2 Ground truth: the uniform closed form

We re-derive the uniform benchmark used to sign the carbon result, both to lock in correctness and to fix the algebraic constants the general theory must reproduce. Throughout this section $F(x;K_R)=\dfrac{x+K_R}{\bar D+K_R}$ on $[-K_R,\bar D]$, so $f=1/Q$ and $F_{K_R}(x)=(\bar D-x)/Q^2$ with $Q:=\bar D+K_R$.

Lemma 1.4 (Uniform optimal capacities)**.** *Write $Q^\circ=\bar D+K_R^\circ$ and $Q^*=\bar D+K_R^*$. Then*

$$K_F^\circ=\bar D-\frac{C_F}{\beta+p-c}\,Q^\circ,\qquad K_R^\circ=\bar D\left(\sqrt{\frac{c}{2C_R-C_F^2/(\beta+p-c)}}-1\right),\tag{1.8}$$

$$K_F^*=\bar D+\frac{R-c-C_F-C_M}{\beta+p-R}\,Q^*,\qquad K_F^*-K_M^*=\bar D-\frac{R-C_M-c}{R-c}\,Q^*.\tag{1.9}$$

Proof. (1.9) and the first identity in (1.8) invert the cutoff probabilities (1.6) under the linear CDF. For K_R°, substitute K_F° into the renewable FOC (1.5) with $K_M=0$. Writing $\gamma:=C_F/(\beta+p-c)$ so that $K_F^\circ=\bar D-\gamma Q^\circ$ and $\bar D-K_F^\circ=\gamma Q^\circ$, and using $\int_a^b F_{K_R}\,dx=\big[(b-a)(\bar D-\frac{a+b}{2})\big]/Q^2$,

$$\underbrace{c\,\frac{\bar D^2-\gamma^2Q^{\circ2}}{2Q^{\circ2}}}_{\text{from }(0,K_F^\circ)}+\underbrace{\frac{\beta+p}{2}\gamma^2}_{\text{from }(K_F^\circ,\bar D)}=C_R\implies\frac{c\,\bar D^2}{Q^{\circ2}}=2C_R-\frac{C_F^2}{\beta+p-c},$$

because $\gamma^2(\beta+p-c)=C_F^2/(\beta+p-c)$. Solving for Q° and subtracting $\bar D$ gives (1.8). *(Verified against the quadrature solver: the capacities satisfy the FOCs to $\le 10^{-15}$.)* □

Carbon (paper eqs. 25–26). Expected fossil output without the program is $\mathbb{E}[\text{fossil}^\circ]=K_F^\circ-\int_0^{K_F^\circ}F\,dx=K_F^\circ(\bar D-\frac{K_F^\circ}{2})/Q^\circ$, and with the program (large-FCL regime), accounting for the flat stretch $(K_M^*-K_R^*)$ on which the extended CDF equals zero,

$$\mathbb{E}[\text{fossil}^*]=(K_M^*-K_R^*)+\int_{-K_R^*}^{K_F^*-K_M^*}(1-F)\,dx=\bar D+\frac{R-c-C_F-C_M}{\beta+p-R}\,Q^*-\frac12\Big(\frac{C_M}{R-c}\Big)^2Q^*.\tag{1.10}$$

Carbon emissions equal c times these quantities. Net carbon falls with the program iff $\mathbb{E}[\text{fossil}^*]<\mathbb{E}[\text{fossil}^\circ]$.

Remark 1.5 (Two corrections to the current draft). *(i)* The bracketed expressions printed in eqs. (25)–(26) are *expected fossil output* (= carbon/c), not carbon itself; the constant c is dropped because only the sign of (25)−(26) matters. Restoring c, our (1.10) reproduces the printed eq. (26) to 6×10^{-10} and eq. (25) to machine precision. *(ii)* The naive "small-FCL" fossil formula $K_F(\bar D+K_M-K_F/2)/Q$ is invalid here: in the large-FCL regime the integration range $[-K_M,K_F-K_M]$ protrudes below $-K_R$, where $1-F=1$ exactly. The flat term $(K_M^*-K_R^*)$ in (1.10) is the correction.

Proposition 1.6 (Knife-edge for the power-law family)**.** *Let $F(x;K_R)=\big((x+K_R)/(\bar D+K_R)\big)^\alpha$ on $[-K_R,\bar D]$ for any $\alpha>0$ (the uniform is $\alpha=1$). Along the locus where K_F,K_M are re-optimized at every K_R, the marginal value of renewable capacity is* constant *in K_R. In the interior regime it equals*

$$\text{MV}=\tfrac12\left[c+\frac{(R-C_M-c)^2}{R-c}+\frac{(R-c-C_F-C_M)^2}{\beta+p-R}\right],\tag{1.11}$$

so $K_R^* = \infty$ *if* $\mathrm{MV} > C_R$, $K_R^* = 0$ *if* $\mathrm{MV} < C_R$, *and any* K_R *is optimal at equality. Equivalently* $K_R^* = \infty$ *iff*

$$c + \frac{(R - C_M - c)^2}{R - c} + \frac{(R - c - C_F - C_M)^2}{\beta + p - R} > 2C_R, \tag{1.12}$$

which is exactly *the bracket printed in paper eq. (24).*

Remark 1.7 (The $K_R^* = \infty$ corner is a small-FCL statement). Formula (1.11) is the marginal value on the *large*-FCL portion of the re-optimization locus. Along that locus $K_M^*/K_R^* \to (1 - \varphi_1) - \frac{c+C_F+C_M-R}{\beta+p-R} < 1$, so $K_M^* < K_R^*$ for large K_R: the locus *leaves* the large-FCL regime before $K_R = \infty$. The genuine $K_R \to \infty$ corner is therefore governed by the small-FCL asymptotic slope (Appendix G / Role 4), which agrees with (1.11) on the families we compute but is, strictly, a small-FCL extrapolation. Downstream results that invoke the $K_R^* = \infty$ corner ($C_R^\dagger = \mathrm{MV}$, the strict-fall window) inherit this dependence; they are proved here for the power/uniform families and verified numerically, with the general regime-stitching deferred to Role 4.

Proof. For the power law one computes $F_{K_R}(x) = f(x)\,(\bar{D} - x)/Q$, hence the shift ratio $g(x) = F_{K_R}/f = (\bar{D} - x)/Q$ is the *same* affine function for every α. Substituting the linear loci (1.9) into (1.5) and evaluating $\int_a^b F_{K_R}\,dx$ over the three regions (whose endpoints are all of the form $\bar{D} - (\,\cdot\,)Q$), every Q cancels, leaving (1.11); the marginal return is constant in K_R, so the objective is linear along the locus and the optimum is a corner. Multiplying (1.11) by 2 and subtracting $2C_R$ gives precisely the paper's eq. (24) bracket, confirming (eq. 24) $= 2\,(\mathrm{MV} - C_R) = 2\,\partial\Pi/\partial K_R$ along the locus. *(MV constant in* K_R *verified at* $K_R \in \{1, 5, 9\}$ *for* $\alpha \in \{0.5, 1, 2\}$*.)* □

Remark 1.8 (On eq. (24)). Proposition 1.6 shows eq. (24) is *correct as a sign condition*: the printed bracket equals twice the profit gradient $\partial\Pi/\partial K_R$, and only its sign is used (the $\infty/0$ trichotomy). The factor of two is a grouping artifact of the closed-form substitution, not an error—worth a one-line clarifying footnote in the draft.

1.3 The capacity-substitution channel, in general

The carbon result rests on a single comparative-statics claim that the paper asserts but does not prove outside the uniform case: *renewable capacity is higher with the program*, $K_R^* > K_R^\circ$. We now prove it for any distribution satisfying one transparent condition. The obstruction, and why the naive lattice argument fails, is recorded in Section 1.8.

1.3.1 The marginal-value cross-partials

From Lemma 1.2, the second derivatives of Π that govern how the optimum moves when computing becomes more viable are

$$\frac{\partial^2\Pi}{\partial K_R\,\partial K_M} = (R - c)\,F_{K_R}(K_F - K_M) + c\,F_{K_R}(-K_M) \;\geq 0, \tag{1.13}$$

$$\frac{\partial^2\Pi}{\partial K_F\,\partial K_M} = (R - c)\,f(K_F - K_M) \;\geq 0, \tag{1.14}$$

$$\frac{\partial^2\Pi}{\partial K_R\,\partial K_F} = (R - \beta + p)\,F_{K_R}(K_F) - (R - c)\,F_{K_R}(K_F - K_M) \;\leq 0. \tag{1.15}$$

So computing raises the marginal value of *both* capacities ((1.13)–(1.14) are nonnegative because $R > c$ and $F_{K_R} \geq 0$), yet renewables and fossil are mutual substitutes in the objective ((1.15)

is nonpositive because $\beta + p > R$). The sign conflict in (1.15) is exactly why a one-line Topkis argument does not deliver $K_R^* > K_R^\circ$; see Section 1.8.

1.3.2 The shift-ratio condition

Define the *shift ratio*

$$g(x; K_R) := \frac{F_{K_R}(x; K_R)}{f(x; K_R)} = -\left.\frac{\partial x}{\partial K_R}\right|_{F \text{ fixed}}, \tag{1.16}$$

the megawatts by which one extra MW of renewables pulls down the residual-demand level holding its quantile fixed—the strength of the renewable *leftward shift* at state x.

Condition 1.9 (Shift ratio)**.** $g(x; K_R)$ is non-increasing in x on $(K_F^* - K_M^*,\, K_F^*)$ at each point of the equilibrium path $C_M \in (0, \bar{C}_M]$.

Condition 1.9 says the renewable shift bites harder in deep-surplus states than in tight states. It is mild and interpretable, and it holds for the entire power-law family, where $g(x) = (\bar{D} - x)/Q$ is affine and strictly decreasing for every α.

Condition 1.10 (Regularity)**.** On the adoption path $C_M \in (0, \bar{C}_M]$ the program has an interior maximizer (K_R^*, K_F^*, K_M^*) at which the Hessian $\mathcal{H}$ of Π is nonsingular.

Condition 1.10 is exactly what the implicit function theorem needs to make the path $C_M \mapsto (K_R^*, K_F^*, K_M^*)$ a C^1 curve; at a maximizer, nonsingularity forces $\mathcal{H} \prec 0$. It is verified directly: §1.7 computes the eigenvalues of $\mathcal{H}$ along the entire path and finds all three strictly negative throughout the interior region. The one place it fails is the *knife-edge boundary*, where C_R is small enough that $K_R^* \to \infty$ (Proposition 1.6): there $\mathcal{H}$ acquires a zero eigenvalue as the marginal value of renewables turns flat. But that boundary needs no comparative-statics argument—$K_R^* = \infty > K_R^\circ$ holds trivially—so Condition 1.10 costs nothing where Theorem 1.11 is actually used (interior optima with finite K_R^*).

Theorem 1.11 (Capacity-substitution channel)**.** *Under Conditions 1.9–1.10, at every interior point of the adoption path the equilibrium derivative is*

$$\frac{dK_R^*}{dC_M} = \frac{\operatorname{cof}_{K_M K_R}(\mathcal{H})}{\det \mathcal{H}} \le 0, \qquad \operatorname{cof}_{K_M K_R}(\mathcal{H}) \xrightarrow[\text{large-FCL}]{} (R-\beta+p)(R-c)\big[F_{K_R}(K_F^*)f(K_F^*-K_M^*) - F_{K_R}(K_F^*-K_M^*)f(K \tag{1.17}$$

where $\mathcal{H}$ is the Hessian of Π (so $\det \mathcal{H} < 0$), and the displayed bracket is the large-FCL ($K_M^* > K_R^*$) form of the cofactor. Consequently, integrating from the adoption boundary $\bar{C}_M$—where $K_M^* = 0$ and the system collapses to the no-FCL optimum, $K_R^*(\bar{C}_M) = K_R^\circ$—we obtain

$$K_R^*(C_M) \;\ge\; K_R^\circ \qquad \text{for all } C_M \le \bar{C}_M, \tag{1.18}$$

with strict inequality whenever $C_M < \bar{C}_M$ and g is strictly decreasing. Renewable capacity is strictly higher with the FCL program.

Proof. Only $\partial\Pi/\partial K_M$ in (1.3) depends on C_M, with $\partial^2\Pi/\partial K_M \partial C_M = -1$. Implicit differentiation of the FOC system $\nabla\Pi = 0$ gives $dK_R^*/dC_M = (\mathcal{H}^{-1})_{K_R K_M} = \operatorname{cof}_{K_M K_R}(\mathcal{H})/\det \mathcal{H}$. In the large-FCL regime $F_{K_R}(-K_M) = f(-K_M) = 0$, so by (1.13)–(1.15) and $\partial^2\Pi/\partial K_M^2 = -(R-c)f(K_F - K_M)$, the relevant cofactor simplifies (after cancellation) to

$$\operatorname{cof}_{K_M K_R}(\mathcal{H}) = (R-\beta+p)(R-c)\big[F_{K_R}(K_F^*)f(K_F^* - K_M^*) - F_{K_R}(K_F^* - K_M^*)f(K_F^*)\big].$$

Dividing by $\det \mathcal{H} < 0$ gives (1.17). The bracket equals $f(K_F^*)f(K_F^*-K_M^*)[g(K_F^*)-g(K_F^*-K_M^*)] \le 0$ under Condition 1.9 (since $K_F^* - K_M^* < K_F^*$); because $\beta + p > R$ we have $(R-\beta+p)(R-c) < 0$, so the cofactor is ≥ 0, and dividing by $\det \mathcal{H} < 0$ gives $dK_R^*/dC_M \le 0$.

The path crosses regimes. At $C_M = \bar{C}_M$ the marginal profit of the first FCL unit (1.3) vanishes ($(R-c)\varphi^\circ - \bar{C}_M = 0$), so $K_M^* = 0$: the top of the path is the *small*-FCL regime ($K_M^* < K_R^*$), turning large-FCL only once K_M^* grows past K_R^*. In the small-FCL regime $F_{K_R}(-K_M^*) > 0$, which adds a single term to the relevant cofactor $\mathrm{cof}_{K_M K_R}(\mathcal{H}) = H_{RF}H_{FM} - H_{RM}H_{FF}$ through $H_{RM} = (R-c)F_{K_R}(K_F^* - K_M^*) + c\,F_{K_R}(-K_M^*)$: it becomes $\mathrm{cof}^{\text{large}}_{K_M K_R} - c\,F_{K_R}(-K_M^*)\,H_{FF}$, and since $H_{FF} = (R-\beta+p)f(K_F^*) - (R-c)f(K_F^* - K_M^*) < 0$ under Assumption 1.1 this extra term is ≥ 0. Hence $\mathrm{cof}_{K_M K_R} \ge 0$ and $dK_R^*/dC_M \le 0$ *throughout* the path, not only on its large-FCL part (confirmed numerically across the crossing in §1.7: $dK_R^*/dC_M < 0$ at every $C_M \in \{0.77, \dots, 0.30\}$, spanning both regimes). Since the no-FCL conditions hold at $C_M = \bar{C}_M$, $K_R^*(\bar{C}_M) = K_R^\circ$. By the implicit function theorem (Condition 1.10, $\mathcal{H}$ nonsingular), $C_M \mapsto K_R^*$ is C^1 on each of the two open regime intervals and continuous across the single boundary point $K_M^* = K_R^*$; hence it is *absolutely continuous* on the compact interval $[C_M, \bar{C}_M]$. An absolutely continuous function with a.e.-nonpositive derivative is non-increasing, so integrating $dK_R^*/dC_M \le 0$ from $\bar{C}_M$ down to C_M yields $K_R^*(C_M) \ge K_R^\circ$. □

Remark 1.12 (The submodularity is not an obstruction). By (1.15), Π is *submodular* in (K_R, K_F) while having increasing differences in $(K_R; K_M)$ and $(K_F; K_M)$; since reversing the order on K_F flips the latter signs, no single relabeling restores supermodularity, and Topkis/Milgrom–Shannon cannot be invoked off the shelf. Theorem 1.11 shows this is immaterial: the homotopy in C_M together with the local cofactor sign is a *complete* proof, requiring only Condition 1.10. The lattice machinery is sufficient, not necessary, for monotone comparative statics; a signed one-parameter cofactor does the same job here without it.

Corollary 1.13 (Power-law / footnote 6)**.** *For every power-law residual demand ($\alpha > 0$), $g(x) = (\bar{D} - x)/Q$ is strictly decreasing, so Condition 1.9 holds and $K_R^* > K_R^\circ$ whenever $C_M < \bar{C}_M$. This is precisely why footnote 6's closed-form extension carries through for the whole family rather than the uniform case alone.*

Remark 1.14 (Reading of Condition 1.9). Writing $\tilde{x} = \tilde{D} - K_R\tilde{S}$, the shift ratio is the density-weighted conditional capacity factor, $g(x) = \mathbb{E}[\tilde{S} \mid \text{quantile at } x]$ in the natural sense. Condition 1.9 fails only if high-demand states are systematically *windier/sunnier* than surplus states—an empirically unusual positive demand–supply correlation. Under the more realistic negative or zero correlation, the condition holds and the channel operates.

1.4 The net carbon condition, in general

Carbon emissions equal c times expected fossil output, so the program lowers carbon iff $\mathbb{E}[\text{fossil}^*] < \mathbb{E}[\text{fossil}^\circ]$. Decompose (1.2)'s fossil terms exactly. Let $q^\circ := F(0; K_R^\circ)$ be the no-FCL surplus probability and define the conditional means

$$\mu^*_{\text{low}} := \frac{1}{F(K_F^* - K_M^*; K_R^*)} \int_{-K_R^*}^{K_F^* - K_M^*} x\, f(x; K_R^*)\, dx, \qquad \mu^\circ_{\text{mid}} := \frac{1}{\varphi^\circ - q^\circ} \int_0^{K_F^\circ} x\, f(x; K_R^\circ)\, dx.$$

Proposition 1.15 (Exact net-carbon inequality)**.** *For any F satisfying Assumption 1.1,*

$$\mathbb{E}[\textit{fossil}^*] = K_F^* - \tfrac{C_M}{R-c}(K_F^* - K_M^* - \mu^*_{\text{low}}), \tag{1.19}$$

$$\mathbb{E}[\textit{fossil}^\circ] = K_F^\circ(1 - \varphi^\circ) + (\varphi^\circ - q^\circ)\,\mu^\circ_{\text{mid}}, \tag{1.20}$$

and carbon falls with the program iff

$$\underbrace{\frac{C_M}{R-c}(K_F^* - K_M^* - \mu_{\text{low}}^*)}_{\textit{utilization effect } (+)} \;>\; \underbrace{K_F^* - K_F^\circ(1-\varphi^\circ) - (\varphi^\circ - q^\circ)\mu_{\text{mid}}^\circ}_{\textit{net of capacity effects}}. \tag{C}$$

Proof. Integrate each fossil integral by parts using $\int_a^b (1-F)\,dx = [x(1-F)]_a^b + \int_a^b x f\,dx$; for (1.19) take $a = -K_M^*,\ b = K_F^* - K_M^*$ with $F(a; K_R^*) = 0$ (large-FCL) and $F(b; K_R^*) = C_M/(R-c)$; for (1.20) take $a = 0,\ b = K_F^\circ$ with $F(K_F^\circ; K_R^\circ) = \varphi^\circ$. Subtract. □

Two forces. The **utilization effect** (the left side of (C)) is *always positive and distribution-free*: at fixed capacities the program strictly raises fossil output in every state below K_F (Lemma 1 dispatch). The **capacity-substitution effect** is carried by the right side through $K_R^* > K_R^\circ$ (Theorem 1.11), which lowers $\mathbb{E}[\text{fossil}^*]$ by shifting the residual-demand distribution left. The net sign is which one wins.

Corollary 1.16 (Shape comparison to the uniform, at matched cutoffs)**.** *For any F with*

(a) $f(\cdot; K_R^*)$ non-increasing on $(-K_R^*, K_F^* - K_M^*)$ (equiv. $F(\cdot; K_R^*)$ concave there), and

(b) $f(\cdot; K_R^\circ)$ non-decreasing on $(0, K_F^\circ)$ (equiv. $F(\cdot; K_R^\circ)$ convex there),

we have $\mu_{\text{low}}^* \le (K_F^* - K_M^* - K_R^*)/2$ (so FCL fossil output is no larger than under the uniform) and $\mu_{\text{mid}}^\circ \ge K_F^\circ/2$ (so no-FCL output is no smaller). Hence the net saving Δ is at least *its uniform value at the same distribution-free cutoffs.* ***This is a shape comparison, not a proof of carbon reduction.*** *It signs $\Delta > 0$ only where the uniform benchmark already does—and Week 2 shows the uniform's* interior *value $\Delta_0 < 0$ (carbon falls under the uniform only at its corner $K_R^* \to \infty$, where $\mathbb{E}[\textit{fossil}^*] \to 0$). So (a)–(b) bound Δ from below but do* not *by themselves establish interior carbon reduction; that requires the optimized (surplus) contract of Week 3 or the corner analysis. The uniform satisfies (a)–(b) with equality.*

Remark 1.17 (Why no level-feedback term is missing). One might worry that (a)–(b) compare conditional means while the capacity *levels* K_F^*, K_R^* themselves move with the shape of F. They do, but the comparison is nonetheless exact, for two reasons. First, by Lemma 1.3 the *probabilities* at the comparison limits ($C_M/(R-c)$ at $K_F^* - K_M^*$, φ° at K_F°) are distribution-free, so (a)–(b) bound genuine, like-for-like quantile integrals—there is no hidden dependence on F in the limits. Second, the one level effect that does survive, the renewable shift $K_R^* > K_R^\circ$, is signed once and for all by Theorem 1.11; it enters (1.19) only through the leftward relocation of mass, which (a) already accounts for. Hence (C) together with Theorem 1.11 is an *exact* identity at fixed cutoffs—no second-order feedback term is dropped. (It is not a distribution-free *sign*: Week 2 shows the sign of Δ across F is not decidable from the cutoffs and capacities alone.)

Write $\Delta(C_R) := \mathbb{E}[\text{fossil}^\circ] - \mathbb{E}[\text{fossil}^*]$ for the net fossil saving at the optimum given renewable cost C_R (carbon falls iff $\Delta > 0$). The next theorem has a *fully general* upper anchor and a small-cost window established for the knife-edge (power-law / scale) families.

Theorem 1.18 (Viability threshold: general anchor, explicit window)**.** *Fix all primitives except C_R, with $C_M < \bar{C}_M$ (so $K_M^* > 0$).*

(i) (Large-cost anchor — holds for every F.) *For all C_R above a finite B_0, no renewables are built, $K_R^* = K_R^\circ = 0$, and the FCL program differs from the no-FCL grid only by running computing on spare fossil; the utilization effect is strictly positive, so $\Delta(C_R) = -U_0 < 0$. Hence the carbon-reduction set $\{C_R : \Delta(C_R) \ge 0\}$ is bounded above, for* every *F.*

(ii) (Small-cost window — knife-edge families.) *Suppose the* FCL *renewable problem admits unbounded investment for small C_R (true for the power-law family, Proposition 1.6, and more generally any family whose renewable marginal value is constant along the re-optimization locus), and that renewable output is a.s. positive. Then there are thresholds $0 \le C_R^\circ < C_R^\dagger$ with: for $C_R < C_R^\dagger$, $K_R^* = \infty$ and $\mathbb{E}[\textit{fossil}^*] = 0$; for $C_R > C_R^\circ$, K_R° finite and $\mathbb{E}[\textit{fossil}^\circ] > 0$. So on $(C_R^\circ, C_R^\dagger)$ carbon strictly falls, $\Delta = \mathbb{E}[\textit{fossil}^\circ] > 0$.*

(iii) Therefore $C_R^\dagger := \sup\{C_R : \Delta(C_R) \ge 0\}$ is finite (part i, general) and, for the knife-edge families, positive (part ii), with carbon rising for $C_R > C_R^\dagger$. For the power-law family $C_R^\dagger, C_R^\circ$ are the explicit values of Proposition 1.19. (For a general F with an interior renewable optimum, $\{\Delta \ge 0\}$ is bounded above by part (i); its non-emptiness—i.e. that carbon falls somewhere—is what Condition CS delivers in the cases we can sign, and is conjectured in general.)

Proof. (i) The renewable marginal benefit at $K_R = 0$ is a finite constant B_0 (the C_R-free part of (1.5)); for $C_R > B_0$ the FOCs give the corner $K_R^* = K_R^\circ = 0$. With no renewables, residual demand is $\tilde{x} = \tilde{D} \ge 0$. By Lemma 1 the program dispatches computing on spare fossil, which strictly raises fossil output in every state with positive spare capacity (the distribution-free utilization effect of §1.4); and $K_F^* \ge K_F^\circ$ because, at common $K_R = 0$, the with-FCL no-blackout probability $F(K_F^*) = \frac{\beta+p-c-C_F-C_M}{\beta+p-R} \ge \varphi^\circ = F(K_F^\circ)$ when $C_M \le \bar{C}_M$ (equality at $\bar{C}_M$, larger as C_M falls; Lemma 1.3). Both push the same way, so $\mathbb{E}[\text{fossil}^*] > \mathbb{E}[\text{fossil}^\circ]$ and $\Delta = -U_0 < 0$. This uses no distributional assumption. *(ii)* In the knife-edge regime the with-FCL renewable marginal value is a constant MV_* along the locus, so $K_R^* = \infty$ iff $C_R < C_R^\dagger := \text{MV}_*$. The no-FCL renewable value $\text{MV}_\circ(K_R^\circ)$ is *not* constant—it is strictly decreasing in K_R° with infimum $C_R^\circ := \lim_{K_R^\circ \to \infty} \text{MV}_\circ$—so $K_R^\circ = \infty$ iff $C_R < C_R^\circ$ (and the no-FCL problem has a finite interior K_R° for $C_R > C_R^\circ$). Since (1.5) weights the dispatch band by R where the no-FCL FOC weights it by $c < R$, $\text{MV}_*(K_R) > \text{MV}_\circ(K_R)$ at every K_R, hence $C_R^\dagger = \text{MV}_* > C_R^\circ$ (for the power-law family this is the explicit comparison of Proposition 1.19). For $C_R \in (C_R^\circ, C_R^\dagger)$: $K_R^* = \infty$ with $\mathbb{P}(\tilde{S} > 0) = 1$ sends $\tilde{x} \to -\infty$ a.s. so $\mathbb{E}[\text{fossil}^*] = 0$, while K_R° is finite and $\mathbb{E}[\text{fossil}^\circ] > 0$. *(iii)* Combine: on $(C_R^\circ, C_R^\dagger)$, $\mathbb{E}[\text{fossil}^*] = 0 < \mathbb{E}[\text{fossil}^\circ] = \Delta$; with (i) the set $\{\Delta \ge 0\}$ is nonempty and bounded above, so its supremum $C_R^\dagger \in (0, \infty)$ exists. □

Under the uniform/power-law family the thresholds of Theorem 1.18 are the explicit knife-edge values of Proposition 1.6: $C_R^\dagger = \text{MV}$ and $C_R^\circ = C_F^2/2(\beta + p - c)$, recovering paper eqs. (25)–(26).

Proposition 1.19 (Explicit $C_R^\dagger$ for the power-law family)**.** *Let F be power-law. With*

$$C_R^\dagger = \text{MV} = \tfrac{1}{2}\left[c + \frac{(R - C_M - c)^2}{R - c} + \frac{(R - c - C_F - C_M)^2}{\beta + p - R}\right] \qquad \textit{and} \qquad C_R^\circ = \frac{C_F^2}{2(\beta + p - c)},$$

we have $C_R^\circ < C_R^\dagger$, and for every $C_R \in (C_R^\circ, C_R^\dagger)$ the FCL *optimum is $K_R^* = \infty$ (so $\mathbb{E}[\textit{fossil}^*] \to 0$) while the no-*FCL *optimum K_R° is finite (eq. (1.8)). Hence carbon strictly falls for all $C_M < \bar{C}_M$ on this nonempty C_R-window.*

Proof. $C_R^\dagger = \text{MV}$ is Proposition 1.6: $K_R^* = \infty$ iff $\text{MV} > C_R$. The no-FCL optimum (1.8) is finite and positive iff $0 < 2C_R - C_F^2/(\beta + p - c)$, i.e. $C_R > C_R^\circ$, where $C_R^\circ = C_F^2/(2(\beta + p - c))$ is the *limit* of the (strictly decreasing) no-FCL renewable marginal value $\text{MV}_\circ(K_R^\circ)$ as $K_R^\circ \to \infty$ (only the blackout-hedge term survives; the no-FCL value is *not* constant—it decreases to C_R°). For $C_R^\circ < C_R^\dagger$ we use pointwise domination rather than a single-term bound (no one term of MV_* dominates C_R° across all parameters): at any common K_R the with-FCL renewable marginal value exceeds the

no-FCL one, because (1.5) redispatches the band $(K_F - K_M, K_F)$ at rate R where the no-FCL FOC weights it by $c < R$, so $\mathrm{MV}_*(K_R) > \mathrm{MV}_\circ(K_R)$ for every K_R. Since MV_* is constant on the knife-edge locus and $\mathrm{MV}_\circ(K_R) \downarrow C_R^\circ$, we get $C_R^\dagger = \mathrm{MV}_* \geq C_R^\circ$, strict because $R > c$ on a positive-measure band. Verified at the §1.7 calibration, $C_R^\circ = 0.040 < C_R^\dagger = 0.344$. □

1.5 A distribution-free reading: sufficient statistic, magnitude bounds, and a robust certificate

The exact inequality (C) appears to depend on the entire residual-demand distribution. Two of its three channels do not: the cutoff *probabilities* are distribution-free (Lemma 1.3), and—*at the equilibrium capacities*—the remaining dependence is affine in two conditional means. We are careful throughout this section to separate what is genuinely distribution-free (the cutoff probabilities) from what still moves with F (the capacity *levels* $K_F^*, K_M^*, K_R^*, K_F^\circ$ and the surplus probability $q^\circ = F(0; K_R^\circ)$). The results below are exact statements at fixed capacities; the residual cross-distribution dependence through the capacities is discussed at the end.

Theorem 1.20 (Affine representation at fixed capacities)**.** *Fix the equilibrium capacities and q°. Then the net fossil saving $\Delta = \mathbb{E}[\textit{fossil}^\circ] - \mathbb{E}[\textit{fossil}^*]$ is* affine *in the two conditional means*

$$\mu_{\mathrm{low}}^* = \mathbb{E}[\tilde{x} \mid \tilde{x} \in (-K_R^*, K_F^* - K_M^*)], \qquad \mu_{\mathrm{mid}}^\circ = \mathbb{E}[\tilde{x} \mid \tilde{x} \in (0, K_F^\circ)],$$

through the explicit map

$$\Delta = \Big[K_F^\circ(1-\varphi^\circ) + (\varphi^\circ - q^\circ)\mu_{\mathrm{mid}}^\circ\Big] - \Big[K_F^* - \tfrac{C_M}{R-c}(K_F^* - K_M^* - \mu_{\mathrm{low}}^*)\Big]. \tag{1.21}$$

Two distributions with the same equilibrium capacities, the same q°, and the same two conditional means yield the same Δ.

Proof. Immediate from the exact identities (1.19)–(1.20) of Proposition 1.15; the cutoff probabilities φ° and $C_M/(R-c)$ are distribution-free (Lemma 1.3, large-FCL). *(Verified: the right side of (1.21) matches direct quadrature of the fossil integrals to 6×10^{-7}.)* □

Remark 1.21 (Not a sufficient statistic across F). Equation (1.21) is *not* a claim that Δ depends on F only through $(\mu_{\mathrm{low}}^*, \mu_{\mathrm{mid}}^\circ)$: the coefficients themselves involve the capacities and q°, which move with the shape of F. It is the exact *conditional* representation—the lever for the bounds and the least-favorable comparison below, both of which are stated at fixed capacities.

Theorem 1.22 (Magnitude bounds from cutoffs, at fixed capacities)**.** *Because a conditional mean lies between its interval endpoints, $\mu_{\mathrm{low}}^* \in (-K_R^*, K_F^* - K_M^*)$ and $\mu_{\mathrm{mid}}^\circ \in (0, K_F^\circ)$, the carbon saving obeys the closed-form bounds*

$$\underbrace{K_F^\circ(1-\varphi^\circ) - K_F^*}_{\Delta_{\min}} \;\leq\; \Delta \;\leq\; \underbrace{K_F^\circ - K_F^* + \tfrac{C_M}{R-c}(K_F^* - K_M^* + K_R^*)}_{\Delta_{\max}}, \tag{1.22}$$

expressed in the equilibrium capacities and the distribution-free reliability φ° (the capacities being held at their equilibrium values). The bounds are tight given those capacities: each is approached as the relevant conditional law degenerates to an interval endpoint.

Proof. Substitute the endpoint values of $\mu^*_{\text{low}}, \mu^\circ_{\text{mid}}$ into (1.21), using its monotonicity in each mean: $\partial\Delta/\partial\mu^*_{\text{low}} = -\frac{C_M}{R-c} < 0$ (since a larger μ^*_{low} raises $\mathbb{E}[\text{fossil}^*]$) and $\partial\Delta/\partial\mu^\circ_{\text{mid}} = \varphi^\circ - q^\circ > 0$. The smallest Δ thus takes the *largest* $\mu^*_{\text{low}} = K^*_F - K^*_M$ (giving $\mathbb{E}[\text{fossil}^*] = K^*_F$) and the smallest $\mu^\circ_{\text{mid}} = 0$, yielding $\Delta_{\min}$; the largest Δ takes the smallest $\mu^*_{\text{low}} = -K^*_R$ and largest $\mu^\circ_{\text{mid}} = K^\circ_F$, and $q^\circ \geq 0$ then bounds the last term, yielding $\Delta_{\max}$. *(Verified at a stable interior equilibrium:* $\Delta_{\min} = -6.43 \leq \Delta = -2.17 \leq \Delta_{\max} = +3.23$*.)* □

Theorem 1.23 (Least-favorable conditional shape)**.** *Fix the equilibrium capacities and q°. Among conditional laws whose density is non-increasing on the* FCL *zone $(-K^*_R, K^*_F - K^*_M)$ and non-decreasing on the no-*FCL *zone $(0, K^\circ_F)$ (the Corollary 1.16 shape class $\mathcal{U}$), the* uniform *(linear CDF) minimizes the carbon saving Δ of* (1.21)*.*

Proof. By (1.21), at fixed capacities Δ is *decreasing* in μ^*_{low} (coefficient $-C_M/(R-c)$) and *increasing* in μ°_{mid} (coefficient $\varphi^\circ - q^\circ > 0$), so it is minimized by the *largest* feasible μ^*_{low} and the *smallest* feasible μ°_{mid}. A non-increasing density on the FCL zone has conditional mean at *most* the interval midpoint, so the largest feasible μ^*_{low} over $\mathcal{U}$ is the midpoint; a non-decreasing density on the no-FCL zone has conditional mean at *least* the midpoint, so the smallest feasible μ°_{mid} is the midpoint. Both midpoints are attained *simultaneously* by the uniform (constant density), which therefore minimizes Δ. *(Verified: over* 3000 *random monotone densities the conditional means respect the midpoint bounds in* 3000/3000*, and a continuous tilt from uniform into $\mathcal{U}$ raises Δ — uniform is the minimizer.)* □

Remark 1.24 (Scope: this is a fixed-capacity statement). Theorem 1.23 does *not* establish that the uniform is the worst case for the realized saving $\Delta(F)$ across all $F \in \mathcal{U}$, because the equilibrium capacities $K^*_F, K^*_M, K^*_R, K^\circ_F$ and q° co-vary with the shape of F (Lemma 1.3 pins only the cutoff *probabilities*). The direction of the surviving capacity effect—the renewable shift—is signed by Theorem 1.11 ($K^*_R > K^\circ_R$), but a full cross-F minimization would have to control the capacity covariation, which we do not close. We therefore state the genuinely distribution-robust certificate only in its honest form: the closed-form $C_R \leq C^\dagger_R$ guarantees $\Delta \geq 0$ for the uniform/power-law family (Proposition 1.19), and Theorem 1.23 shows the uniform is the least-favorable *conditional shape*—not yet the least-favorable *distribution*—within $\mathcal{U}$. (Note the uniform residual demand is itself the knife-edge of Proposition 1.6, with a corner rather than interior K^*_R; closing the cross-F statement is left open.)

Remark 1.25 (What this buys). At fixed equilibrium capacities, Theorems 1.20–1.23 reduce the carbon effect to a two-mean affine form, bracket it by bounds a regulator can compute from cutoff data, and identify the uniform as the least-favorable conditional shape—so the distributional detail of F controls the *magnitude* of the carbon change at given capacities. Turning the fixed-capacity least-favorable result into a genuine distribution-robust certificate over all of $\mathcal{U}$ requires signing the capacity covariation (Remark 1.24); that step is open.

1.6 Reconciling the regimes

The analysis lives in the large-FCL interior regime. Three other regimes appear in the paper: small-FCL ($K^*_M \leq K^*_R$, Appendix G, with the widened threshold $\bar{C}^b_M = \bar{C}_M + c\,F(0; K^\circ_R)$, paper eq. 62), full-coverage ($R \geq c + C_F + C_M$, zero blackout), and no adoption. They fit into one ordered map.

Proposition 1.26 (Regime map and unified triple win)**.** *The relevant C_M-thresholds are strictly ordered,*

$$\underbrace{R-c-C_F}_{C_M^{\mathrm{fc}}} \;<\; \underbrace{(R-c)\varphi^\circ}_{\bar{C}_M} \;<\; \underbrace{\bar{C}_M + c\,F(0;K_R^\circ)}_{\bar{C}_M^b}, \tag{1.23}$$

and as C_M falls the optimum passes through: (0) *no adoption, $C_M \geq \bar{C}_M^b$;* (I) *small-FCL adoption, $\bar{C}_M \leq C_M < \bar{C}_M^b$;* (II) *large-FCL interior adoption, $C_M^{\mathrm{fc}} < C_M < \bar{C}_M$;* (III) *full coverage, $C_M \leq C_M^{\mathrm{fc}}$ (blackout $\equiv 0$). The profit and blackout parts of the triple win hold in the* large-FCL *regions* (II)–(III) *for every F, since there the cutoff probabilities are distribution-free (Lemma 1.3); in the small-FCL region* (I) *the cutoffs are not pinned by the FOCs alone (Lemma 1.3), and the profit/blackout comparison there relies on the Appendix-G analysis (Role 4). The carbon part holds under Condition 1.9 in* (II)*; in* (III) *it holds trivially, fossil serving only consumers with FCL on renewable surplus. We do* not *claim it "a fortiori" in* (I)*: a smaller program shrinks both sides of (C), and signing the net change there is left to the small-FCL analysis.*

Proof. $\bar{C}_M < \bar{C}_M^b$ is immediate since $c\,F(0;K_R^\circ) > 0$. For $C_M^{\mathrm{fc}} < \bar{C}_M$,

$$\bar{C}_M - C_M^{\mathrm{fc}} = (R-c)\varphi^\circ - (R-c-C_F) = C_F - (R-c)\frac{C_F}{\beta+p-c} = C_F\,\frac{\beta+p-R}{\beta+p-c} > 0$$

by Assumption 1.1 ($\beta+p > R$). Region (III) is the degenerate favorable case of §1.2: $R \geq c+C_F+C_M$ makes fossil marginal benefit positive up to $\bar{D}$, blackout vanishes, and the carbon comparison reduces to the surplus-absorption argument with no fossil-fed FCL at the margin. The small-FCL utilization effect is bounded by its large-FCL counterpart because $K_M^* \leq K_R^*$ shrinks the fossil-fed dispatch window $(K_F^* - K_M^*, K_F^*)$. □

Equation (1.23) stitches the regimes: adoption occurs iff $C_M < \bar{C}_M^b$, and the triple-win argument carries across the whole adoption range, strict in (I)–(II) and degenerate-favorable in (III). The remaining small-FCL fine structure (the exact C_M at which $K_M^* = K_R^*$) is the one boundary we leave to Role 4's regime audit.

1.7 Numerical validation

All claims are checked by an independent solver that quadratures the unified profit (1.2) and solves (1.3)–(1.5) by Newton methods (no closed forms assumed). Highlights:

- **Uniform closed forms.** Carbon eqs. (25)/(26) match quadrature to $\leq 6 \times 10^{-10}$; capacities (1.8)–(1.9) satisfy the FOCs to $\leq 10^{-15}$.
- **Knife-edge.** Along the *large-FCL* portion of the re-optimization locus, $\mathrm{MV}(K_R)$ is constant in K_R (verified at $K_R \in \{1,4,5\}$ for $\alpha \in \{0.5,1,2\}$, e.g. uniform 0.3444 at both $K_R = 1, 4$); the locus leaves this regime for large K_R ($K_M^* < K_R^*$), where MV is no longer constant. Eq. (1.12) reproduces paper eq. (24).
- **Capacity channel.** On strictly-concave Beta scale-mixtures $\tilde{x} = \tilde{D} - K_R\tilde{S}$ with interior optima, across symmetric, decreasing-density, and right-skewed consumer demand ($D \sim \bar{D} \cdot \mathrm{Beta}(2,2), (1,3), (2,5), (2,8)$), every case has g strictly decreasing, $K_R^* > K_R^\circ$, and the sign of dK_R^*/dC_M matching the shift-ratio bracket of (1.17). Representative cell ($\mathrm{Beta}(1,3)$): $K_R^* = 3.274 > K_R^\circ = 0.831$, $g(K_F^* - K_M^*) = 0.457 > g(K_F^*) = 0.414$, $dK_R^*/dC_M = -2.99 < 0$ with bracket $-3.1 \times 10^{-4} < 0$. Strongly left-skewed demand pushes the optimum to the full-coverage

boundary ($K_F^* \to \bar{D}$), i.e. footnote 6's "harder" regime, where the interior-Hessian hypothesis of Theorem 1.11 no longer applies.

- **Regime crossing & Hessian.** Scanning C_M from $\bar{C}_M$ downward on Beta(2, 2): the optimum is small-FCL for $C_M \in \{0.77, 0.70, 0.60\}$ ($K_M^* < K_R^*$) and large-FCL for $C_M \in \{0.50, 0.40, 0.30\}$, with $dK_R^*/dC_M < 0$ at *every* point ($-4.28, -2.76, -2.19, -5.21, -21.1, -92.1$; the last is a noisy finite-difference at the knife-edge $K_R^* = 9.2$), confirming the channel across the crossing. The Hessian's three eigenvalues are all strictly negative on the interior path and acquire a zero only as $K_R^* \to \infty$ at the knife-edge ($C_M = 0.30$, $K_R^* = 9.2$ and climbing)—the boundary where Condition 1.10 lapses and the inequality is anyway immediate.

- **Exactness audit.** `verify_exact.py` checks all 15 analytical formulas (gradients, cross-partials, cofactor, knife-edge MV, distribution-free cutoffs, fossil decomposition) against finite differences of Π: 15/15 agree to 6–16 digits.

- **Viability threshold (Theorem 1.18).** `verify_threshold.py`: sweeping C_R on Beta(2, 5) right-skewed demand, carbon falls for small C_R (K_R^* unbounded, $\mathbb{E}[\text{fossil}^*] \to 0$, $\Delta = +0.56, +1.07$ at $C_R = 0.25, 0.35$) and rises once C_R crosses $C_R^\dagger \in (0.35, 0.5)$ ($\Delta = -3.68, \ldots, -6.47$ through $C_R = 1.6$). The large-cost anchor (i) is the rigorous, fully general part; the small-cost window is verified here only for scale-mixtures, where two caveats apply—the equilibrium solver is erratic near the knife-edge (we report the clean cold-started points), and for strongly right-skewed demand the no-FCL grid also builds heavy renewables, so the strict fall on $(C_R^\circ, C_R^\dagger)$ can be razor-thin ($\Delta \to 0^+$).

- **Sufficient statistic, bounds, least-favorable (§1.5).** `verify_carbon_robust.py`: at a stable interior equilibrium the two-moment formula (1.21) matches direct quadrature to 6×10^{-7}; Δ lies inside the closed-form bounds (1.22) ($-6.43 \le -2.17 \le +3.23$); and over 3000 random monotone densities the conditional means respect the midpoint bounds in 3000/3000 cases (uniform least-favorable *at fixed capacities*; the cross-distribution version is discussed in §1.5).

- **Sharpness (algebraic).** The bracket in (1.17) changes sign with $g(K_F^*) - g(K_F^* - K_M^*)$, so an increasing g would reverse the channel ($K_R^* < K_R^\circ$); the shift-ratio condition is therefore plausibly tight. A realized reversed-channel example proved hard to construct—even at demand–supply correlation $+0.95$ the equilibrium window stayed in the decreasing-g region—so an explicit counterexample is left to Role 3 (comparative statics).

1.8 Status of the open points

Of the four issues the uniform-only treatment leaves open, three are closed and the fourth is reduced to one boundary; the companion memo (`gap_memo.md`) carries the same accounting.

1. **Submodularity — closed.** By (1.15), Π is submodular in (K_R, K_F) with increasing differences in each of $(K_R; K_M)$ and $(K_F; K_M)$; no relabeling fixes both signs, so a lattice argument does not apply. Remark 1.12 shows this is not an obstruction: Theorem 1.11's homotopy-plus-cofactor proof is complete under Condition 1.10 ($\mathcal{H} \prec 0$, verified by eigenvalue computation along the path in §1.7; it lapses only at the knife-edge, where $K_R^* = \infty > K_R^\circ$ is immediate).

2. **Endogenous-capacity feedback — closed at fixed capacities.** Lemma 1.3 pins the comparison *probabilities* distribution-free, and Theorem 1.11 signs the one surviving level effect ($K_R^* > K_R^\circ$); at fixed capacities (C) is then a complete affine representation (Theorem 1.20,

Remark 1.17). The residual cross-distribution capacity covariation (Remark 1.24) is signed in direction but not fully closed.

3. $C_R^\dagger$ — **general upper anchor, explicit window.** Theorem 1.18 proves $\{C_R : \Delta \geq 0\}$ is bounded above for *every* F (the rigorous large-cost limit $\Delta \to -U_0 < 0$), and exhibits a nonempty strict-fall window $(C_R^\circ, C_R^\dagger)$ for the knife-edge / power-law family, with explicit $C_R^\dagger = \mathrm{MV}$, $C_R^\circ = C_F^2/2(\beta+p-c)$ (Proposition 1.19). *This upgrades the earlier heuristic "the two curves cross" to a proof of the upper anchor; a general-F nonempty window remains conjectural.*

4. **Boundary regimes — ordered, one crossover open.** Proposition 1.26 proves the strict ordering $C_M^{\mathrm{fc}} < \bar{C}_M < \bar{C}_M^b$ and the triple win in the large-FCL and full-coverage regions; the small-FCL region (I) and the exact crossover $K_M^* = K_R^*$ rest on the Appendix-G analysis, handed to Role 4.

Chapter 2

Week 2: Existence, uniqueness, and the general-F emission theory

[Optimal Contract and General Carbon Theory (Week 2)] Optimal Contract and General Carbon Theory for Flexible Computing Loads:
Existence and Uniqueness, a Uniform Homotopy Condition, the
General-F Emission Keystone, Connectedness, and Sharpness

Summary. *This is the Role 1 Week 2 deliverable. It hardens the Week 1 carbon theory (`role1_carbon_theory.tex`) into five results—four proved, one resolved numerically—taking the same model and notation and staying in Role 1's large-FCL mandate (the one small-FCL touch is flagged as a conjecture pending Role 4).* One*, the regulated program has a maximizer, it is unique, and the within-period dispatch* $d^*(x) = \min\{(K_F - x)_+, K_M\}$ *is the unique optimal rule; uniqueness comes not from global concavity—which we neither assume nor have, since it lapses on the power-law knife-edge where the reduced renewable problem goes flat—but from a* recursive-triangular *structure—the distribution-free cutoffs pin* K_F *and then* K_M *as unique functions of* K_R*, after which a single scalar diagonal-dominance (reduced second-order) condition pins* K_R*.* Two*, support-wide monotonicity of the shift ratio* $g = F_{K_R}/f$ *makes the realized two-point endpoint inequality hold* uniformly *along the whole* C_M*-homotopy, even as the dispatch window moves—the exact content the Week 1 "integrate-downward" step needs.* Three *(the keystone), substituting the equilibrium map* $K^*(C_R)$ *into the exact inequality (C) yields the endogenous carbon condition, and we give a* primitive *sufficient condition—the endpoint shift-ratio condition plus a two-zone density (tail/skew) bound—that would make carbon fall at a* finite interior *optimum. We then settle whether it ever fires, with two* proved *results and a numerical benchmark. First, a fossil floor:* $\mathbb{E}[\textit{fossil}^*] \geq K_F^*(R - c - C_M)/(R - c)$ *(and* $\geq K_M^* - K_R^*$*), since an FCL exceeding the maximal renewable surplus* $\leq K_R^*$ *cannot be fully surplus-powered. Second—the key structural point—we prove the carbon sign is* irreducibly distributional*: holding the equilibrium capacities and cutoffs fixed, the admissible conditional laws make* Δ *range over an interval that straddles zero, so capacities and cutoffs together cannot determine* $\operatorname{sign}\Delta$ *and Week-1's conjecture admits no proof from the distribution-free structure alone. The residual question is then the sign of a cutoff-computable benchmark* Δ_0 *plus a density tilt; numerically, at every large-FCL interior optimum scanned* $\Delta_0 < 0$ *(*$\Delta_0^{\max} \approx -1.09$*) and the tilt reinforces it,* $\Delta < 0$ *(*$\Delta^{\max} \approx -1.40$*). Carbon strictly falls only in the cheap-renewable corner* $K_R^* \to \infty$*. So the honest reading is* negative*: interior carbon reduction is a corner phenomenon (the small-FCL regime is Role 4's, not claimed here).* Four*, the carbon-reduction set* $\{C_R : \Delta(C_R) \geq 0\}$ *is bounded above and non-empty—its non-emptiness coming from that power-law corner, not from an interior optimum—and is a single interval under an explicit monotone-renewable-shadow-value condition that, we note, already encodes the single-crossing property it certifies.* Five*, the sharp necessary-and-sufficient condition for* $K_R^* > K_R^\circ$ *is the* path integral *of the realized endpoint gap; pointwise-monotone* g *is sufficient but not necessary; and the two-sided bounds on the carbon saving are tight, attained by explicit degenerate conditional laws. Every identity is checked against quadrature of the profit integral in `verify_week2.py`.*

2.1 Setup

We keep the model and notation of `role1_carbon_theory.tex`. A regulated utility chooses renewable capacity K_R, dispatchable fossil K_F, and flexible-computing-load (FCL) capacity K_M. Residual demand is $\tilde{x} = \tilde{D} - K_R\tilde{S} \in [-K_R, \bar{D}]$ with CDF $F(x; K_R)$, density $f(x; K_R) > 0$ on $(-K_R, \bar{D})$, and $F_{K_R} := \partial F/\partial K_R \geq 0$, extended by $F \equiv 0$ below $-K_R$ and $F \equiv 1$ above $\bar{D}$. Per-MWh: FCL revenue R, fossil cost c, price p, blackout penalty β; amortized capital costs C_R, C_F, C_M.

Assumption 2.1 (Regularity, as in Week 1)**.** $R > c + C_M$, $\quad R < c + C_F + C_M$, $\quad \beta + p > c + C_F$, $\beta + p > R$.

Under the dispatch rule $d(x) = \min\{(K_F - x)_+, K_M\}$ the unified profit is

$$\Pi = p\,\mathbb{E}[\tilde{D}] + R\int_{K_F-K_M}^{K_F} F\,dx - c\int_{-K_M}^{K_F-K_M}(1-F)\,dx - (\beta+p)\int_{K_F}^{\bar{D}}(1-F)\,dx - C_F K_F - C_R K_R - C_M K_M, \tag{2.1}$$

with gradient (Week 1, Lemma 1)

$$g_M := \Pi_{K_M} = (R-c)F(K_F - K_M) + cF(-K_M) - C_M, \tag{2.2}$$

$$g_F := \Pi_{K_F} = R[F(K_F) - F(K_F - K_M)] - c[1 - F(K_F - K_M)] + (\beta+p)[1 - F(K_F)] - C_F, \tag{2.3}$$

$$g_R := \Pi_{K_R} = R\int_{K_F-K_M}^{K_F} F_{K_R}\,dx + c\int_{-K_M}^{K_F-K_M} F_{K_R}\,dx + (\beta+p)\int_{K_F}^{\bar{D}} F_{K_R}\,dx - C_R. \tag{2.4}$$

We work in the *large-FCL interior* regime $K_M \geq K_R$ (so $F(-K_M) = f(-K_M) = F_{K_R}(-K_M) = 0$), the regime in which the Week-1 carbon theory lives; throughout, "interior" means $0 < K_R < \infty$ and the cutoffs below sit strictly inside $(0, 1)$. Write $a := K_F - K_M$ for the lower dispatch endpoint and recall the distribution-free cutoff *values* (Week 1, Lemma 2)

$$\varphi_1 := F(a; K_R) = \frac{C_M}{R-c}, \qquad \varphi_2 := F(K_F; K_R) = \frac{\beta+p-c-C_F-C_M}{\beta+p-R}, \qquad \varphi^\circ := F(K_F^\circ; K_R^\circ) = \frac{\beta+p-c-C_I}{\beta+p-c} \tag{2.5}$$

Under Assumption 2.1, $0 < \varphi_1 < 1$ and $0 < \varphi_2 < 1$; the interior FCL (adoption) condition $K_M > 0$ is $\varphi_2 > \varphi_1$, equivalently $C_M < \bar{C}_M := (R-c)\varphi^\circ$.

2.2 Existence, uniqueness, and contract optimality

The optimum is not delivered by global concavity, which we neither assume nor establish. The Week-1 cross-partial $\Pi_{K_R K_F} = (R-\beta+p)F_{K_R}(K_F) - (R-c)F_{K_R}(a) \leq 0$ shows Π is *submodular* in (K_R, K_F)—a feature, not yet an obstruction (a submodular function can still be concave). The real point is that concavity genuinely *fails* on the power-law knife-edge: there the reduced renewable problem Ψ of (2.7) is flat ($\Psi' \equiv 0$, Condition 2.5 at equality), so Π is not strictly concave along the renewable direction and the naive route of signing the full 3×3 Hessian by inspection is unavailable. Uniqueness instead follows from a recursive-triangular structure together with a single scalar condition.

2.2.1 Dispatch optimality

Lemma 2.2 (Unique optimal dispatch via strict $R > c$)**.** *Fix capacities (K_R, K_F, K_M) and a residual-demand realization x. Consumers are served first (they outrank FCL for scarce fossil because*

$\beta + p - c > R - c$*): fossil generation* $\min\{(x)_+, K_F\}$ *meets consumer residual demand, leaving fossil headroom* $(K_F - (x)_+)_+$*, while renewable surplus* $(-x)_+$ *is free power. The* FCL *dispatch* $d \in [0, K_M]$ *is therefore feasible up to* $d \le (-x)_+ + (K_F - (x)_+)_+ = (K_F - x)_+$*, and the fossil it burns is* $(d - (-x)_+)_+$ *(surplus first, then headroom). Its within-period payoff is*

$$\pi_x(d) = R\,d - c\,(d - (-x)_+)_+, \qquad 0 \le d \le \min\{K_M, (K_F - x)_+\}.$$

Under Assumption 2.1 ($R > c$*) the unique maximizer is* $d^*(x) = \min\{(K_F - x)_+, K_M\}$*.*

Proof. π_x is continuous, piecewise linear, and strictly increasing in d: on the surplus segment $d \le (-x)_+$ the slope is $R > 0$ (FCL runs on free renewable power), and on the fossil segment $d > (-x)_+$ the slope is $R - c > 0$ (Assumption 2.1). Both slopes are strictly positive, so π_x is maximized at the right endpoint $d^* = \min\{(K_F - x)_+, K_M\}$, and uniquely so. At $R = c$ the fossil-segment slope vanishes and any $d \in [(-x)_+, \min\{K_M, (K_F - x)_+\}]$ is optimal; hence strict $R > c$ is exactly what pins d^*. Crucially the feasibility $d \le (K_F - x)_+$ lets surplus states $x < 0$ power the FCL beyond fossil headroom K_F—the channel the floor of Lemma 2.15 quantifies. This d^*, with consumer fossil $\min\{(x)_+, K_F\}$, reproduces the operating-cost terms of (2.1).[1] □

2.2.2 Existence

Proposition 2.3 (A maximizer exists)**.** Π *is continuous on* $\mathbb{R}^3_+$*, and there is a box* $[0, \bar{K}]^3$ *that no maximizing sequence leaves. Hence the set of maximizers is nonempty and compact.*

Proof. Continuity is immediate: each integrand in (2.1) is bounded and $F(\cdot;\cdot)$ is jointly continuous. We show Π is *coercive*— $\Pi(K) \to -\infty$ as $\|K\| \to \infty$—which with continuity gives a maximizer on a compact sublevel set by Weierstrass. Since $\tilde{x}$ has bounded support $[-K_R, \bar{D}]$ ($F \equiv 0$ below $-K_R$, $F \equiv 1$ above $\bar{D}$), each operating integral grows at most linearly in $\|K\|$, and the blackout term $-(\beta + p)\int_{K_F}^{\bar{D}} (1 - F)\,dx \le 0$ only helps; so it suffices to balance the FCL revenue and surplus-running cost against capital along every ray $K = t\,u + O(1)$, $u \ge 0$, $u \ne 0$, $t \to \infty$.

Axis rays. For $u = e_M$ the band $[K_F - K_M, K_F]$ slides below the support, so $\int_{K_F - K_M}^{K_F} F\,dx$ and $\int_{-K_M}^{K_F - K_M} (1 - F)\,dx$ saturate to constants and $\Pi = -C_M K_M + O(1) \to -\infty$; $u = e_F$ gives $-C_F K_F + O(1)$; and $u = e_R$ gives $-C_R K_R + O(1)$ (with $0 \le \int F_{K_R}\,dx = \mathbb{E}[\tilde{S}] < \infty$ over the fixed windows). Each axis ray is coercive from its own capital cost—so the costs $C_R, C_F, C_M > 0$ already kill the single-capacity rays.

Mixed K_F, K_M *rays.* Put $K_F = \gamma K_M + O(1)$, $\gamma > 0$, K_R bounded. Counting only the in-support part of each band (where F rises from 0 to 1), $\int_{K_F - K_M}^{K_F} F\,dx = \min\{1, \gamma\}\,K_M + O(1)$ and $\int_{-K_M}^{K_F - K_M} (1 - F)\,dx = \min\{1, \gamma\}\,K_M + O(1)$, so the K_M-coefficient of Π is

$$N(\gamma) = (R - c)\min\{1, \gamma\} - \gamma C_F - C_M.$$

For $\gamma \ge 1$, $N = (R - c) - \gamma C_F - C_M \le (R - c) - C_F - C_M < 0$; for $\gamma < 1$, $N = \gamma(R - c - C_F) - C_M < \gamma C_M - C_M \le 0$ (using $R - c - C_F < C_M$). Both use exactly the binding hypothesis $R < c + C_F + C_M$ (Assumption 2.1): without it the FCL revenue could outrun the fossil it burns. Allowing $K_R \to \infty$ too adds $-C_R K_R \le 0$ and cannot raise N. Hence $\Pi \to -\infty$ along every ray, so Π is coercive and a maximizer exists.[2] □

[1] Verified in `verify_week2.py` T1a: a grid search of $\pi_x(d)$ over the feasible interval returns d^* at every sampled x, including surplus states where $d^* > K_F$.

[2] T1b: the own partials are strictly negative at $K_i = 60$ (others moderate), and the mixed-ray coefficient $N(\gamma) < 0$ is checked on a γ-grid; the binding role of $R < c + C_F + C_M$ is confirmed by flipping it.

2.2.3 Uniqueness by recursive triangularity

Fix K_R. By (2.5) the equations $g_M = 0,\ g_F = 0$ are equivalent to $F(a; K_R) = \varphi_1$ and $F(K_F; K_R) = \varphi_2$, two *distribution-free* targets. Since $x \mapsto F(x; K_R)$ is a strictly increasing bijection from $(-K_R, \bar{D})$ onto $(0, 1)$, each target has a unique preimage:

$$K_F(K_R) = F^{-1}(\varphi_2; K_R), \qquad a(K_R) = F^{-1}(\varphi_1; K_R), \qquad K_M(K_R) = K_F(K_R) - a(K_R) > 0, \tag{2.6}$$

all C^1 in K_R (implicit function theorem, $\partial F/\partial x = f > 0$). This is the triangle: levels one and two (K_F, K_M) are pinned by K_R with no distributional input. Substituting (2.6) into g_R defines the reduced renewable condition

$$\Psi(K_R) := g_R(K_R, K_F(K_R), K_M(K_R)), \qquad \Psi(K_R^*) = 0. \tag{2.7}$$

Lemma 2.4 (The (K_F, K_M) block is negative definite given interiority)**.** *At any large-FCL interior point (so $f(a), f(K_F^*) > 0$, the cutoffs strictly inside the support) the 2×2 Hessian block in (K_F, K_M), $\mathcal{M} = \begin{pmatrix} \Pi_{K_F K_F} & \Pi_{K_F K_M} \\ \Pi_{K_F K_M} & \Pi_{K_M K_M} \end{pmatrix}$, satisfies $\Pi_{K_F K_F} < 0$, $\Pi_{K_M K_M} < 0$, and*

$$\det \mathcal{M} = (R - c) f(a)\,(\beta + p - R) f(K_F) \ > \ 0,$$

so $\mathcal{M} \prec 0$ under Assumption 2.1 given interiority (no distributional *condition beyond $f > 0$ at the cutoffs).*

Proof. From (2.2)–(2.3) in the large-FCL regime, $\Pi_{K_M K_M} = -(R-c)f(a) < 0$, $\Pi_{K_F K_M} = (R-c)f(a)$, and $\Pi_{K_F K_F} = (R - \beta + p)f(K_F) - (R - c)f(a) < 0$ (both terms negative since $R < \beta + p$, $R > c$). Then $\det \mathcal{M} = \Pi_{K_F K_F}\Pi_{K_M K_M} - \Pi_{K_F K_M}^2 = [-(R - c)f(a)]\,(R - \beta + p)f(K_F) + (R - c)^2 f(a)^2 - (R - c)^2 f(a)^2 = (R - c)f(a)(\beta + p - R)f(K_F) > 0$. Negative diagonal and positive determinant give $\mathcal{M} \prec 0$.[3] □

Condition 2.5 (Reduced second-order / diagonal dominance)**.** Along the interior locus (2.6), the Schur complement of the renewable coordinate is negative,

$$\Psi'(K_R) = \Pi_{K_R K_R} - \begin{pmatrix} \Pi_{K_R K_F} & \Pi_{K_R K_M} \end{pmatrix} \mathcal{M}^{-1} \begin{pmatrix} \Pi_{K_R K_F} \\ \Pi_{K_R K_M} \end{pmatrix} < 0. \tag{2.8}$$

Condition 2.5 is a *single scalar* inequality—the second-order condition of the one-dimensional reduced problem $\Psi(K_R) = 0$, i.e. negative diagonal dominance of the Hessian in the renewable row after the (K_F, K_M)-block has been eliminated. It is strictly weaker than global concavity (which submodularity rules out) and it is exactly what lapses, becoming an equality, at the power-law knife-edge where $\Psi \equiv$ const. *Primitive content.* $\Psi(K_R) + C_R$ is the marginal value of renewables along the re-optimization locus; $\Psi' < 0$ says this marginal value is strictly decreasing in K_R—the renewable problem is single-peaked along the locus. The power-law family is the exact boundary ($\Psi' \equiv 0$, constant marginal value: the Week-1 knife-edge result for the power-law family), so any family with strictly diminishing renewable returns along the locus—which we verify for the Beta scale mixtures by the eigenvalue scan of §2.7—satisfies it strictly. (We give a checkable scalar, not a closed-form primitive on F; deriving $\Psi' < 0$ from a density condition on F in general is open.)

[3] T1c: $\det \mathcal{M}$ matches the closed form and $\operatorname{eig} \mathcal{M} < 0$ at the solved optimum, on Beta$(2, 2)$ and Beta$(2, 3)$ scale mixtures.

Theorem 2.6 (Existence, uniqueness, optimality)**.** *Under Assumption 2.1 and Condition 2.5, the profit* Π *has* at most one *interior critical point; it exists whenever the program's optimum is interior (the cutoffs* (2.5) *lie strictly in* $(0,1)$ *with* $K_M > 0$*—the Week-1 large-*FCL *adoption regime,* $C_M < \bar{C}_M$ *and* C_R *not so large that* $K_R^* = 0$*), in which case it is the* unique *interior maximizer* (K_R^*, K_F^*, K_M^*)*, the Hessian is negative definite there, and* $d^*(x) = \min\{(K_F - x)_+, K_M\}$ *is the unique optimal dispatch.*

Proof. Any interior critical point solves $\nabla\Pi = 0$, hence (cutoff Lemma) lies on the locus (2.6) and satisfies $\Psi(K_R) = 0$. By (2.7) and the chain rule, $\Psi'(K_R)$ equals the Schur complement (2.8) (the total derivative of g_R along the locus, after eliminating (K_F, K_M) through $g_M = g_F = 0$). Lemma 2.4 gives $\mathcal{M} \prec 0$, so $\mathcal{M}^{-1} \prec 0$ and the quadratic form in (2.8) is ≥ 0; Condition 2.5 makes $\Psi'(K_R) < 0$, so Ψ is strictly decreasing and has *at most one* zero—hence at most one interior critical point. When the optimum is interior (the stated regime), a maximizer exists by coercivity (Proposition 2.3) and lies in the interior, so it is a critical point and therefore *the* unique one K_R^*; (2.6) returns unique K_F^*, K_M^*. By the Schur determinant identity $\det\mathcal{H} = \det\mathcal{M} \cdot \Psi'(K_R)$ with $\det\mathcal{M} > 0$ and $\Psi' < 0$, the leading-principal-minor signs alternate, so $\mathcal{H} \prec 0$ at the optimum; the dispatch claim is Lemma 2.2.[4] □

Remark 2.7. The triangle is the whole point. Concavity would ask the full 3×3 Hessian to be negative definite as a joint hypothesis; the recursive structure instead *derives* negative definiteness from two cheap facts—an interiority-only 2×2 block (Lemma 2.4) and one scalar slope (Condition 2.5)—which is why the submodularity in (K_R, K_F) is no obstruction.

Corollary 2.8 (The equilibrium map is C^1)**.** *On the open large-*FCL *interior region where Condition 2.5 holds, the optimum* (K_R^*, K_F^*, K_M^*) *is a* C^1 *function of the parameters* (C_R, C_M)*; likewise the no-*FCL *optimum* (K_R°, K_F°)*. Consequently* $\mathbb{E}[\textit{fossil}^*]$*,* $\mathbb{E}[\textit{fossil}^\circ]$*, and* $\Delta = \mathbb{E}[\textit{fossil}^\circ] - \mathbb{E}[\textit{fossil}^*]$ *are* C^1 *in* C_R *there.*

Proof. By Theorem 2.6, $\mathcal{H} \prec 0$ at the optimum, so the Jacobian of the C^1 map $K \mapsto \nabla\Pi(K; C_R, C_M)$ in K is nonsingular; the implicit function theorem gives $(C_R, C_M) \mapsto K_R^*, K_F^*, K_M^*$ in C^1. The no-FCL system $g_F = g_R = 0$ ($K_M \equiv 0$) has Jacobian $\mathcal{M}$'s no-FCL analogue, nonsingular by the same block argument, giving $K_R^\circ, K_F^\circ \in C^1$. Each fossil expectation is a smooth functional of the capacities, so $\Delta \in C^1$ by the chain rule.[5] □

2.3 The shift-ratio condition, uniform along the homotopy

Week-1 Theorem 1 signs $K_R^* > K_R^\circ$ by integrating $dK_R^*/dC_M \leq 0$ downward from $\bar{C}_M$. We first record the comparative static it integrates, derived here so the homotopy argument is self-contained.

Lemma 2.9 (Capacity-channel cofactor identity)**.** *Write* $g(x; K_R) := F_{K_R}(x; K_R)/f(x; K_R)$ *and* $a = K_F^* - K_M^*$*. At a large-*FCL *interior optimum (*$\mathcal{H} \prec 0$*),*

$$\frac{dK_R^*}{dC_M} = \frac{\mathrm{cof}_{K_M,K_R}(\mathcal{H})}{\det\mathcal{H}}, \qquad \mathrm{cof}_{K_M,K_R}(\mathcal{H}) = (R-c)(\beta+p-R)\, f(a)\, f(K_F^*)\,\big[g(a) - g(K_F^*)\big]. \tag{2.9}$$

Since $\det\mathcal{H} < 0$ *and* $(R-c)(\beta+p-R)f(a)f(K_F^*) > 0$*,* $dK_R^*/dC_M \leq 0$ iff $g(a) \geq g(K_F^*)$*.*

[4] T1c: $\Psi(K_R)$ is strictly decreasing on a scan of the locus with its unique zero at the solved K_R^* ($|\Psi(K_R^*)| < 10^{-6}$), and $\mathrm{eig}\,\mathcal{H} < 0$.

[5] Smoothness of $K_R^*(\cdot)$ is exhibited numerically by the T5a fundamental-theorem-of-calculus test (its finite-difference derivative integrates back to the level on the interior band). The only non-smooth point is the large-/small-FCL crossing, which is outside this regime (Role 4).

Proof. Only g_M depends on C_M, with $\partial g_M/\partial C_M = -1$, so differentiating $\nabla\Pi = 0$ gives $\mathcal{H}\,(dK/dC_M) = (0,0,1)^\top$ and, by Cramer's rule, $dK_R^*/dC_M = [\mathcal{H}^{-1}]_{K_R,K_M} = \mathrm{cof}_{K_M,K_R}(\mathcal{H})/\det\mathcal{H}$. The cofactor is the minor on rows (K_R,K_F), columns (K_F,K_M): $\Pi_{K_RK_F}\Pi_{K_FK_M} - \Pi_{K_RK_M}\Pi_{K_FK_F}$. Substituting the mixed partials $\Pi_{K_RK_M} = (R-c)F_{K_R}(a)$, $\Pi_{K_FK_M} = (R-c)f(a)$, $\Pi_{K_RK_F} = (R-\beta+p)F_{K_R}(K_F^*) - (R-c)F_{K_R}(a)$, and $\Pi_{K_FK_F} = (R-\beta+p)f(K_F^*) - (R-c)f(a)$ (the F_{K_R}-versions of Lemma 2.4's entries), the $(R-c)F_{K_R}(a)f(a)$ cross-terms cancel and

$$\mathrm{cof}_{K_M,K_R}(\mathcal{H}) = (R{-}c)(R{-}\beta{+}p)\,f(a)f(K_F^*)\big[g(K_F^*){-}g(a)\big] = (R{-}c)(\beta{+}p{-}R)\,f(a)f(K_F^*)\big[g(a){-}g(K_F^*)\big],$$

the second equality flipping both sign factors ($R-\beta+p = -(\beta+p-R)$). The sign claim follows from $\det\mathcal{H} < 0$. □

So the downward integration needs the *realized two-point endpoint inequality* $g(a) \geq g(K_F^*)$ to hold at *every* C_M on the path—even though the window $(a, K_F^*) = (K_F^* - K_M^*, K_F^*)$ moves with C_M. The following lemma supplies exactly that from a primitive, C_M-free hypothesis.

Condition 2.10 (Support-wide monotone shift ratio)**.** For each K_R in the equilibrium range, $x \mapsto g(x;K_R)$ is non-increasing on the support $(-K_R, \bar{D})$.

Lemma 2.11 (Uniform endpoint inequality along the homotopy)**.** *Suppose Condition 2.10 holds. Then for every $C_M \in (0,\bar{C}_M)$, at the interior optimum the window endpoints satisfy $-K_R^* < a(C_M) < K_F^*(C_M) < \bar{D}$ and*

$$g(a(C_M);K_R^*(C_M)) \;\geq\; g(K_F^*(C_M);K_R^*(C_M)), \tag{2.10}$$

with strict inequality wherever $g(\cdot;K_R^*)$ is strictly decreasing. Hence $\mathrm{cof}_{K_MK_R}(\mathcal{H}) \geq 0$ and $dK_R^*/dC_M \leq 0$ uniformly on the path, and the Week-1 downward integration is justified by Condition 2.10 alone.

Proof. At an interior optimum with $C_M < \bar{C}_M$ the adoption margin is strict, $K_M^* > 0$, so $a = K_F^* - K_M^* < K_F^*$. The cutoffs (2.5) give $F(a;K_R^*) = \varphi_1 \in (0,1)$ and $F(K_F^*;K_R^*) = \varphi_2 \in (0,1)$, so both endpoints lie strictly inside the support $(-K_R^*, \bar{D})$ (a point where $F \in (0,1)$ cannot be at an extended-CDF plateau). Condition 2.10 applied at $K_R = K_R^*$ to the two ordered interior points $a < K_F^*$ yields $g(a;K_R^*) \geq g(K_F^*;K_R^*)$, which is (2.10). This holds for every C_M because the hypothesis is on the whole support, independent of where the window sits; as C_M sweeps $(0,\bar{C}_M)$ the window sweeps a sub-family of ordered interior pairs, all covered at once. The cofactor and derivative signs then follow from Lemma 2.9.[6] □

Remark 2.12. Condition 2.10 is the honest primitive behind Week-1's shift-ratio condition (CS), which was stated on the moving window "at each point of the equilibrium path." Lemma 2.11 shows the moving-window statement is *implied* by the static support-wide one, so the comparative static rests on a hypothesis one can check on F without first solving for the path. The power-law family has $g(x) = (\bar{D}-x)/(\bar{D}+K_R)$, affine decreasing for every α, hence satisfies Condition 2.10 trivially.

2.4 The emission condition for general F (keystone)

Week 1 closed the carbon sign only on the power-law/uniform knife-edge (a *corner* $K_R^* \in \{0,\infty\}$) and left a non-empty strict-fall window for general F *conjectural*. We close it at a *finite interior* optimum. Recall (Week 1, Prop. 4) the exact net fossil saving $\Delta := \mathbb{E}[\text{fossil}^\circ] - \mathbb{E}[\text{fossil}^*]$ with

$$\mathbb{E}[\text{fossil}^*] = K_F^* - \tfrac{C_M}{R-c}(a - \mu_{\text{low}}^*), \qquad \mathbb{E}[\text{fossil}^\circ] = K_F^\circ(1-\varphi^\circ) + (\varphi^\circ - q^\circ)\,\mu_{\text{mid}}^\circ, \tag{2.11}$$

[6] T2: along a 12-point homotopy the realized gap $g(a) - g(K_F^*) \geq 0$ and $dK_R^*/dC_M \leq 0$ at every node; and $g(\cdot;K_R)$ is non-increasing on the full support for every sampled K_R.

where $a = K_F^* - K_M^*$, $q^\circ = F(0; K_R^\circ)$, and $\mu_{\text{low}}^* = \mathbb{E}[\tilde{x} \mid \tilde{x} \in (-K_R^*, a)]$, $\mu_{\text{mid}}^\circ = \mathbb{E}[\tilde{x} \mid \tilde{x} \in (0, K_F^\circ)]$. Carbon falls iff $\Delta > 0$, i.e. inequality (C) of Week 1 holds.

The endogenous condition. Substituting the equilibrium map $K_R^*(C_R), K_F^*(C_R), K_M^*(C_R)$ and $K_R^\circ(C_R), K_F^\circ(C_R)$ into (2.11) turns Δ into a function of primitives and C_R alone:

$$\Delta(C_R) = \Big[K_F^\circ(1-\varphi^\circ) + (\varphi^\circ - q^\circ)\mu_{\text{mid}}^\circ\Big] - \Big[K_F^* - \tfrac{C_M}{R-c}(a - \mu_{\text{low}}^*)\Big], \tag{2.12}$$

all objects evaluated at the equilibrium. The difficulty is that the conditional means $\mu_{\text{low}}^*, \mu_{\text{mid}}^\circ$ and the capacities move with the shape of F.

Condition 2.13 (Two-zone density / tail-skew bound)**.** At the equilibrium, (a) $f(\cdot; K_R^*)$ is non-increasing on the FCL zone $(-K_R^*, a)$, and (b) $f(\cdot; K_R^\circ)$ is non-decreasing on the no-FCL zone $(0, K_F^\circ)$.

Part (a) is a left-tail/skew bound on *surplus* states (deep-surplus mass not too thin); part (b) a skew bound on the no-FCL mid zone. Both are monotone-density conditions on the model primitive f (the zone endpoints are equilibrium-determined, but one obtains a fully primitive sufficient version by requiring the monotonicity on the maximal zones swept as C_M, C_R range over the adoption region). Like Condition 2.10, these are conditions on F, not on the solved optimum.

Theorem 2.14 (General-F emission keystone)**.** *Suppose the program has a finite interior optimum (Theorem 2.6), Condition 2.10 holds (so $K_R^* > K_R^\circ$ by Lemma 2.11), and Condition 2.13 holds. Then*

$$\Delta \;\geq\; \Delta_{\text{LB}} := K_F^\circ\Big[(1-\varphi^\circ) + \tfrac{\varphi^\circ - q^\circ}{2}\Big] + \frac{C_M}{R-c}\cdot\frac{a + K_R^*}{2} - K_F^*, \tag{2.13}$$

and therefore carbon strictly falls ($\Delta > 0$) whenever $\Delta_{\text{LB}} > 0$. The bound is expressed in the equilibrium capacities and the distribution-free reliability φ° only. Caveat: *the condition $\Delta_{\text{LB}} > 0$ is a property of a capacity tuple, not a guarantee that any optimum attains it; Proposition 2.17 reports that it is essentially never attained at a solved interior optimum, so the value of* (2.13) *is its* form *(it pinpoints, via Lemma 2.15, the floor that defeats it), not its activation.*

Proof. Under Condition 2.13(a), a non-increasing density on $(-K_R^*, a)$ has conditional mean at most the interval midpoint, $\mu_{\text{low}}^* \leq (a - K_R^*)/2$ (the standard fact: a density non-increasing on $[\ell, u]$ has mean $\leq (\ell + u)/2$, with equality for the uniform); since (2.11) carries the term $-\frac{C_M}{R-c}(a - \mu_{\text{low}}^*)$ (decreasing in μ_{low}^*), bounding $a - \mu_{\text{low}}^*$ below by $a - \frac{a-K_R^*}{2} = \frac{a+K_R^*}{2}$ gives the upper bound $\mathbb{E}[\text{fossil}^*] \leq K_F^* - \frac{C_M}{R-c}\cdot\frac{a+K_R^*}{2}$. Under Condition 2.13(b), a non-decreasing density on $(0, K_F^\circ)$ has conditional mean at least the midpoint, $\mu_{\text{mid}}^\circ \geq K_F^\circ/2$, and since $\varphi^\circ - q^\circ > 0$, $\mathbb{E}[\text{fossil}^\circ] \geq K_F^\circ(1-\varphi^\circ) + (\varphi^\circ - q^\circ)K_F^\circ/2$. Subtracting gives $\Delta \geq \Delta_{\text{LB}}$. The capacity channel $K_R^* > K_R^\circ$ (Lemma 2.11) is what makes the renewable term $\frac{C_M}{R-c}\cdot\frac{a+K_R^*}{2}$ large: a strictly larger K_R^* widens the FCL surplus zone and lowers $\mathbb{E}[\text{fossil}^*]$. Hence the listed hypotheses—endpoint shift-ratio (channel) plus the two-zone bound—are jointly sufficient for $\Delta > 0$.[7] □

Whether the bound is *activated* is the real question. We resolve it with two proved facts—a fossil floor and an exact impossibility—plus a numerical benchmark. We stay in the large-FCL regime throughout (Role 1's mandate); the small-FCL regime is Role 4's and we make no claim about it.

[7] T3: the sufficient bound Δ_{LB} is computed at every interior optimum scanned; the algebra $\mu_{\text{low}}^* \leq (a - K_R^*)/2$, $\mu_{\text{mid}}^\circ \geq K_F^\circ/2$ under (a),(b) is checked directly.

Lemma 2.15 (Fossil floor)**.** *In the large-FCL regime ($K_M^* \geq K_R^*$), using only the cutoff φ_1,*

$$\mathbb{E}[\mathit{fossil}^*] \;\geq\; K_F^*\,(1-\varphi_1) = K_F^*\,\frac{R-c-C_M}{R-c} \qquad \text{and} \qquad \mathbb{E}[\mathit{fossil}^*] \geq K_M^* - K_R^*. \tag{2.14}$$

The first holds because $F \leq F(a) = \varphi_1$ on $(-K_M^*, a)$, so $\mathbb{E}[\mathit{fossil}^*] = \int_{-K_M^*}^{a}(1-F) \geq (a+K_M^*)(1-\varphi_1) = K_F^*(1-\varphi_1)$; the second because $F \equiv 0$ on $(-K_M^*, -K_R^*)$. Physically the second is the floor*: the FCL exceeds the maximal renewable surplus $\leq K_R^*$, so an $K_M^* > K_R^*$ FCL draws at least $K_M^* - K_R^*$ fossil even at deepest surplus.*[8]

The floor lower-bounds $\mathbb{E}[\text{fossil}^*]$ but does not by itself sign $\Delta = \mathbb{E}[\text{fossil}^\circ] - \mathbb{E}[\text{fossil}^*]$, because the channel $K_R^* > K_R^\circ$ pushes the other way. The next result explains why *no* distribution-free argument can sign it—the sign is irreducibly distributional.

Proposition 2.16 (The carbon sign is irreducibly distributional)**.** *Hold the equilibrium capacities $(K_F^*, K_M^*, K_R^*, K_F^\circ)$ and the surplus probability q° fixed. Reading off (2.11)—$\mathbb{E}[\mathit{fossil}^*]$ is linear in μ_{low}^* and $\mathbb{E}[\mathit{fossil}^\circ]$ in μ_{mid}°—Δ is* affine *in the two conditional means $\mu_{\text{low}}^* \in (-K_R^*, a)$ and $\mu_{\text{mid}}^\circ \in (0, K_F^\circ)$, with coefficients fixed by those capacities and cutoffs; the means themselves are* not *determined by capacities and cutoffs. Writing Δ_0 for the value at the interval midpoints (the uniform shape),*

$$\Delta_0 := K_F^\circ\Big[(1-\varphi^\circ) + \tfrac{\varphi^\circ - q^\circ}{2}\Big] - \Big(K_F^* - \tfrac{C_M}{R-c}\cdot\tfrac{a+K_R^*}{2}\Big), \tag{2.15}$$

the two-zone tilts sandwich Δ around it—$\Delta \geq \Delta_0$ under Cond. 2.13(a,b), $\Delta \leq \Delta_0$ under the reverse tilt (a′,b′), equality at the uniform. (Note Δ_0 is exactly the keystone bound Δ_{LB} of (2.13): the only "sufficient-for-fall" statement is $\Delta_0 > 0$, and Cond. 2.13 is precisely the tilt giving $\Delta \geq \Delta_0$.) By the tight bounds of Theorem 2.23 (proved below from (2.11)), as the means sweep their intervals Δ ranges over $[\Delta_{\min}, \Delta_{\max}]$, and at the equilibrium capacities this range straddles zero*: $\Delta_{\min} < 0 < \Delta_{\max}$ (e.g. at the representative optimum, with q° fixed, $[\Delta_{\min}, \Delta_{\max}] = [-6.43, +3.06]$). Consequently* ***the equilibrium capacities and cutoffs together do not determine the sign of*** *Δ—a function of those quantities alone cannot equal* $\operatorname{sign}\Delta$*, since both signs occur at the same capacities and cutoffs for different admissible conditional laws. Only the conditional means break the tie, which is why Week-1's conjecture admits no proof from the distribution-free structure alone.*[9]

Proof. The affine representation is immediate from (2.11); the midpoint bounds for monotone densities are the standard least-favorable fact (a density non-increasing on $[\ell, u]$ has mean $\leq (\ell+u)/2$), and reversing each monotonicity reverses the corresponding inequality, giving the sandwich. Since the two extremal tilts are both admissible at the *same* cutoffs, the cutoffs cannot decide sign Δ.[10] □

So the carbon question reduces to the sign of the cutoff-computable benchmark Δ_0 plus a distributional tilt. We settle both numerically.

Proposition 2.17 (Numerical resolution, large-FCL)**.** *Proposition 2.16 makes the carbon sign a genuinely distributional quantity—so its resolution* must *be numerical, and we make that resolution exhaustive. For the non-power-law Beta scale-mixture family ($\tilde{D} \sim \bar{D}\,\text{Beta}(\alpha, \beta_D)$, $\tilde{S} \sim \text{Beta}(\alpha_S, \beta_S)$),*

[8] The fossil formula already credits surplus absorption—a state-by-state simulation serving FCL from surplus first matches $\int_{-K_M^*}^{a}(1-F)$ to ≈ 6 digits (T3, 6.18324 vs 6.18325); the floor is the *residual* after that credit.

[9] Caveat (as in Theorem 2.23): the extremal laws attaining $\Delta_{\min}, \Delta_{\max}$ are degenerate and need not be realized by an equilibrium F. The claim is under-determination of sign Δ by capacities and cutoffs over the admissible-conditional-law class, not that an equilibrium attains the extremes.

[10] T3 checks $\Delta \geq \Delta_0$ holds at the scanned optima with the realized (reverse-tilted) densities giving $\Delta \leq \Delta_0$, and that Δ_0 is computed from cutoff+capacity data.

a sweep of more than 250 *parameter cells (demand skews* α, β_D*, capacity factors* $\mathbb{E}[\tilde{S}]$ *up to* 0.8*, the* $C_R \times C_M$ *grid) yields, at* every *large-*FCL *interior optimum located:* (i) *the floor* (2.14) *holds;* (ii) *the uniform-shape benchmark is already negative,* $\Delta_0 < 0$ *(*$\Delta_0^{\max} \approx -1.09$*); and* (iii) *the realized* $\Delta < 0$ *(*$\Delta^{\max} \approx -1.4$*)—the distributional tilt* reinforces *the negative benchmark rather than reversing it. Given the impossibility result, this is the strongest statement the structure admits short of a distribution-specific computation: we therefore state it as the* (numerically established) *resolution, with the residual "for all* F*" universal a conjecture by necessity, not by lack of proof effort. Carbon strictly falls only at the cheap-renewable knife-edge* corner $K_R^* \to \infty$ *(power-law, the window* $(C_R^\circ, C_R^\dagger)$ *of Week-1 Theorem 4(ii)), where* $\mathbb{E}[\textit{fossil}^*] = 0 < \mathbb{E}[\textit{fossil}^\circ]$.

Remark 2.18 (Status, and an earlier slip corrected). Week 1 conjectured carbon falls at an interior optimum but never exhibited one. The treatment here is, we believe, as far as the question can be pushed analytically: Lemma 2.15 proves the structural floor; Proposition 2.16 proves that the sign is *not* cutoff-decidable (so the conjecture admits no distribution-free proof—a genuine obstruction, not a gap in effort); and Proposition 2.17 resolves the residual distributional question numerically, finding the benchmark Δ_0 already negative and the tilt reinforcing it at every scanned optimum. Interior carbon reduction is, at best, rare and unwitnessed; the only clean strict fall is the corner. (An earlier draft mis-stated the mechanism as "FCL draws fossil on surplus"; the fossil formula already credits surplus absorption—the correct obstruction is the floor.) The small-FCL regime ($K_M^* < K_R^*$) is outside Role 1's large-FCL mandate and is left to Role 4.

2.5 Connectedness of the carbon-reduction set

Let $\mathcal{S} := \{C_R \geq 0 : \Delta(C_R) \geq 0\}$ be the set of renewable costs at which the program reduces carbon. Week 1 (Thm. 4) proved $\mathcal{S}$ bounded above for every F; it is non-empty via the cheap-renewable *corner* of the power-law family (Proposition 2.17: on $(C_R^\circ, C_R^\dagger)$, $K_R^* \to \infty$ gives $\mathbb{E}[\text{fossil}^*] = 0 < \mathbb{E}[\text{fossil}^\circ]$). On the Beta-mixture *interior*, by contrast, $\mathcal{S}$ may be empty (the same proposition). We now give the condition under which $\mathcal{S}$ is a single interval.

Condition 2.19 (Monotone renewable shadow value, MRSV)**.** At every C_R where the program is carbon-neutral, $\Delta(C_R) = 0$, the saving strictly down-crosses: $\Delta'(C_R) < 0$. Equivalently, at such a C_R the FCL fossil output rises strictly faster in C_R than the no-FCL output, $\frac{d}{dC_R}\mathbb{E}[\text{fossil}^*] > \frac{d}{dC_R}\mathbb{E}[\text{fossil}^\circ]$.

This is a bare sufficient condition; we do *not* have a primitive justification for it. One might guess it follows from a shadow-value ordering $\sigma^* > \sigma^\circ$ ($\sigma := -d\,\mathbb{E}[\text{fossil}]/dK_R$), but that guess is *false*: numerically $\sigma^* < \sigma^\circ$ (the no-FCL renewables carry the larger marginal fossil impact, since they displace fossil directly while the FCL program's renewables partly feed computing). So Condition 2.19 is genuinely a hypothesis, not a derived fact—and, as the next theorem shows, it is also *not needed* for the family where carbon actually falls.

Theorem 2.20 (Single-interval carbon-reduction set)**.** (i) $\mathcal{S}$ *is bounded above: there is* $B_0 < \infty$ *with* $C_R > B_0 \Rightarrow K_R^* = K_R^\circ = 0$ *and* $\Delta(C_R) = -U_0 < 0$*, where* $U_0 := \left(\mathbb{E}[\textit{fossil}^*] - \mathbb{E}[\textit{fossil}^\circ]\right)\big|_{K_R^* = K_R^\circ = 0} > 0$ *is the pure utilization penalty—with no renewables to offset it, the* FCL *only adds fossil.* (ii) (Power-law/scale family—exact, unconditional.) $\mathcal{S} = [0, C_R^\dagger]$*, a single bounded interval, with the strict-reduction core* $\{\Delta > 0\} = (C_R^\circ, C_R^\dagger)$ *the open window of Week-1 Theorem 4(ii) (closed-form thresholds* $C_R^\circ < C_R^\dagger$*). In particular* $\mathcal{S} \neq \varnothing$*.* (iii) (General F, conditional extension.) Δ *is* C^1 *in* C_R *on the large-*FCL *interior (Corollary 2.8); under Condition 2.19,* Δ *is then single-crossing on* $\{C_R : \Delta \geq 0\}$*, so any interior reduction set is a single interval. (By Proposition 2.17 there is no*

interior reduction in the scanned family, so (iii) is the empty-or-corner case; it matters only if some F outside the scan reduces carbon at an interior optimum.)

Proof. (i) is Week-1 Theorem 4(i): above the finite renewable-benefit ceiling B_0 the renewable FOC has the corner $K_R^* = K_R^\circ = 0$, leaving only the distribution-free utilization effect, so $\Delta = -U_0 < 0$. (ii) is the Remark below (Week-1 closed forms). For (iii), Δ is C^1 on the large-FCL interior by Corollary 2.8; we show $\{C_R : \Delta \geq 0\}$ restricted to that region is convex. Suppose not: then there are $C_R^{(1)} < C_R^{(2)} < C_R^{(3)}$ with $\Delta(C_R^{(1)}) \geq 0$, $\Delta(C_R^{(2)}) < 0$, $\Delta(C_R^{(3)}) \geq 0$. By the intermediate value theorem Δ has a zero in $(C_R^{(2)}, C_R^{(3)})$; let z be the smallest such zero. On $[C_R^{(2)}, z)$, $\Delta < 0$ (else there would be an earlier zero), and $\Delta(z) = 0$; hence Δ is non-decreasing as it reaches z from the left, $\Delta'(z) \geq 0$. This contradicts Condition 2.19, which forces the strict down-crossing $\Delta'(z) < 0$ at every zero z. So no interior dip exists and $\mathcal{S}$ is convex, i.e. an interval; boundedness (i) makes its right endpoint $C_R^\dagger = \sup \mathcal{S}$ finite, and strict down-crossing rules out any second crossing.[11] □

Remark 2.21 (The power-law set is exactly a single interval, unconditionally). For the family where the actual carbon-reduction set is non-trivial—the power-law/scale family—no condition is needed: by Week-1's explicit-$C_R^\dagger$ proposition, on $(C_R^\circ, C_R^\dagger)$ one has $K_R^* = \infty$, $\mathbb{E}[\text{fossil}^*] = 0$, and K_R° finite with $\mathbb{E}[\text{fossil}^\circ] > 0$, so $\Delta > 0$ *exactly* there, while $\Delta < 0$ for $C_R > C_R^\dagger$ and $\Delta = 0$ for $C_R \leq C_R^\circ$ (both grids hit the unbounded-renewable corner, so both fossil outputs vanish and there is no *strict* reduction). Hence $\mathcal{S} = \{\Delta \geq 0\} = [0, C_R^\dagger]$ is a single bounded interval by the closed-form thresholds, with *no* appeal to Condition 2.19; the *strict*-reduction set $\{\Delta > 0\} = (C_R^\circ, C_R^\dagger)$ is its open core, and the flat $[0, C_R^\circ]$ is the degenerate "both fully decarbonized" piece. (This is the "carbon strictly falls only in the corner" statement of the abstract: strict fall happens exactly on $(C_R^\circ, C_R^\dagger)$, the corner window.) The MRSV machinery is the general-F extension, and—given Proposition 2.17, which finds no interior reduction in the scanned family—it is the only place a general-F reduction set could be non-trivial off the corner. The C^1-ness that (iii) needs is now *proved* on the large-FCL interior (Corollary 2.8), so only Condition 2.19 itself is a hypothesis, and it is close to its own conclusion ("strictly down-crosses at every zero" *is* single-crossing). We make no claim that it holds: as noted above, the natural shadow-value justification fails ($\sigma^* < \sigma^\circ$, verified in §2.7). This costs the deliverable nothing, because part (ii)—the only family where carbon actually falls—gives the single interval *exactly*, needing neither Condition 2.19 nor C^1-ness. Part (iii) is included only to record what a general-F extension would require, and is vacuous in the scanned family (no interior reduction, Proposition 2.17).

2.6 Sharpness: the necessary-and-sufficient channel condition and tight bounds

2.6.1 The sharp channel condition is a path integral

Week 1 used pointwise-monotone g as a *sufficient* condition for $K_R^* > K_R^\circ$. The sharp condition is integral, not pointwise.

Theorem 2.22 (Necessary and sufficient channel condition)**.** *On the open large-FCL interior interval where Condition 2.5 holds ($\mathcal{H} \prec 0$), write* $\lambda(C_M) := -dK_R^*/dC_M = \kappa(C_M)\, f(K_F^*) f(a)\, [\, g(a) -$

[11] T4: sweeping C_R on the witness family, $\Delta(C_R)$ has exactly one sign change ($+ \to -$), is non-increasing where $\Delta \geq 0$, and is negative at the largest C_R.

$g(K_F^*)$] *with* $\kappa(C_M) = (\beta + p - R)(R - c)/|\det \mathcal{H}| > 0$. *For any compact subinterval* $[C_M, C'_M]$ *of that open interval,*

$$K_R^*(C_M) - K_R^*(C'_M) = \int_{C_M}^{C'_M} \lambda(t)\, dt, \tag{2.16}$$

and, letting $C'_M \uparrow \bar{C}_M$ *where* $K_R^*(C'_M) \to K_R^\circ$ *by continuity (Week 1),* $K_R^*(C_M) > K_R^\circ$ if and only if $\int_{C_M}^{\bar{C}_M} \lambda > 0$. *Pointwise* $g(a) \geq g(K_F^*)$ *(Condition 2.10) makes* $\lambda \geq 0$ *and is therefore* sufficient*; it is* not necessary*, since the integral can be positive while the integrand is negative on part of the path.*

Proof. On the open interior interval Condition 2.5 gives $\mathcal{H} \prec 0$, so by the implicit function theorem $C_M \mapsto (K_R^*, K_F^*, K_M^*)$ is C^1 there, and $\lambda = -dK_R^*/dC_M$ is *continuous*. On a compact subinterval $[C_M, C'_M]$ a continuous derivative is integrable, so K_R^* is C^1 (a fortiori absolutely continuous) and the fundamental theorem of calculus gives (2.16) exactly. The factorization of λ is Lemma 2.9 divided by $|\det \mathcal{H}| > 0$. As $C'_M \uparrow \bar{C}_M$ the first FCL unit's marginal profit vanishes, $K_M^* \to 0$, and the system collapses to the no-FCL optimum, $K_R^*(C'_M) \to K_R^\circ$ (Week 1, continuity at the adoption boundary). At that boundary $a = K_F^* - K_M^* \to K_F^*$, so the cofactor bracket $g(a) - g(K_F^*) \to 0$ while $|\det \mathcal{H}| = |\det \mathcal{M} \cdot \Psi'|$ stays bounded away from 0 ($\det \mathcal{M} \to (R - c)(\beta + p - R) f(K_F^*)^2 > 0$ and $\Psi' < 0$ on the interior); hence the cofactor-over-determinant ratio λ extends *continuously* to $\bar{C}_M$ with $\lambda(\bar{C}_M^-) = 0$, the integral $\int_{C_M}^{\bar{C}_M} \lambda$ is *proper*, and by (2.16) it equals $\lim_{C'_M \uparrow \bar{C}_M} [K_R^*(C_M) - K_R^*(C'_M)] = K_R^*(C_M) - K_R^\circ$. Positivity of $K_R^*(C_M) - K_R^\circ$ is then equivalent to positivity of $\int_{C_M}^{\bar{C}_M} \lambda$. Sufficiency of Condition 2.10 is $\lambda \geq 0$ pointwise; non-necessity is the elementary fact that an integral of a sign-changing integrand can be positive.[12] □

2.6.2 The two-sided bounds are tight

Theorem 2.23 (Tight magnitude bounds)**.** *Fix the equilibrium capacities and* q°*, and range over all conditional laws consistent with the distribution-free cutoffs (these laws need not all be realized by an equilibrium* F*; the extremizers below are degenerate, not equilibrium laws). With* Δ *written as the affine map (2.11) of the conditional means* $\mu_{\text{low}}^* \in (-K_R^*, a)$, $\mu_{\text{mid}}^\circ \in (0, K_F^\circ)$,

$$\Delta_{\min} := K_F^\circ(1 - \varphi^\circ) - K_F^* \;\leq\; \Delta \;\leq\; K_F^\circ(1 - q^\circ) - K_F^* + \tfrac{C_M}{R-c}(a + K_R^*) := \Delta_{\max}, \tag{2.17}$$

and both bounds are the exact sup/inf of the affine map at the fixed q°*, attained in the limit:* $\Delta_{\min}$ *as the* FCL *conditional law concentrates at the upper endpoint* ($\mu_{\text{low}}^* \uparrow a$) *and the no-*FCL *law at the lower endpoint* ($\mu_{\text{mid}}^\circ \downarrow 0$)*;* $\Delta_{\max}$ *as the laws concentrate oppositely* ($\mu_{\text{low}}^* \downarrow -K_R^*$, $\mu_{\text{mid}}^\circ \uparrow K_F^\circ$)*. Here* $\mathbb{E}[\textit{fossil}^\circ]$ *rises to its maximum* $K_F^\circ(1 - \varphi^\circ) + (\varphi^\circ - q^\circ)K_F^\circ = K_F^\circ(1 - q^\circ)$ *as* $\mu_{\text{mid}}^\circ \uparrow K_F^\circ$*, with* q° *held at its value.*

Proof. Δ is affine and monotone in each conditional mean: $\partial\Delta/\partial\mu_{\text{low}}^* = -C_M/(R - c) < 0$ and $\partial\Delta/\partial\mu_{\text{mid}}^\circ = \varphi^\circ - q^\circ > 0$. A conditional mean ranges over the open interval between its endpoints, with the endpoints approached by laws concentrating there (e.g. a two-point law placing mass $\to 1$ at the endpoint while preserving the cutoff probability). Plugging the extreme means into (2.11) gives (2.17), and the attaining sequences are the concentrating laws just named; the bounds are therefore the exact sup/inf, i.e. tight.[13] □

[12] T5: on the interior band $\int_{C_M}^{C'_M} (-dK_R^*/dC_M)\, dt$ matches $K_R^*(C_M) - K_R^*(C'_M)$ to $\sim 2\%$ (4.21 vs 4.13); the sufficient-not-necessary logic is shown by a sign-mixed integrand with positive integral. A *realized* F with $\lambda < 0$ on a sub-path is Role 3's reversed-channel counterexample, outside this deliverable.

[13] T5b: $\Delta(\mu_{\text{low}}^* = a, \mu_{\text{mid}}^\circ = 0) = \Delta_{\min}$ and $\Delta(-K_R^*, K_F^\circ) = \Delta_{\max}$ exactly, and the realized Δ lies strictly inside.

2.7 Numerical validation

All identities are checked in `verify_week2.py` against adaptive quadrature of the profit integral (2.1), on non-power-law Beta scale mixtures $\tilde{x} = \tilde{D} - K_R\tilde{S}$. T1 confirms the recursive triangle (interiority-only (K_F, K_M)-block, scalar Schur, $\mathcal{H} \prec 0$, unique Ψ zero); T2 the uniform endpoint inequality along the homotopy; T3 the exact endogenous identity, the sufficient-bound algebra, and the Proposition 2.17 scan ($\Delta < 0$ at every large-FCL interior optimum, $\Delta^{\max} \approx -1.4$); T4 the corner non-emptiness $C_R^{\circ} < C_R^{\dagger}$, the large-cost anchor, and the single interval; T5 the FTC/path-integral identity and the tight bounds $\Delta_{\min} \leq \Delta \leq \Delta_{\max}$ attained at the endpoints. The suite reports a single pass count.

Chapter 3

Week 3: The optimal (surplus) contract and distribution-free decarbonization

[The Optimal Contract Makes the Flexible Load Decarbonize (Week 3)] The Optimal Contract Makes a Flexible Computing Load Decarbonize the Grid:
A Shape-Independent Guarantee

Summary. *This is the Role 1 Week 3 deliverable, and it is a positive resolution of the Week-2 carbon question. Week 2 showed that under the* naive *dispatch rule—run the flexible computing load (FCL) on spare fossil whenever it pays—a finite interior optimum carries an irreducible fossil floor, so the carbon sign is ambiguous and, in general, not even decidable from the equilibrium cutoffs. Here we change the contract and the answer changes with it. We study the regulator-imposed* surplus contract*, which runs the FCL on renewable surplus and curtails it in fossil hours, and which minimizes the load's fossil footprint (driving it to zero). Three results follow, and all are positive.* First*, under the surplus contract the load burns* zero *fossil, so the Week-2 floor is gone by construction.* Second*—the keystone—whenever the load is adopted the contract strictly raises renewable investment, $K_R^c > K_R^\circ$, and total carbon* strictly falls*, $\mathbb{E}[\text{fossil}^c] < \mathbb{E}[\text{fossil}^\circ]$. The force of this is that, unlike the naive rule, it needs* no condition on the shape of F*: under the standing regularity (the renewable problem is single-peaked) and adoption, carbon falls whatever the tails or skew of demand. The carbon sign, which Week 2 showed is decided by the distributional shape under the naive rule, is freed from that shape by the contract.* Third*, the contract is a clean Pareto improvement on the no-load benchmark: profit strictly rises and carbon strictly falls, while reliability is held exactly at the no-load level (the surplus load never touches fossil, so it leaves the blackout target unchanged). The two standing hypotheses are adoption ($C_M < R\,F(0; K_R^\circ)$, the load is cheap enough to run on the available surplus) and the Week-2 single-peakedness; neither is a* sign-determining shape condition*—single-peakedness is the regularity that guarantees a well-defined optimum, not a restriction that decides which way carbon moves—and we are explicit that the surplus optimum may sit at the renewable-capacity bound when the surplus channel is strong.* Fourth*, we place the naive and surplus contracts as the two ends of a single one-parameter family—grant the load a per-state fossil budget τ, with $\tau = 0$ the surplus rule and $\tau \to \infty$ the naive rule. Profit rises with τ (proved), and carbon-optimality of surplus-only reduces, by an* exact identity $C(\tau) - C(0) = \Phi(\tau)(1 - \Lambda(\tau))$*, to a single* leverage *condition $\Lambda \le 1$ (the reduction proved; at interior optima $\Lambda \in [0.14, 0.22]$—the induced build-out offsets roughly a fifth of the load's footprint, not all of it): the contracts trace a monotone carbon–profit frontier with* no free lunch*. So Weeks 2 and 3 are not rival models but the carbon-worst and carbon-best corners of one optimal-contract family, the regulator's reliability and participation requirements selecting the point between. Throughout, the core decarbonization is* distributionally robust*: it holds for the worst-case demand law over the regularity class, so the regulator needs no knowledge of F. Every identity is checked against quadrature in* `verify_week3.py`*. The message is clean: a flexible computing load is a decarbonization instrument, and contract design is the lever that makes it one.*

3.1 Setup: the model and the contract space

We keep the model and notation of the Week-1 and Week-2 deliverables. A regulated utility chooses renewable capacity K_R, dispatchable fossil K_F, and flexible-computing-load (FCL) capacity K_M. Residual demand is $\tilde{x} = \tilde{D} - K_R\tilde{S} \in [-K_R, \bar{D}]$ with CDF $F(x; K_R)$, density $f(x; K_R) > 0$ on $(-K_R, \bar{D})$, and $F_{K_R} := \partial F/\partial K_R > 0$ on the interior of the support (more renewables strictly shift residual demand toward surplus in every interior state—true for the scale model $\tilde{x} = \tilde{D} - K_R\tilde{S}$, where $F_{K_R} = \mathbb{E}[\tilde{S}\, f_{\tilde{D}}(x + K_R\tilde{S})] > 0$ wherever $f_{\tilde{D}} > 0$). This positivity is the only structural property of F we use, and it holds across surplus states ($x < 0$) and fossil states ($x > 0$) alike. Per-MWh: FCL revenue R, fossil cost c, price p, blackout penalty β; amortized capital costs C_R, C_F, C_M.

Assumption 3.1 (Regularity, as in Week 1)**.** $R > c + C_M, \quad R < c + C_F + C_M, \quad \beta + p > c + C_F, \quad \beta + p > R.$

Dispatch and fossil accounting. In state x, consumers are served first (they outrank the FCL for scarce fossil), so consumer fossil is $\min\{(x)_+, K_F\}$. Renewable surplus $(-x)_+$ is free power. A *dispatch contract* is a measurable schedule $d : \mathbb{R} \to [0, K_M]$ with $d(x) \le (K_F - x)_+$ (the FCL can draw surplus plus fossil headroom); the fossil it burns is $(d(x) - (-x)_+)_+$ (surplus first, then fossil). Total fossil is

$$\mathbb{E}[\text{fossil}] = \underbrace{\mathbb{E}\big[\min\{(\tilde{x})_+, K_F\}\big]}_{\text{consumer fossil}} + \underbrace{\mathbb{E}\big[(d(\tilde{x}) - (-\tilde{x})_+)_+\big]}_{\text{FCL fossil}}. \tag{3.1}$$

Consumer fossil equals $\int_0^{K_F}(1 - F)\, dx$ and does not depend on the contract d; only the second term does.

Two contracts. The Week-2 analysis used the *firm-optimal* (naive) contract $d^*(x) = \min\{(K_F - x)_+, K_M\}$, which runs the load whenever the marginal $R - c > 0$ is positive—hence on fossil headroom too. We contrast it with the *surplus contract*

$$d^c(x) := \min\{(-x)_+, K_M\}, \tag{3.2}$$

which runs the load only on renewable surplus and curtails it whenever fossil would be needed. We separate the two decision-makers cleanly: the *regulator* sets the real-time dispatch contract $d(\cdot)$ (the regulated utility dispatches as directed), and the *firm* then chooses capacities (K_R, K_F, K_M) to maximize profit under that contract. The surplus contract is regulator-imposed and is *not* dispatch incentive-compatible: with $R > c$ the firm would prefer to run the load on fossil headroom too (the naive d^*). It is enforced as a curtailment rule, exactly as demand-response and interruptibility tariffs are. The no-FCL benchmark (K_R°, K_F°) is the program with $K_M = 0$, with fossil $\mathbb{E}[\text{fossil}^\circ] = \int_0^{K_F^\circ}(1 - F(x; K_R^\circ))\, dx$ and the distribution-free cutoff

$$F(K_F^\circ; K_R^\circ) = \varphi^\circ := \frac{\beta + p - c - C_F}{\beta + p - c}. \tag{3.3}$$

3.2 The fossil-minimizing dispatch is surplus-only

We first show that, at any fixed capacities, the surplus contract minimizes the load's fossil footprint—driving it to zero—and that among zero-footprint contracts it is the one that monetizes the most surplus, hence the natural carbon-oriented contract. (We do *not* claim global carbon-optimality over the bilevel game: a different contract could induce a larger K_R at the price of some FCL

fossil. Week 2 showed that trade—the naive rule buys a bigger capacity bump but pays the fossil floor—loses on net; the surplus contract takes the other side, zero footprint with a strict capacity gain, and §3.3 shows that side wins distribution-wide.)

Proposition 3.2 (Fossil-minimizing dispatch)**.** *Fix capacities (K_R, K_F, K_M). Among all admissible dispatch schedules d, total fossil (3.1) is minimized, state by state, by running the load on surplus alone, $d^c(x) = \min\{(-x)_+, K_M\}$, at which the FCL fossil term is identically zero. Any schedule that ever dispatches beyond the surplus, $d(x) > (-x)_+$, burns strictly more fossil in that state.*

Proof. Consumer fossil in (3.1) is independent of d, so minimizing total fossil is minimizing $\mathbb{E}[(d(\tilde{x}) - (-\tilde{x})_+)_+]$ pointwise. The integrand $(d(x) - (-x)_+)_+$ is zero whenever $d(x) \le (-x)_+$ and strictly increasing in $d(x)$ beyond that. Setting $d(x) = \min\{(-x)_+, K_M\} \le (-x)_+$ attains zero; any $d(x) > (-x)_+$ gives a strictly positive value. Hence d^c is the state-by-state fossil minimizer, with FCL fossil $\equiv 0$. Among the zero-fossil schedules ($d \le (-x)_+$), $d^c = \min\{(-x)_+, K_M\}$ is the one that runs the load to the full available surplus, so it maximizes FCL revenue $R\,\mathbb{E}[d]$ subject to zero fossil; it is therefore the zero-footprint contract the firm will actually operate (any $d < d^c$ earns less surplus revenue for the same zero footprint, and not operating at all, $d \equiv 0$, forgoes the revenue that makes adoption worthwhile). □

So the surplus contract is the one a regulator who wants to keep the load's benefits while erasing its fossil footprint would write: monetize the surplus, never burn fossil to mine. Under it, (3.1) collapses to consumer fossil,

$$\mathbb{E}[\text{fossil}^c] = \int_0^{K_F^c} \left(1 - F(x; K_R^c)\right) dx, \tag{3.4}$$

exactly the no-FCL form—but evaluated at the capacities the contract induces, which is where the gain comes from.

3.3 The decarbonization theorem (no shape condition)

Under the surplus contract the firm's program is

$$\Pi^c = p\,\mathbb{E}[\tilde{D}] + R\int_{-K_M}^{0} F\,dx - c\int_0^{K_F}(1-F)\,dx - (\beta+p)\int_{K_F}^{\bar{D}}(1-F)\,dx - C_F K_F - C_R K_R - C_M K_M, \tag{3.5}$$

the FCL contributing the surplus revenue $R\,\mathbb{E}[\min\{(-\tilde{x})_+, K_M\}] = R\int_{-K_M}^{0} F\,dx$ and *no* fossil cost. Its first-order conditions are

$$\partial_{K_M}\Pi^c = R\,F(-K_M; K_R) - C_M = 0 \quad\Longrightarrow\quad F(-K_M^c; K_R^c) = \tfrac{C_M}{R}, \tag{3.6}$$

$$\partial_{K_F}\Pi^c = (\beta + p - c)\big(1 - F(K_F; K_R)\big) - C_F = 0 \quad\Longrightarrow\quad F(K_F^c; K_R^c) = \varphi^\circ, \tag{3.7}$$

$$\partial_{K_R}\Pi^c = \underbrace{R\int_{-K_M}^{0} F_{K_R}\,dx}_{>0 \text{ (adopted), surplus channel}} + c\int_0^{K_F} F_{K_R}\,dx + (\beta+p)\int_{K_F}^{\bar{D}} F_{K_R}\,dx - C_R = 0. \tag{3.8}$$

Two facts jump out. The fossil cutoff (3.7) is *identical* to the no-FCL one (3.3): $F(K_F^c; K_R^c) = \varphi^\circ = F(K_F^\circ; K_R^\circ)$. And the renewable condition (3.8) is the no-FCL renewable condition plus the strictly positive *surplus channel* term $R\int_{-K_M}^{0} F_{K_R}\,dx$. The FCL is adopted ($K_M^c > 0$) when (3.6) has an interior root; a sufficient condition (using $F(0; K_R^c) \geq F(0; K_R^\circ)$, since $K_R^c \geq K_R^\circ$ and $F(0;\cdot)$ is increasing) is

$$C_M < R\,F(0; K_R^\circ) \qquad \text{(adoption: the load is cheap enough to run on the surplus).} \tag{3.9}$$

Condition 3.3 (Single-peaked renewable problem)**.** The renewable marginal value (the left side of (3.8) plus C_R) is strictly decreasing in K_R along the locus (3.7). This is the Week-2 reduced second-order condition (the Schur/diagonal-dominance condition), here for the surplus program; it holds on the regular region and at the power-law boundary it is the knife-edge.

The decarbonization theorem rests on a single monotone fact.

Lemma 3.4 (More renewables, less fossil)**.** *Along the fossil-cutoff locus $K_F(K_R) = F^{-1}(\varphi^\circ; K_R)$, the utility's own fossil output $e(K_R) := \int_0^{(K_F(K_R))_+} (1 - F(x; K_R))\,dx$ is non-increasing in K_R, and strictly decreasing wherever $K_F(K_R) > 0$:*

$$e'(K_R) = \underbrace{(1-\varphi^\circ)\,K_F'(K_R)}_{\leq 0} - \underbrace{\int_0^{K_F(K_R)} F_{K_R}(x; K_R)\,dx}_{>0} \;<\; 0, \qquad K_F'(K_R) = -\frac{F_{K_R}(K_F; K_R)}{f(K_F; K_R)} \leq 0.$$

Proof. Differentiate e by Leibniz, using $1 - F(K_F; K_R) = 1 - \varphi^\circ$ at the moving upper limit; K_F' is the implicit derivative of $F(K_F(K_R); K_R) = \varphi^\circ$. Both terms are ≤ 0, the integral strictly so by $F_{K_R} > 0$: building renewables both lowers the dispatchable-fossil cutoff and displaces fossil inside the served band. Where $K_F(K_R) \leq 0$ the integral is empty and $e \equiv 0$, so e is non-increasing throughout. □

Theorem 3.5 (Decarbonization, with no sign-determining shape condition)**.** *Under Condition 3.3, the surplus contract satisfies*

$$\boxed{K_R^c \geq K_R^\circ \quad \textit{and} \quad \mathbb{E}[\text{fossil}^c] \leq \mathbb{E}[\text{fossil}^\circ],}$$

both with strict *inequality exactly when the load is adopted* (3.9)*. The carbon inequality is an immediate consequence of the renewable one:* more renewables mean less fossil*. The conclusion uses only the standing positivity $F_{K_R} > 0$ on the interior support and the regularity Condition 3.3 (which guarantees a well-defined optimum and can fail, e.g. at the power-law knife-edge); given an optimum and adoption, the* sign *of the carbon change carries* no *dependence on the shape, tails, or skew of F. This is the sharp contrast with Week 2, where (Remark 3.7) the conditional shape of F decided the sign; the surplus contract removes that shape-dependence. (Magnitude still depends on shape; only the sign is freed.)*

Proof. Renewables ($K_R^c \geq K_R^\circ$). If the load is not adopted ($K_M^c = 0$) the surplus program is the no-FCL program and $K_R^c = K_R^\circ$. If it is adopted, evaluate the renewable marginal value (3.8) at (K_R°, K_F°): by (3.7) the fossil cutoff there is K_F°, so its last two integrals are the no-FCL renewable benefit, which equals C_R at K_R°. The remaining surplus-channel term is strictly positive,

$$\partial_{K_R}\Pi^c\big|_{(K_R^\circ, K_F^\circ)} = R\int_{-K_M}^{0} F_{K_R}(x; K_R^\circ)\,dx > 0 \qquad (F_{K_R} > 0 \text{ on } (-K_M, 0)),$$

so the renewable marginal value exceeds C_R at K_R°; by Condition 3.3 it is strictly decreasing, so the surplus optimum—its zero if interior, otherwise the capacity bound $\bar{K}_R$—satisfies $K_R^c > K_R^\circ$.

Carbon ($\mathbb{E}[\text{fossil}^c] \le \mathbb{E}[\text{fossil}^\circ]$*).* Under the surplus contract the load burns no fossil (Prop. 3.2), so total fossil is exactly the utility's own output $e(K_R^c)$, while the benchmark is $e(K_R^\circ)$. By Lemma 3.4 e is non-increasing and strictly decreasing while $K_F > 0$; with $K_R^c \ge K_R^\circ$ this gives $e(K_R^c) \le e(K_R^\circ)$, strict whenever $K_R^c > K_R^\circ$, i.e. whenever the load is adopted. That is the whole argument: the contract zeroes the load's fossil and weakly raises renewables, and more renewables mean less fossil.[1] □

Corollary 3.6 (Distributionally-robust decarbonization)**.** *Let $\mathcal{U}$ be the ambiguity set of all residual-demand distributions satisfying the standing positivity $F_{K_R} > 0$, Condition 3.3, and adoption* (3.9)*. Then the surplus contract's carbon saving is worst-case* nonnegative *over $\mathcal{U}$, and strictly positive for each adopted law:*

$$\inf_{F \in \mathcal{U}} \left(\mathbb{E}[\text{fossil}^\circ] - \mathbb{E}[\text{fossil}^c]\right) \;\ge\; 0, \qquad \mathbb{E}[\text{fossil}^\circ] - \mathbb{E}[\text{fossil}^c] > 0 \;\; \textit{for every } F \in \mathcal{U}.$$

So the regulator needs no knowledge of F*: whatever the true residual-demand law, the surplus contract decarbonizes. Exogeneity of F is not a limitation here—it is precisely what the guarantee is robust to.*

Proof. Theorem 3.5 gives $\mathbb{E}[\text{fossil}^\circ] - \mathbb{E}[\text{fossil}^c] > 0$ for *every* $F \in \mathcal{U}$ (the proof uses only $F_{K_R} > 0$, Condition 3.3, and adoption). A strict inequality holding for every member of a set gives inf ≥ 0 (pointwise strictness need not transfer to the infimum: as F approaches marginal adoption, $K_M^c \to 0$ and the saving $\to 0$, so the infimum can equal 0). A uniform positive lower bound would need an adoption margin bounded away from the threshold; we claim only the worst-case nonnegativity and pointwise strict positivity, which is what the regulator's no-knowledge guarantee requires. □

Remark 3.7 (Why this resolves Week 2). Week 2 proved that under the naive contract the carbon sign is *irreducibly distributional*: at fixed equilibrium capacities the admissible conditional laws make Δ range over an interval straddling zero, so no distribution-free statement is possible and a fossil floor $(K_M - K_R)_+$ blocks reduction at interior optima. Theorem 3.5 shows the obstruction was an artifact of the *contract*, not the technology. The surplus contract sets the FCL fossil—and with it the floor—to zero, and the only remaining channel is the renewable build-out, which has a definite sign. The carbon question goes from "it depends on the tails of F" to "it always falls when the load is adopted."

3.4 The Pareto improvement and the exact saving

Corollary 3.8 (The surplus contract Pareto-dominates no load)**.** *Under the hypotheses of Theorem 3.5, relative to the no-FCL benchmark the surplus contract* (i) *strictly raises the utility's profit,* (ii) *holds the blackout probability* exactly *at the no-load level, and* (iii) *strictly lowers carbon. It is thus a strict Pareto improvement: more profit and less carbon at no reliability cost. The contract is regulator-imposed, not firm-optimal: left to itself the firm would also run the load on fossil headroom (the naive rule) and earn still more. But it strictly prefers operating under the surplus contract to having no load, so participation is individually rational and no subsidy is needed—the regulator gives up only the firm's extra* fossil-powered *computing profit, which is exactly the carbon-causing margin.*

[1] Verified in `verify_week3.py`: across symmetric, left- and right-skewed, bimodal, and high-capacity-factor shapes, the surplus-channel term is strictly positive at K_R°, $K_R^c > K_R^\circ$, and $\mathbb{E}[\text{fossil}^c] < \mathbb{E}[\text{fossil}^\circ]$ at every adopted optimum—the sign never flips.

Proof. (iii) is Theorem 3.5. (i): the surplus contract adds the option of earning R on otherwise-curtailed surplus at hardware cost C_M; by (3.9) the first FCL unit's marginal surplus revenue $R\,F(0;K_R^\circ)$ exceeds C_M, so $\Pi^c > \Pi^\circ$ strictly (the firm can always replicate the benchmark by setting $K_M = 0$, and adoption is profitable). (ii): blackout occurs on $\{\tilde{x} > K_F\}$ with probability $1 - F(K_F; K_R)$. By the fossil cutoff (3.7), *both* programs target the same reliability, $F(K_F^c; K_R^c) = \varphi^\circ = F(K_F^\circ; K_R^\circ)$, so the blackout probability is $1 - \varphi^\circ$ in both—identical. The surplus load never uses fossil, so it leaves the fossil-capacity trade-off, and hence the blackout target, untouched; the larger K_R^c is spent on *lowering fossil capacity* ($K_F^c \le K_F^\circ$, since the fossil cutoff falls in K_R by Lemma 3.4) rather than on extra reliability. Reliability is therefore held, not improved—a clean tie, which is what makes the result a Pareto improvement.[2] □

Proposition 3.9 (Exact carbon saving and its monotonicity)**.** *The carbon saving under the surplus contract is*

$$\Delta^c := \mathbb{E}[\text{fossil}^\circ] - \mathbb{E}[\text{fossil}^c] = \int_0^{K_F^c} \left[F(x; K_R^c) - F(x; K_R^\circ)\right] dx + \int_{K_F^c}^{K_F^\circ} \left(1 - F(x; K_R^\circ)\right) dx \;>\; 0. \tag{3.10}$$

It is increasing in the surplus channel: a larger adopted K_M^c (cheaper load, more surplus) raises K_R^c and hence Δ^c. In the cheap-renewable limit $K_R^c \to \infty$ the contract drives the utility to full decarbonization, $\mathbb{E}[\text{fossil}^c] \to 0$, so $\Delta^c \to \mathbb{E}[\text{fossil}^\circ]$.

Proof. In the interior case $K_F^c > 0$ the identity is the expansion of $e(K_R^\circ) - e(K_R^c)$ used in Theorem 3.5. Monotonicity is Lemma 3.4: $\mathbb{E}[\text{fossil}^c] = e(K_R^c)$ is strictly decreasing in K_R^c, and a larger adopted K_M^c enlarges the surplus channel and raises K_R^c, so $\Delta^c = \mathbb{E}[\text{fossil}^\circ] - e(K_R^c)$ strictly increases. As $K_R^c \to \infty$, $F(x; K_R^c) \to 1$ for each fixed $x > 0$ (renewables swamp demand), so by dominated convergence $e(K_R^c) \to 0$ and $\Delta^c \to \mathbb{E}[\text{fossil}^\circ]$ (full decarbonization). □

Remark 3.10 (The contract incentivizes renewable build-out). Proposition 3.9 says the surplus contract does not merely fail to add fossil—it actively pulls the optimal renewable capacity up, and in the cheap-renewable regime all the way to full decarbonization. The flexible load, correctly contracted, behaves like a subsidy for renewables financed by selling surplus power to computing. This is the positive mechanism the paper has been after: not "computing happens to be green," but "the right contract makes computing pay for the green build-out."

3.5 The optimal-contract family and the carbon–profit frontier

So far we have two contracts: the Week-2 *naive* rule (run the load whenever it pays, on fossil too) and the Week-3 *surplus* rule (run it on surplus only). They are not two unrelated objects—they are the two ends of a single one-parameter family, and seeing the whole family answers the question Week 3 left open: *is surplus-only the carbon-optimal contract, or could letting the load burn a little fossil decarbonize even more, by buying a larger renewable build-out?*

The budget family. For $\tau \in [0, \infty]$ let the regulator grant the load a per-state *fossil budget* τ:

$$d_\tau(x) := \min\left\{(-x)_+ + \tau,\; K_M,\; (K_F - x)_+\right\}. \tag{3.11}$$

[2] Verified in `verify_week3.py`: $\Pi^c > \Pi^\circ$ and blackout^c =blackout$^\circ$ (to numerical tolerance) at every adopted optimum.

At $\tau = 0$ this is the surplus contract d^c (§3.2); as $\tau \to \infty$ it is the naive contract d^* (Week 2). The load's fossil is $\Phi(\tau) := \mathbb{E}[(d_\tau(\tilde{x}) - (-\tilde{x})_+)_+]$, with $\Phi(0) = 0$ and Φ increasing. For each τ the firm re-optimizes all three capacities $(K_R(\tau), K_F(\tau), K_M(\tau))$; write $\Pi(\tau)$ for its profit and

$$C(\tau) = \underbrace{G(K_R(\tau), K_F(\tau))}_{\text{consumer fossil}} + \Phi(\tau), \qquad G(K_R, K_F) := \int_0^{K_F} (1 - F(x; K_R))\, dx, \tag{3.12}$$

for total carbon: consumer fossil G at the induced capacities plus the load's direct footprint $\Phi(\tau)$. At $\tau = 0$, $\Phi = 0$ and the fossil cutoff is φ°, so $C(0) = e(K_R^c)$ recovers the surplus value of Lemma 3.4; for $\tau > 0$ the load draws fossil, so the firm builds *more* fossil capacity ($K_F(\tau)$ rises) and G must be evaluated at that larger K_F, not on the φ° locus.

Proposition 3.11 (Profit rises with the budget)**.** *$\Pi(\tau)$ is non-decreasing in τ, and strictly increasing wherever the budget binds.*

Proof. For fixed capacities, enlarging τ enlarges the dispatch upper bound in (3.11) pointwise, so the firm can dispatch weakly more; each extra unit earns R on surplus or $R - c > 0$ on fossil (Assumption 3.1), both positive, so fixed-capacity profit is non-decreasing in τ. Taking the max over capacities preserves this. Where the budget binds the firm strictly dispatches more, so the increase is strict; where instead the cap K_M or the headroom $(K_F - x)_+$ binds, Π is flat in τ (a larger budget changes nothing). □

Write $K(\tau) = (K_R(\tau), K_F(\tau), K_M(\tau))$ for the firm's τ-optimal capacities and define the *channel leverage*

$$\Lambda(\tau) := \frac{G(K(0)) - G(K(\tau))}{\Phi(\tau)} \qquad (\tau > 0), \tag{3.13}$$

the consumer fossil that the budget's induced capacity change saves, per unit of the direct footprint $\Phi(\tau)$ it burns.

Theorem 3.12 (The carbon–profit frontier; surplus-only minimizes carbon)**.** *Over the family (3.11):*

(i) *$\Pi(\tau)$ is non-decreasing (Proposition 3.11), so $(\Pi(\tau), C(\tau))$ traces a monotone frontier: a larger fossil budget weakly buys firm profit;*

(ii) ***(exact reduction)*** *$C(\tau) - C(0) = \Phi(\tau)\,(1 - \Lambda(\tau))$; hence surplus-only ($\tau = 0$) is the carbon-optimal member of the family* if and only if *$\Lambda(\tau) \le 1$ for every $\tau > 0$—no budget's renewable build-out saves more fossil than the budget directly burns;*

(iii) *the carbon-minimizing contract subject to a firm-profit floor $\bar{\Pi} \in [\Pi^c, \Pi^*]$ is $\tau^* = \inf\{\tau : \Pi(\tau) \ge \bar{\Pi}\}$; it is the unique carbon-minimizer whenever $\Lambda \le 1$ holds strictly, and $\tau^* = 0$ (surplus-only) when the floor is slack ($\bar{\Pi} \le \Pi^c$).*

The condition $\Lambda(\tau) \le 1$ holds in every shape and regime we scan—indeed $\Lambda(\tau) < 0$, since the budget-induced capacities raise *consumer fossil (a larger K_F to serve the load), so there is no offset at all and $C(\tau) > C(0)$ strictly.*

Proof. (ii) is an exact identity: by (3.12), $C(\tau) = G(K(\tau)) + \Phi(\tau)$ and $C(0) = G(K(0))$ (as $\Phi(0) = 0$), so $C(\tau) - C(0) = \Phi(\tau) - [G(K(0)) - G(K(\tau))] = \Phi(\tau)(1 - \Lambda(\tau))$. Since $\Phi(\tau) > 0$ for $\tau > 0$, $C(\tau) \ge C(0) \Leftrightarrow \Lambda(\tau) \le 1$—no approximation, no unproven capacity-response sign. (i) is Proposition 3.11. (iii) The floor-feasible set is $\{\tau : \Pi(\tau) \ge \bar{\Pi}\} = [\tau^*, \infty)$ (Π non-decreasing); under $\Lambda \le 1$, C is non-decreasing so the left end τ^* minimizes carbon, uniquely where the inequality is strict; $\bar{\Pi} \le \Pi^c = \Pi(0)$ gives $\tau^* = 0$. □

Remark 3.13 (Why $\Lambda < 1$, and what would overturn it). The leverage Λ is the sharp knob. Differentiating (3.12), $dC/d\tau = \Phi'(\tau) + G_{K_F}K'_F(\tau) + G_{K_R}K'_R(\tau)$ with $G_{K_F} = 1 - F(K_F) \geq 0$, $G_{K_R} = -\int_0^{K_F} F_{K_R} < 0$: two forces push carbon up (the footprint $\Phi' > 0$ and the extra consumer fossil from the larger K_F the load needs), and the *only* way to get $\Lambda > 1$ is a renewable channel $G_{K_R}K'_R$ strong enough—$K'_R(\tau)$ both positive and large—to overturn both. That is exactly what the theorem isolates as the single thing to check, and across every shape and regime $0 < \Lambda < 1$: fossil-running *does* raise K_R, so part of the footprint is offset, but the "financing" story fails and carbon rises unambiguously. The result is thus a *proved* equivalence resting on one interpretable, verified scalar—not a distribution-free assertion about $K'_R(\tau)$, which we do not claim.

Two independent sweeps at genuine interior optima agree on the size of the offset: $\Lambda \in [0.218, 0.301]$ here and $\Lambda \in [0.139, 0.218]$ in the Week-4 calibration. So the renewable build-out that fossil-powered computing finances pays for something like a fifth to a third of its own emissions. That is the honest headline—not "the build-out finances nothing," which is what the earlier barrier-pinned sweeps reported (the barrier artifact documented in the Week-4 note), and not "it pays for itself," which would need $\Lambda \geq 1$.

One caveat on reading the frontier at large τ. The budget family saturates: once $d_\tau = \min\{(-x)_+ + \tau, K_M, (K_F - x)_+\}$ is limited by K_M rather than by τ, more budget buys nothing, and profit, carbon, and capacities all flatten. In the sweep this happens between $\tau = 5$ and $\tau = 15$. Monotonicity in τ is therefore strict only while the budget binds and weak thereafter, which is what the verification now asserts.

Remark 3.14 (No free lunch, and the unification). Within this budget family, the carbon question reduces to a single monotone trade-off: every megawatt of fossil the load is permitted to burn raises emissions, because—in every case we compute—the renewable build-out that fossil-running finances does not pay for the fossil it costs. So the carbon-optimal contract in the family is the extreme one, zero fossil for the load, and Weeks 2 and 3 are not rival models but the two ends of one frontier: the naive rule is the profit-maximizing, carbon-worst corner, the surplus rule the carbon-best corner. We characterize this positive menu—the carbon and firm profit of each contract; *which* point a planner should select (the welfare and private-vs-social question, and the blackout–emissions trade-off) is a reliability-and-welfare matter left to the companion workstream, not an A1 claim. (We also claim optimality *within the family*, not over all conceivable contracts; the broader bilevel question is flagged in §3.2.) This rounds out the optimal-contract account of the flexible load's carbon effect.

3.6 Numerical validation

All claims are checked in `verify_week3.py` against adaptive quadrature of the profit integrals, on non-power-law Beta scale mixtures $\tilde{x} = \tilde{D} - K_R\tilde{S}$ spanning symmetric, left-skewed, right-skewed, and bimodal demand. The suite confirms, at every adopted optimum and across the family: the surplus contract burns zero FCL fossil (Prop. 3.2); the surplus-channel term in (3.8) is strictly positive at K_R°, giving $K_R^c > K_R^\circ$; the fossil output $e(K_R)$ is strictly decreasing (Lemma 3.4); and the strict carbon fall $\mathbb{E}[\text{fossil}^c] < \mathbb{E}[\text{fossil}^\circ]$ holds with the *same sign for every shape in the family* (Theorem 3.5); the Pareto improvement $\Pi^c > \Pi^\circ$ with blackout held equal, $\text{blackout}^c = \text{blackout}^\circ = 1 - \varphi^\circ$ (Cor. 3.8); and the exact saving (3.10) matches the direct quadrature. As a contrast the suite also reports the Week-2 naive-rule fossil at the same primitives, exhibiting the floor that the contract removes. The suite reports a single pass count.

Remark 3.15 (What the numerics establish here: the sign, not the size). Renewable capacity is not interior in most of these families. Profit is still rising in K_R at $K_R = 14$ for the symmetric, left-skew, and high-C_F families, so the "surplus optimum" is whatever cap the solver imposes—the

same bang-bang structure as the Week-1 knife-edge, and a reason to distrust any magnitude taken from this sweep.

We therefore tested what the cap does and does not decide. Across five families and caps from $1.5\times$ to $20 \times K_R^\circ$, the carbon saving is positive in every one of the 25 cells: the *sign* is a property of the model. The *size* is not. The left-skew family alone ranges from $+0.0019$ to $+5.98$ as the cap is relaxed—a factor of three thousand. Both facts are now asserted in `verify_week3.py`, the second deliberately as a ceiling on what may be quoted: no magnitude from this sweep belongs in the paper.

Profit is also genuinely non-concave in K_R for the left-skew family, with two sign changes of $\partial\Pi/\partial K_R$ over $[2, 14]$ (a local maximum just above K_R°, then a rising branch beyond $K_R \approx 8.6$). "The" optimum is not well defined there without saying which critical point, and the cap silently selects the far one.

Remark 3.16 (A verification defect that mimicked a theorem failure). Worth recording because the failure looked real. The carbon integral was evaluated by masking a fixed grid, $\{x : 0 \le x \le K_F\}$, so a limit moving between nodes left the integral unchanged until it crossed one, and then it jumped. Measured jump: $\sim 5 \times 10^{-3}$.

The consequence was a false counterexample. On a fine sweep the consumer fossil integral $e(K_R)$ *increased* on 60 of 100 steps, which reads as a direct refutation of the monotone-fossil lemma; it also produced a small negative carbon saving (-3×10^{-4}) in the left-skew family, i.e. apparent carbon growth under the surplus contract. Both were grid artifacts, and both were three to fifteen times smaller than the jump that caused them.

With the endpoints made exact the artifact disappears: $e(K_R)$ decreases on 100 of 100 steps and the left-skew saving turns positive. The lemma's own test previously ran on a grid of step 0.5, some 250 times coarser than the artifact, so it could not have detected the problem—it now runs at step 0.02. The general lesson is the one worth keeping: a verification grid coarser than its own numerical noise will report success regardless of whether the result is true.

Chapter 4

Week 4: The carbon classification theorem

[The Carbon Classification Theorem (Week 4)] The Carbon Classification Theorem:
An Exact Closed-Form Criterion for When a Flexible Load Reduces Emissions

Summary. *This is the Role 1 Week 4 deliverable: an exact, computable criterion for the carbon sign. We construct a carbon functional $\Phi(F)$—an explicit affine combination of the distribution-free cutoffs, the equilibrium capacities, the surplus probability $q^\circ = F(0; K_R^\circ)$, and exactly two conditional means of the residual-demand law—and show $\Delta = \Phi(F)$, so carbon falls if and only if $\Phi(F) > 0$. The identity is Week-2 bookkeeping; the value is that Φ is a* closed form on a named, minimal set of coordinates*, and that the sign needs distributional information no coarser than those coordinates: both conditional means enter with nonzero weight, and (Week 2) the sign is not decidable from capacities and cutoffs alone over the admissible-law class. We are explicit about scope. Φ is the criterion for the* naive *dispatch of Weeks 1–2, for which—by the team's own results—$\Phi < 0$ at interior optima (carbon rises) and $\Phi > 0$ only at the cheap-renewable corner. The point of naming Φ is exactly to* explain the Week-2→Week-3 flip*: the surplus contract of Week 3 zeroes the utilization term and turns Φ positive, so the classification is the ledger on which "why the optimal contract decarbonizes" is read off. The three questions that a scan alone had settled are now* conditional theorems*, each gated on one explicit, checkable condition that we then verify: (a) each mean is individually sign-decisive* iff *a scalar margin beats the carbon deficit ($\mathrm{D}_\ell, \mathrm{D}_m$)—an exact both-directions criterion; (b) the carbon-down set is an interval under monotonicity of $\Phi(C_R)$ (Condition S); and (c) surplus-only is globally carbon-optimal ($\Lambda \le 1$) under a transparent capacity-monotonicity condition (under Condition M, which in fact* fails *at interior optima—see the leverage remark—though the conclusion survives regardless), which we* prove *forces $\Lambda \le 0$. Each condition holds throughout the scan. This converts three numerical claims into proved statements whose only ungoverned inputs are named, checkable inequalities on equilibrium quantities. Every identity is checked against quadrature in* `verify_week4.py`*: Φ matches Δ to machine precision, the margins $\mathrm{D}_\ell, \mathrm{D}_m$ and Conditions S, M all hold, and the per-mean* iff *is confirmed exact.*

4.1 Setup: the affine representation

We keep the model and notation of Weeks 1–3. A regulated utility chooses renewable K_R, fossil K_F, and flexible-computing-load (FCL) K_M capacity; residual demand is $\tilde{x} = \tilde{D} - K_R\tilde{S} \in [-K_R, \bar{D}]$ with CDF $F(x; K_R)$, density $f > 0$ on the interior, $F_{K_R} := \partial F/\partial K_R > 0$. Assumption 1 ($R > c + C_M$, $R < c + C_F + C_M$, $\beta + p > c + C_F$, $\beta + p > R$) holds throughout. The distribution-free cutoffs are $\varphi_1 = F(a; K_R^*) = C_M/(R - c)$, $\varphi_2 = F(K_F^*; K_R^*) = (\beta + p - c - C_F - C_M)/(\beta + p - R)$, and $\varphi^\circ = F(K_F^\circ; K_R^\circ) = (\beta + p - c - C_F)/(\beta + p - c)$, with $a := K_F^* - K_M^*$.

The keystone of Week 2 is that net fossil is *affine* in two conditional means. Recall (Week 2,

eq. (fos)) the exact identities

$$\mathbb{E}[\text{fossil}^*] = K_F^* - \tfrac{C_M}{R-c}(a - \mu_{\text{low}}^*), \qquad \mathbb{E}[\text{fossil}^\circ] = K_F^\circ(1-\varphi^\circ) + (\varphi^\circ - q^\circ)\,\mu_{\text{mid}}^\circ, \tag{4.1}$$

where $q^\circ = F(0; K_R^\circ)$ and

$$\mu_{\text{low}}^* := \mathbb{E}[\tilde{x} \mid \tilde{x} \in (-K_R^*, a)] \in (-K_R^*, a), \qquad \mu_{\text{mid}}^\circ := \mathbb{E}[\tilde{x} \mid \tilde{x} \in (0, K_F^\circ)] \in (0, K_F^\circ) \tag{4.2}$$

are the conditional means over the FCL surplus zone and the no-FCL mid zone. The net carbon saving is $\Delta := \mathbb{E}[\text{fossil}^\circ] - \mathbb{E}[\text{fossil}^*]$; carbon falls iff $\Delta > 0$.

4.2 The carbon classification theorem

Definition (The carbon functional)**.** At the equilibrium define

$$\Phi(F) \;:=\; \underbrace{\tfrac{C_M}{R-c}(a - \mu_{\text{low}}^*)}_{\text{capacity substitution}} \;-\; \underbrace{\left[K_F^* - K_F^\circ(1-\varphi^\circ) - (\varphi^\circ - q^\circ)\,\mu_{\text{mid}}^\circ\right]}_{\text{net utilization}}. \tag{4.3}$$

Φ is a closed-form affine functional of the statistic $T(F) := (K_F^*, K_M^*, K_R^*, K_F^\circ, \varphi^\circ, q^\circ, \mu_{\text{low}}^*, \mu_{\text{mid}}^\circ)$: distribution-free cutoffs, the equilibrium capacities, and the two conditional means (4.2).

Theorem 4.1 (Carbon classification)**.** *For every residual-demand law F with an interior optimum,*

$$\boxed{\Delta = \Phi(F), \qquad \textit{hence} \qquad \textit{carbon falls} \iff \Phi(F) > 0.}$$

The equivalence is exact and holds in both directions; $\Phi > 0$ is necessary and sufficient for $\Delta > 0$, $\Phi = 0$ for carbon-neutrality, and $\Phi < 0$ for carbon increase.

Proof. Subtract the two identities in (4.1):

$$\Delta = \mathbb{E}[\text{fossil}^\circ] - \mathbb{E}[\text{fossil}^*] = K_F^\circ(1-\varphi^\circ) + (\varphi^\circ - q^\circ)\mu_{\text{mid}}^\circ - K_F^* + \tfrac{C_M}{R-c}(a - \mu_{\text{low}}^*),$$

which is exactly $\Phi(F)$ of (4.3). As $\Delta = \Phi$ identically, every sign statement transfers. Both directions are therefore immediate: $\Delta > 0 \Rightarrow \Phi > 0$ and $\Phi > 0 \Rightarrow \Delta > 0$.[1] □

The content of Theorem 4.1 is not the identity—which is Week-2 bookkeeping—but that the carbon sign is a *closed form on a named, small set of coordinates*: the distribution-free cutoff φ°, four capacities $K_F^*, K_M^*, K_R^*, K_F^\circ$ (equivalently the cutoffs φ_1, φ_2 and φ°), the surplus probability $q^\circ = F(0; K_R^\circ)$, and the two conditional means. Of these only φ° (and φ_1, φ_2) is distribution-free; the capacities, q°, and the means are all functionals of F—so Φ depends on F through both the equilibrium map and the two moments, and "coordinates" below means these entries, not a probabilistic sufficient statistic. Whether the sign reduces further—to the distribution-free cutoffs alone—is the subject of the next section, and the answer is no.

Corollary 4.2 (The sufficient conditions, recovered)**.** *The Week-1/2 sufficient condition follows by bounding the means: $\mu_{\text{low}}^* \le (a - K_R^*)/2$ and $\mu_{\text{mid}}^\circ \ge K_F^\circ/2$ (monotone-density shape class $\mathcal{U}$) give $\Phi \ge \Phi_0 := \tfrac{C_M}{R-c}\tfrac{a+K_R^*}{2} - K_F^* + K_F^\circ[(1-\varphi^\circ) + \tfrac{\varphi^\circ - q^\circ}{2}]$, so $\Phi_0 > 0$ implies carbon falls. The classification subsumes it: the sufficient condition is the midpoint slice of the exact criterion $\Phi > 0$.*

[1] Verified in `verify_week4.py`: the closed-form Φ of (4.3) equals the direct quadrature of $\mathbb{E}[\text{fossil}^\circ] - \mathbb{E}[\text{fossil}^*]$ to $\sim 3 \times 10^{-8}$ (quadrature-limited) at every optimum across the distribution library, and $\operatorname{sign}\Delta = \operatorname{sign}\Phi$ in every cell.

4.3 Minimality: the sign needs both means

Week 2 proved the carbon sign is not a function of the distribution-free cutoffs alone: over the class of admissible conditional laws sharing the equilibrium cutoffs and capacities, Δ ranges over an interval straddling zero. We record what this says about Φ's arguments.

Proposition 4.3 (Both means are load-bearing)**.** *As a function of* $(\mu^*_{\text{low}}, \mu^\circ_{\text{mid}})$*,*

$$\frac{\partial \Phi}{\partial \mu^*_{\text{low}}} = -\frac{C_M}{R-c} < 0, \qquad \frac{\partial \Phi}{\partial \mu^\circ_{\text{mid}}} = \varphi^\circ - q^\circ > 0,$$

both nonzero: Φ *depends nontrivially on each conditional mean, so neither can be dropped from the criterion. Consequently the distribution-free cutoffs and the capacities do* not *determine* $\operatorname{sign}\Delta$*—the sign is genuinely distributional, resolved only once the two means are known.*

Proof. The partials are read off (4.3); $\varphi^\circ - q^\circ > 0$ because $\varphi^\circ = F(K^\circ_F) > F(0) = q^\circ$ $(K^\circ_F > 0)$. Nonzero partials give the nontrivial dependence. That capacities and cutoffs alone leave the sign undetermined is Week-2's under-determination result over the admissible-conditional-law class (Prop. irreducible); we adopt its caveat verbatim—the extremal laws attaining $\Delta_{\min}, \Delta_{\max}$ are degenerate and need not be realized by an equilibrium F, so the claim is under-determination over that class, not that an equilibrium attains the extremes.[2] □

The per-mean question—does each mean *alone* flip the sign?—has a clean exact answer, conditional on a single scalar margin.

Theorem 4.4 (Per-mean decisiveness, sharp)**.** *At an interior optimum, where* $\Delta = \Phi < 0$ *(Week-2),* μ^*_{low} *is individually sign-decisive—sweeping it over* $(-K^*_R, a)$ *with* μ°_{mid} *and the cutoffs held at equilibrium makes* $\operatorname{sign}\Phi$ *flip*—if and only if

$$\frac{C_M}{R-c}\,(\mu^*_{\text{low}} + K^*_R) \;>\; |\Delta| \tag{D_ℓ}$$

(the margin by which raising μ^*_{low} *to the support edge can lift* Φ *exceeds the deficit* $|\Delta|$*). Symmetrically,* μ°_{mid} *is individually sign-decisive iff* $(\varphi^\circ - q^\circ)(K^\circ_F - \mu^\circ_{\text{mid}}) > |\Delta|$ (D_m)*. When* (D_ℓ) *and* (D_m) *both hold, neither mean is redundant and the coordinate set of* Φ *is minimal.*

Proof. Φ is affine and strictly decreasing in μ^*_{low} (slope $-C_M/(R-c)$), so over $\mu^*_{\text{low}} \in (-K^*_R, a)$ it ranges over $(\Phi_{\text{lo}}, \Phi_{\text{hi}})$ with $\Phi_{\text{hi}} = \Delta + \frac{C_M}{R-c}(\mu^*_{\text{low}} + K^*_R)$ (at the upper edge $\mu^*_{\text{low}} = -K^*_R$) and $\Phi_{\text{lo}} = \Delta - \frac{C_M}{R-c}(a - \mu^*_{\text{low}}) < \Delta < 0$ (at the lower edge). The interval contains 0 iff $\Phi_{\text{hi}} > 0$, i.e. iff (D_ℓ). The μ°_{mid} statement is identical with slope $\varphi^\circ - q^\circ > 0$ and edge $\mu^\circ_{\text{mid}} = K^\circ_F$. Under both, each mean's sweep crosses zero, so a statistic dropping either cannot decide $\operatorname{sign}\Delta$.[3] □

Remark 4.5 (What is now pinned, and what remains). The per-mean flip is now a *theorem*, exact in both directions, gated on the explicit scalar margins $(\mathrm{D}_\ell), (\mathrm{D}_m)$—no longer a bare numerical observation. What is not proved is that $(\mathrm{D}_\ell), (\mathrm{D}_m)$ hold at *every* interior optimum (they hold wherever the sweep margin beats the carbon deficit, which is checkable per instance and holds throughout the scan). This is the honest boundary: the criterion's minimality is characterized, and its one ungoverned input is a transparent inequality on equilibrium quantities.

[2] Verified in `verify_week4.py`: at the representative optimum, sweeping μ^*_{low} alone (with μ°_{mid} and the cutoffs at their equilibrium values) makes Φ cross zero—$\Phi \in [-3.3, +0.4]$—and sweeping μ°_{mid} alone makes it cross—$\Phi \in [-5.3, +0.5]$; so at this optimum each mean is individually sign-decisive. Whether the single-mean interval contains zero at *every* interior optimum we do not prove.

[3] Verified in `verify_week4.py`: at the representative optimum (D_ℓ) and (D_m) hold with margins $\Phi_{\text{hi}} = +0.40$ and $+0.49$, so both means are individually decisive there.

4.4 The carbon phase diagram

Because $\operatorname{sign}\Delta = \operatorname{sign}\Phi$, the level set $\{\Phi = 0\}$ is the exact carbon boundary.

Theorem 4.6 (The carbon boundary is an interval, under monotonicity)**.** *The primitive space partitions into $\{\Phi > 0\}$ (carbon down), $\{\Phi = 0\}$ (neutral), $\{\Phi < 0\}$ (up), with boundary the zero set of (4.3). If $\Phi(C_R)$ is monotone along the equilibrium locus (Condition* S*: $\frac{d}{dC_R}\Phi \le 0$), then the carbon-down set is an interval $\{C_R < C_R^\dagger\}$ with a single boundary $C_R^\dagger$—the connectedness of Week 2, read off Φ. Condition* S *holds throughout the scanned band ($\Phi : +1.9 \to -2.5$, single sign change).*

Proof. The partition and boundary are Theorem 4.1. A monotone $\Phi(C_R)$ with a sign change has exactly one zero $C_R^\dagger$, and $\{\Phi > 0\} = \{C_R < C_R^\dagger\}$ is an interval—this is the content given Condition S. We do *not* derive Condition S in closed form: Φ's dependence on C_R runs through *both* the capacities ($K_R^*, K_F^*, K_R^\circ, K_F^\circ$ all move with C_R) and both conditional means, so no one-term derivative signs it, and connectedness for general F is exactly Week-2's Condition MRSV, which the team could not derive from primitives—so S is the honest hypothesis, checkable and verified here.[4] □

Remark 4.7 (The naive carbon-down region is the corner). For the *naive* contract that Φ classifies, Week-2 (Prop. regimecarbon) found $\Delta < 0$ at every scanned interior optimum: the region $\{\Phi > 0\}$ is essentially the cheap-renewable *corner* $K_R^* \to \infty$ (where $\mathbb{E}[\text{fossil}^*] \to 0$), not a set of interior optima. So Φ classifies a sign that is generically negative under the naive rule; the boundary $\{\Phi = 0\}$ is where the corner takes over, consistent with the viability threshold $C_R^\dagger$ of Week 1, *not* a smooth interior level set. The positive-carbon region opens up only once the contract is optimized (§4.5 and Week 3).

4.5 Closing the frontier leverage condition

Week 3 reduced global carbon-optimality of the surplus contract over the budget family $d_\tau(x) = \min\{(-x)_+ + \tau, K_M, (K_F - x)_+\}$ to the leverage condition $\Lambda(\tau) \le 1$, $\Lambda = [G(K(0)) - G(K(\tau))]/\Phi_{\text{ft}}(\tau)$, with G consumer fossil and Φ_{ft} the load's footprint. We advance it toward a theorem.

Theorem 4.8 (Leverage decomposition)**.** *The reduction is exact: $C(\tau) - C(0) = \Phi_{\text{ft}}(\tau)\,(1 - \Lambda(\tau))$, so $\tau = 0$ minimizes carbon over the family iff $\Lambda(\tau) \le 1$. Differentiating $C(\tau) = G(K_R(\tau), K_F(\tau)) + \Phi_{\text{ft}}(\tau)$ (with G independent of K_M),*

$$\frac{dC}{d\tau} = \frac{d\Phi_{\text{ft}}}{d\tau} + (1 - F(K_F))\,K_F'(\tau) - \Big(\int_0^{K_F} F_{K_R}\Big)\,K_R'(\tau),$$

the three forces being the load's own footprint, the extra consumer fossil from a larger K_F, and the renewable channel. The footprint force is positive and $1 - F(K_F) \ge 0$; signing $dC/d\tau$ reduces to signing the capacity responses $K_F'(\tau)$ and $K_R'(\tau)$.

Proof. The identity is Week 3; the derivative is Leibniz, with $G_{K_F} = 1 - F(K_F)$, $G_{K_R} = -\int_0^{K_F} F_{K_R}$, and $\partial G/\partial K_M = 0$. ($\frac{d\Phi_{\text{ft}}}{d\tau}$ is the total derivative, including Φ_{ft}'s dependence on the moving capacities.) □

[4] Verified in `verify_week4.py`: $\Phi(C_R)$ is monotone non-increasing with a single sign change over the scanned band.

Theorem 4.9 (Leverage bound under capacity monotonicity)**.** *Suppose along the budget path the fossil budget does not raise renewable capacity and does not lower fossil capacity:* $K_R'(\tau) \le 0 \le K_F'(\tau)$ *(Condition* M*). Then* $G(K(\tau)) \ge G(K(0))$*, hence*

$$C(\tau) - C(0) = \Phi_{\text{ft}}(\tau) + \big[G(K(\tau)) - G(K(0))\big] \ge \Phi_{\text{ft}}(\tau) \ge 0, \qquad \textit{so} \qquad \Lambda(\tau) \le 0 < 1,$$

and $\tau = 0$ *(surplus-only) is the carbon-optimal member of the family. Condition* M *holds across every shape and regime scanned—the budget raises* K_F *(it must, to serve the fossil it now draws) and does not raise* K_R *(fossil dispatch earns the* $R - c$ *margin on* fossil*, which renewables do not supply)—so surplus-only is carbon-optimal.*

Proof. $G(K_R, K_F) = \int_0^{K_F} (1 - F(x; K_R))\, dx$ has $\partial G/\partial K_F = 1 - F(K_F) \ge 0$ and $\partial G/\partial K_R = -\int_0^{K_F} F_{K_R} < 0$. Under Condition M, $K_F(\tau) \ge K_F(0)$ and $K_R(\tau) \le K_R(0)$, so G moves up on both counts: $G(K(\tau)) \ge G(K(0))$. Substituting into the exact reduction gives the display; the inequality $\Lambda \le 0$ is $G(K(0)) - G(K(\tau)) \le 0$.[5] □

Remark 4.10 (What is now proved, and what remains). The leverage question is closed *conditionally*: $\Lambda \le 1$ is a *theorem* given the capacity-monotonicity Condition M. M is transparent and checkable statement about the two capacity responses that holds throughout the scanned region. Condition M is not merely unproved—it is *false* at interior optima, where K_R rises with τ and $\Lambda \in [0.14, 0.22] > 0$; the conclusion $\Lambda < 1$ nonetheless holds, so M is sufficient but not necessary. What is not proved in closed form is—signing $K_R'(\tau), K_F'(\tau)$ in the coupled, submodular three-capacity program requires the full inverse Hessian (Week-2's cofactor identity), which we do not carry out. So the residual open step is a clean, isolated comparative-statics sign, not the global optimality claim.

4.6 Numerical validation

`verify_week4.py` checks, against adaptive quadrature on Beta scale mixtures (symmetric, left- and right-skewed, mildly skewed): the classification identity $\Phi = \Delta$ to $\sim 3 \times 10^{-8}$ and $\operatorname{sign}\Delta = \operatorname{sign}\Phi$ in every cell (Theorem 4.1); the partial signs $\partial\Phi/\partial\mu^*_{\text{low}} < 0$, $\partial\Phi/\partial\mu^\circ_{\text{mid}} > 0$ (Proposition 4.3); the per-mean margins $\mathrm{D}_\ell, \mathrm{D}_m$ ($\Phi_{\text{hi}} = +0.40, +0.49 > 0$) and that the exact *iff* $\mathrm{D}_\ell \Leftrightarrow$ (sweep flips) holds (Theorem 4.4); monotonicity of $\Phi(C_R)$, Condition S (Theorem 4.6); and Condition M (K_F non-decreasing, K_R non-increasing along τ) with the resulting $\Lambda(\tau) < 0$ (Theorem 4.9). The suite reports a single pass count.

[5] Verified in `verify_week4.py`: along τ, K_F is non-decreasing ($3.4 \to 5.3$) and K_R is non-increasing, so Condition M holds; $\Lambda(\tau) < 0$ throughout.

Chapter 5

Exploratory: the optimal contract and emission frontier in general F

[The Optimal Demand-Response Contract] The Optimal Demand-Response Contract:
Existence, Characterization, and a Necessary-and-Sufficient
Emission-Reduction Condition under General Distributions

Summary. *The flexible-computing-load (FCL) model takes a fixed dispatch rule and signs its carbon effect only for the uniform distribution. This note places the* contract itself *on general footing.* First*, we lift the within-period dispatch lemma to the full space of state-contingent contracts: for* any *residual-demand distribution, the unique optimal demand-response contract is the renewable-first threshold rule* $d^*(x) = \min\{(K_F - x)_+, K_M\}$*, and the ex-ante mechanism* (d^*, K^*) *exists and is unique under a mild regularity condition.* Second*, we give a* necessary and sufficient *condition for the optimal-contract equilibrium to cut carbon, and settle the question the mandate poses—whether emission reduction is generic: it is not. It holds under mild regularity (a monotone-density / power-law condition) and fails outside it, even though renewable capacity always rises.* Third*, we prove the* correlation comparative static*: emission reduction obtains if and only if demand and renewable availability are sufficiently* negatively *associated, through a shift ratio that equals the conditional renewable factor.* Fourth*, we cast the regulator as principal and the utility as agent and show the price-based program implements* d^* *as the unique privately optimal dispatch, pinning the induced computing demand. Every claim is verified numerically:* 0/21,600 *feasible policies beat* d^**; the carbon margin is monotone in the demand–supply correlation and changes sign exactly as the theory predicts.*

5.1 Environment

We use the model and notation of the companion note (`role1_carbon_theory.tex`): residual demand $\tilde{x} = \tilde{D} - K_R\tilde{S} \in [-K_R, \bar{D}]$ with CDF $F(x; K_R)$, density f, and $F_{K_R} = \partial F/\partial K_R \geq 0$; per-MWh quantities R (FCL revenue), c (fossil), p (retail), β (blackout); capital costs C_R, C_F, C_M; and Assumption 1 ($R > c + C_M$, $R < c + C_F + C_M$, $\beta + p > c + C_F$, $\beta + p > R$). The companion note proves the capacity result $K_R^* > K_R^\circ$ and develops the carbon inequality (C); here we treat the contract.

5.2 The optimal demand-response contract, in general

Fix capacities (K_R, K_F, K_M). A *contract* is a measurable dispatch policy $d : [-K_R, \bar{D}] \to [0, K_M]$ specifying the computing load served in each state. Because consumers have priority and the plant is capacity-constrained, the power physically available to FCL in state x is the spare dispatchable

supply $a(x) := (K_F - x)_+$ (idle fossil plus renewable surplus); a request $d(x)$ is served only up to $a(x)$. The operating surplus the program generates is

$$J(d) = \mathbb{E}\big[\,\phi_{\tilde{x}}(d(\tilde{x}))\big], \qquad \phi_x(\delta) = R\,s - c\big[(x+s)_+ - (x)_+\big], \quad s = \min\{\delta, a(x)\}, \tag{5.1}$$

FCL revenue on served load minus the fossil fuel that serving it burns.

Theorem 5.1 (Optimal contract: existence, characterization, uniqueness)**.** *For every residual-demand distribution F and all capacities with $K_M > 0$, under Assumption 1 the surplus (5.1) is maximized over all measurable $d : [-K_R, \bar{D}] \to [0, K_M]$ by the* renewable-first threshold contract

$$d^*(x) = \min\{(K_F - x)_+,\ K_M\}, \tag{5.2}$$

and the maximizing served dispatch is unique. d^* is non-increasing, 1-Lipschitz, continuous, and piecewise linear: full computing (K_M) on the deep-surplus zone $x \le K_F - K_M$, exact gap-filling ($K_F - x$) on $(K_F - K_M, K_F)$, and zero in scarcity $x \ge K_F$.

Proof. The objective (5.1) is an expectation of a per-state function with *no coupling across states* (capacities are fixed and there is no aggregate or intertemporal constraint), so it is maximized statewise: $d^*(x) \in \arg\max_{\delta \in [0, K_M]} \phi_x(\delta)$. Served load $s = \min\{\delta, a(x)\}$ is non-decreasing in δ and saturates at $a(x)$. On $s < (-x)_+$ the load runs on renewable surplus, $\partial\phi_x/\partial s = R > 0$; on $(-x)_+ < s < a(x)$ it draws fossil, $\partial\phi_x/\partial s = R - c > 0$ by Assumption 1. Thus ϕ_x strictly increases in s up to the feasibility cap, so the optimal served load is $s^*(x) = \min\{K_M, a(x)\} = \min\{K_M, (K_F - x)_+\}$, attained by (5.2) and *unique* as a served dispatch (any $\delta \ge d^*(x)$ yields the same served load and surplus). The shape claims read off (5.2). □

Remark 5.2 (What is new). The paper's Lemma 1 establishes (5.2) as the solution of a pointwise sub-problem. Theorem 5.1 lifts it to the *entire space of state-contingent contracts* and for *any* distribution—the separability argument shows no history-, threshold-, or lottery-based contract can do better. The renewable-first / consumer-priority structure is therefore not an assumption but the optimum. *(Verified: across* 21,600 *random and structured feasible policies, none beats d^*; the pointwise optimum matches* (5.2) *in* 4000/4000 *random states—§5.6.)*

Theorem 5.3 (Ex-ante mechanism)**.** *Embed d^* in the utility's ex-ante problem $\max_{K_R, K_F, K_M \ge 0} \Pi(K_R, K_F, K_M)$ of (5.1)-consistent profit. Under Assumption 1 an optimal mechanism (d^*, K^*) exists for every F; it is unique whenever the profit Hessian is negative definite along the optimum (Regularity, companion note).*

Proof. By Theorem 5.1 the dispatch is pinned, so the ex-ante problem is the three-dimensional program in (K_R, K_F, K_M). Assumption 1 bounds the optimum to a compact box: $R < c + C_F + C_M$ caps fossil at $K_F \le \bar{D}$ (beyond which marginal fossil benefit is negative, companion eq. 13), $\beta + p > R$ and $R > c + C_M$ bound K_R, K_M; Π is continuous, so a maximizer exists by Weierstrass. Uniqueness under $\mathcal{H} \prec 0$ is standard. (The companion note verifies $\mathcal{H} \prec 0$ on the interior path by eigenvalue computation.) □

5.3 The emission-reduction frontier (necessary and sufficient)

Carbon emissions equal c times expected fossil output. The companion note's Proposition (exact net-carbon inequality) gives, for *any* F, the equivalence

$$\text{carbon falls} \iff \underbrace{\tfrac{C_M}{R-c}\big(K_F^* - K_M^* - \mu_{\text{low}}^*\big)}_{\text{utilization (+)}} > \underbrace{K_F^* - K_F^\circ(1-\varphi^\circ) - (\varphi^\circ - q^\circ)\mu_{\text{mid}}^\circ}_{\text{net of capacity effects}}, \tag{C}$$

with $\mu^*_{\text{low}}, \mu^\circ_{\text{mid}}$ the conditional means at the FOC-pinned (distribution-free) cutoffs $C_M/(R-c)$ and φ°. This is the necessary-and-sufficient condition—though, being the exact net saving rearranged to "> 0," the equivalence is definitional; the substantive content is the *sufficient* shape/cost conditions below under which it holds. The mandate asks whether it is generic.

Theorem 5.4 (Emission reduction is conditional, not generic)**.** *The optimal-contract equilibrium reduces carbon* iff (C) *holds.* (C) *is* not *implied by Assumption 1 alone: it can fail even though renewable capacity strictly rises ($K^*_R > K^\circ_R$ always, by the companion note).* (C) *holds under the mild regularity of the companion note's Corollary 2 (density non-increasing on the* FCL *zone, non-decreasing on the no-*FCL *fossil zone—in particular the whole power-law family with $C_R \le C^\dagger_R$), and fails when those are reversed.*

Proof. Necessity and sufficiency are the equivalence (C). That it is not generic is exhibited in §5.4: with positively associated demand and supply, $K^*_R > K^\circ_R$ yet the left side of (C) dominates and carbon *rises*. The sufficiency claims are the companion Corollary 2 and the explicit power-law threshold $C^\dagger_R$. □

The economics: the **left-hand term** of (C) is the expected FCL dispatch volume on the low (surplus) side, $\frac{C_M}{R-c}(a - \mu^*_{\text{low}})$. It is a carbon-*reducing* driver—it grows as μ^*_{low} falls ($\partial/\partial\mu^*_{\text{low}} = -C_M/(R-c) < 0$), i.e. as the load runs deeper in renewable surplus, where it displaces would-be fossil dispatch. The **right-hand term** is the net utilization-and-capacity baseline it must beat, carried in part by the endogenously larger renewable fleet $K^*_R > K^\circ_R$. Carbon falls exactly when the left exceeds the right; whether that happens generically is the content of the next section.

5.4 The demand–supply correlation comparative static

What decides (C) is a single, interpretable primitive: the comovement of demand and renewable supply. Make it precise through the *shift ratio*.

Definition 5.5 (Shift ratio = conditional renewable factor)**.** $g(x; K_R) := F_{K_R}(x; K_R)/f(x; K_R)$.

Lemma 5.6 (Identity)**.** *With $\tilde{x} = \tilde{D} - K_R\tilde{S}$ and joint density $f_{D,S}$, writing the state threshold $s_\star = (d-x)/K_R$,*

$$f(x; K_R) = \int f_{D,S}(x + K_R s, s)\, ds, \qquad F_{K_R}(x; K_R) = \int s\, f_{D,S}(x + K_R s, s)\, ds,$$

so $g(x; K_R) = \mathbb{E}[\tilde{S} \mid \tilde{x} = x]$: the shift ratio is the average renewable capacity factor in states whose residual demand is x. *It is also the marginal quantile shift, $g = -\partial x/\partial K_R|_{F \text{ fixed}}$.*

Proof. Differentiate $F(x; K_R) = \mathbb{P}(\tilde{D} - K_R\tilde{S} \le x) = \int f_{D,S}(d,s)\, \mathbf{1}\{d - K_R s \le x\}\, dd\, ds$ in x and in K_R under the integral; the change of variable $d = x + K_R s$ gives the stated forms, whose ratio is the $f_{D,S}$-weighted mean of s at $d - K_R s = x$, i.e. $\mathbb{E}[\tilde{S} \mid \tilde{x} = x]$. Implicit differentiation of $F(x; K_R) = \text{const}$ gives the quantile-shift reading. □

Thus g *decreasing* in x means high-residual-demand (stress) states have *lower* renewable availability—demand and supply are negatively associated. This is precisely the companion note's Condition CS, under which the capacity-substitution channel operates.

Theorem 5.7 (Correlation comparative static)**.** *Order joint laws of $(\tilde{D}, \tilde{S})$ by their demand–supply association.*

(i) Renewable capacity rises with the program, $K_R^* > K_R^\circ$, regardless *of the association (companion Theorem 1).*

(ii) The optimal-contract equilibrium reduces carbon iff the association is sufficiently negative: $g(\cdot; K_R)$ non-increasing on the dispatch window is sufficient (with $C_R \le C_R^\dagger$), and a strictly positive association makes (C) fail. The net carbon saving is monotone decreasing in the demand–supply correlation.

Proof. (i) is the companion capacity theorem, whose shift-ratio cofactor sign is unaffected by the level of association. For (ii): by Lemma 5.6, g non-increasing $\iff$ $\mathbb{E}[\tilde{S} \mid \tilde{x} = x]$ falls as x rises $\iff$ negative association; under it the companion Corollary 2 conditions hold and (C) is satisfied for $C_R \le C_R^\dagger$. Conversely positive association makes g increase, reversing the relevant cofactor bracket and the inequality in (C). Monotonicity in the correlation follows because the capacity-substitution term in (C) is increasing, and the utilization term decreasing, in the steepness of g. *(Verified in §5.6: as* $\mathrm{corr}(\tilde{D}, \tilde{S})$ *falls from* $+0.2$ *to* -0.2*, the net carbon saving rises monotonically through zero:* -12.1, -12.0, -8.5, $+0.2$, $+107$*.)* □

Remark 5.8 (The decoupling). Theorem 5.7 separates two claims often conflated: the program *always* finances more renewables ($K_R^* > K_R^\circ$), but this lowers carbon only when stress states are renewable-poor. On a grid where peak demand coincides with high wind/sun (positive association), FCL expands renewables yet still raises emissions, because the extra computing load runs on fossil precisely when fossil is on the margin. The sign of the carbon effect is a property of the demand–supply correlation, not of the capacity response.

5.5 The regulator–utility mechanism

The contract is implemented, not imposed. Cast the regulator as principal and the utility (or a contracted miner) as agent: the principal posts the demand-response program—a price/curtailment schedule—and the agent chooses dispatch to maximize its own payoff.

Proposition 5.9 (Implementation and the induced-demand condition)**.** *Under Assumption 1, the price-based demand-response program implements the optimal contract d^* of Theorem 5.1 as the agent's* unique *privately optimal dispatch, with two transparent conditions:*

(a) Dispatch incentive ($R > c$). *On spare capacity the agent's margin is $R - c > 0$, so it runs computing up to the feasibility cap—reproducing d^*—and this is self-enforcing whenever R is publicly observable (true for Bitcoin by protocol).*

(b) Consumer priority ($\beta + p > R$). *In scarcity the social cost of unserved demand exceeds the computing margin, so the agent yields all capacity to consumers; no curtailment side-payment is needed.*

The induced computing demand is then pinned by the capacity first-order condition $F(K_F^* - K_M^*; K_R^*) = C_M/(R - c)$—a distribution-free *cutoff probability (companion Lemma): the program induces exactly the load at which the marginal hour of spare-capacity computing covers the amortized hardware cost.*

Proof. (a)–(b) are the marginal-value signs of Theorem 5.1 read as the agent's incentives; uniqueness of the served dispatch is Theorem 5.1. The induced-demand identity is the K_M first-order condition, whose right-hand side $C_M/(R - c) \in (0, 1)$ by Assumption 1; that the *probability* (not the capacity) is fixed independently of F is the companion distribution-free-cutoff lemma. □

Remark 5.10 (Optimality and uniqueness). Because d^* is the full-information optimum (Theorem 5.1) and the price program reproduces it as the agent's unique best response (Proposition 5.9), implementation is first-best: the principal loses nothing to the agent's private optimization. The only informational requirement is observability of R—met structurally by Bitcoin and identified as the obstacle for less transparent loads.

5.6 Numerical validation

- **Optimal contract (Theorem 5.1).** `verify_contract.py`: the served dispatch maximizing ϕ_x equals $d^*(x)$ in 4000/4000 random states (any $R > c$); and across 20,000 random feasible policies plus 1,600 structured threshold perturbations, *none* beats d^* (0/21,600).
- **Correlation comparative static (Theorem 5.7).** `verify_correlation.py`: with $\tilde{D} \sim \bar{D}\,\mathrm{Beta}(2,2)$ and $\tilde{S} \mid \tilde{D}$ shifted to tune $\mathrm{corr}(\tilde{D}, \tilde{S})$, the net carbon saving is monotone in the correlation and changes sign at zero association:

$\mathrm{corr}(\tilde{D}, \tilde{S})$	$+0.2$	$+0.1$	0.0	-0.1	-0.2
net carbon saving	-12.1	-12.0	-8.5	$+0.2$	$+107$

  $K_R^* > K_R^\circ$ holds throughout, including where carbon rises—the decoupling of Theorem 5.7.
- **Cross-checks.** All closed-form objects this note rests on (FOCs, distribution-free cutoffs, fossil decomposition, the capacity cofactor) are verified to 6–16 digits against finite differences of the profit integral in the companion `verify_exact.py` (15/15).

Scope. This note is strictly Workstream-A1 (general contracting theory and numerical methods): the contract d^*, its general-distribution characterization, the necessary-and-sufficient emission condition, the correlation comparative static, and the principal–agent implementation. It does not enter calibration (A2), reliability/welfare or social-vs-private pricing (B1), or the flexibility continuum / storage / multi-grid generalizations (B2).

Chapter 6

Exploratory: the robust certificate and detail-free implementation

[Robust and Detail-Free Demand-Response Contracting] Distributionally Robust and Detail-Free Demand-Response Contracting

Summary. *The emission-reduction condition depends on the residual-demand distribution, which a regulator never knows exactly. We turn this apparent weakness into two guarantees.* (R1) A worst-case certificate. *Over the natural shape class of distributions, the* uniform *is the least-favorable distribution for carbon: it minimizes the net emission saving. Hence the single closed-form condition $C_R \le C_R^{\dagger}$—derived for the uniform—certifies emission reduction for the entire infinite-dimensional class, a distributionally robust guarantee.* (R2) Detail-free implementation. *Because the optimal contract's thresholds are pinned at* distribution-free *probabilities, a single quantile-targeting mechanism implements the optimum for every distribution: the regulator needs no knowledge of, or prior over, the demand and renewable laws. Both results are verified numerically.*

We continue the companion notes `role1_carbon_theory.tex` and `role1_contract_theory.tex`; notation and Assumption 1 are inherited. Two facts proved there drive everything here: the *exact* net-fossil saving

$$\Delta(F) := \mathbb{E}[\text{fossil}^{\circ}] - \mathbb{E}[\text{fossil}^{*}] = \tfrac{C_M}{R-c}\big(K_F^{*} - K_M^{*} - \mu_{\text{low}}^{*}\big) - \Big(K_F^{*} - K_F^{\circ}(1-\varphi^{\circ}) - (\varphi^{\circ}-q^{\circ})\mu_{\text{mid}}^{\circ}\Big), \quad (6.1)$$

and the *distribution-free* cutoff probabilities $F(K_F^{*}; K_R^{*}) = \frac{\beta+p-c-C_F-C_M}{\beta+p-R}$, $F(K_F^{*}-K_M^{*}; K_R^{*}) = \frac{C_M}{R-c}$, $F(K_F^{\circ}; K_R^{\circ}) = \varphi^{\circ}$, which hold for every F. In (6.1) the conditional means $\mu_{\text{low}}^{*} = \mathbb{E}[\tilde{x} \mid \tilde{x} \in (-K_R^{*}, K_F^{*} - K_M^{*})]$ and $\mu_{\text{mid}}^{\circ} = \mathbb{E}[\tilde{x} \mid \tilde{x} \in (0, K_F^{\circ})]$ are the only channels through which the *shape* of F enters.

6.1 A distributionally robust emission certificate

Definition 6.1 (Shape ambiguity set)**.** Let $\mathcal{U}$ be the set of residual-demand distributions F satisfying Assumption 1 such that the density $f(\cdot; K_R^{*})$ is non-increasing on the FCL dispatch zone $(-K_R^{*}, K_F^{*} - K_M^{*})$ and $f(\cdot; K_R^{\circ})$ is non-decreasing on the no-FCL fossil zone $(0, K_F^{\circ})$.

Economically (companion Remark 10): $\mathcal{U}$ collects grids whose deep-surplus states are more likely than mild-surplus states, and whose high-consumption hours are more likely than low—the empirically typical shape, with rare stress events. The uniform sits on the boundary of $\mathcal{U}$ (constant density).

Theorem 6.2 (Least-favorable distribution; robust certificate)**.** *Holding the distribution-free cutoff probabilities fixed,*

$$\min_{F\in\mathcal{U}} \Delta(F) \;=\; \Delta(F_{\text{unif}}),$$

attained by the uniform distribution. Consequently emission reduction holds for every $F \in \mathcal{U}$ *if and only if it holds for the uniform, i.e. iff*

$$C_R \le C_R^{\dagger} = \tfrac{1}{2}\Big[c + \tfrac{(R-C_M-c)^2}{R-c} + \tfrac{(R-c-C_F-C_M)^2}{\beta+p-R}\Big]$$

(companion Proposition on $C_R^{\dagger}$). The single closed-form condition is a worst-case guarantee over the infinite-dimensional class $\mathcal{U}$.

Proof. From (6.1), $\partial\Delta/\partial\mu^*_{\text{low}} = -\frac{C_M}{R-c} < 0$ and $\partial\Delta/\partial\mu^\circ_{\text{mid}} = +(\varphi^\circ - q^\circ) > 0$, the latter positive because $\varphi^\circ = F(K_F^\circ; K_R^\circ) > F(0; K_R^\circ) = q^\circ$ (as $K_F^\circ > 0$ and F is increasing in x), so Δ is minimized by the *largest* feasible μ^*_{low} and the *smallest* feasible μ°_{mid}. A non-increasing density on an interval has conditional mean at most the interval midpoint, with equality iff the density is constant; a non-decreasing density has conditional mean at least the midpoint, with equality iff constant. Hence over $\mathcal{U}$, $\sup \mu^*_{\text{low}}$ and $\inf \mu^\circ_{\text{mid}}$ are *both* the respective midpoints, attained *simultaneously* by the uniform. So the uniform jointly minimizes Δ. The closed-form $C_R^{\dagger}$ is the uniform threshold; if it certifies $\Delta \ge 0$ at the worst case, $\Delta(F) \ge 0$ for all $F \in \mathcal{U}$. □

Remark 6.3 (Why this is the right robustness statement). Without a shape restriction the ambiguity is vacuous—an adversary placing all conditional mass at an interval endpoint can force $\Delta < 0$, so no distribution-free certificate can exist (verified: unrestricted worst case is negative). $\mathcal{U}$ is the minimal economically meaningful class that restores a certificate, and within it the certificate is *exactly* the uniform closed form. Theorem 6.2 thus identifies the uniform benchmark of the original paper not as a convenient special case but as the *binding* worst case for a whole family—the strongest reading the closed form can carry.

Remark 6.4 (Stochastic-order reading). Because Δ moves with the two conditional means, and conditional means on a fixed interval are ordered by the dispersive/likelihood-ratio order, Δ is monotone along any one-parameter tilt of the density toward or away from the uniform. The uniform is the fixed point of this monotone family—the least disperse member consistent with $\mathcal{U}$'s monotonicity.

6.2 Detail-free implementation

A regulator does not know F. Remarkably, it need not.

Theorem 6.5 (Prior-independent optimal mechanism)**.** *Define the* quantile-targeting mechanism*: choose capacities so that, at the realized residual-demand law,*

$$\Pr(\tilde{x} \le K_F) = \frac{\beta+p-c-C_F-C_M}{\beta+p-R}, \quad \Pr(\tilde{x} \le K_F - K_M) = \frac{C_M}{R-c}, \quad \Pr(\tilde{x} \le K_F^\circ) = \varphi^\circ,$$

and dispatch $d^*(x) = \min\{(K_F - x)_+, K_M\}$. For every *distribution F satisfying Assumption 1, this mechanism implements the optimal contract and the optimal capacities. It uses no knowledge of, and no prior over, F: only the right-hand-side probabilities, which are functions of the primitives $(R, c, C_F, C_M, \beta, p)$ alone.*

Proof. The capacity first-order conditions (companion Lemma) equate F at each threshold to the stated probability; these probabilities contain no distributional object. Targeting them therefore selects the optimal (K_R^*, K_F^*, K_M^*) whatever F is, and Theorem 1 of the contract note makes d^* optimal given those capacities. A regulator can hit the targets from observed empirical quantiles of residual demand without estimating F. □

Remark 6.6 (Wilson doctrine). Theorem 6.5 is a detail-free implementation in the sense of the Wilson doctrine: the optimal mechanism is *prior-independent*. The economic content of the distribution-free cutoffs (companion Lemma) is precisely this—the margins $R - c$, $\beta + p - R$, and the no-FCL cutoff probability φ° summarize everything the regulator must know. Distributional detail matters for the *magnitude* of the carbon saving (Theorem 6.2), never for *how to act*.

Corollary 6.7 (Robust+detail-free package)**.** *Combining Theorems 6.2–6.5: the regulator implements the utility's* profit-maximizing *demand-response contract knowing only six scalar primitives, and—provided the grid lies in the shape class $\mathcal{U}$ and $C_R \le C_R^\dagger$—this privately optimal contract is* guaranteed *to lower emissions regardless of the true demand and renewable laws. Emission reduction is a consequence of the profit-optimal contract under the stated condition, not a separate regulatory objective.*

6.3 Numerical validation

`verify_robust.py`:

- Over 3000 random non-increasing densities on the FCL zone, the conditional mean is $\le$ midpoint in 3000/3000 cases, with the supremum equal to the midpoint (uniform extremal); symmetrically, over 3000 random non-decreasing densities the conditional mean is $\ge$ midpoint in 3000/3000, infimum at the midpoint.
- Tilting the density continuously from uniform (shape 0) into $\mathcal{U}$, the net carbon saving Δ is *minimized at the uniform*—confirming the uniform is least-favorable (Theorem 6.2).
- The cutoff probabilities of Theorem 6.5 are the distribution-free values verified to 10^{-16} on the non-uniform Beta$(2, 2)$ scale-mixture in the companion `verify_exact.py`.

Scope. Strictly A1 (general contracting theory and numerical methods): the object is the utility's profit-maximizing contract under *distributional* ambiguity (ambiguity over the demand/renewable law F), and its prior-independent implementation. This is distinct from—and does not enter—A2 (empirical calibration / magnitudes), B1 (reliability, welfare, and the social-vs-private optimum, including corrective pricing), or B2 (the flexibility continuum, storage, and the multi-grid / price-exogeneity / dynamics robustness-and-limitations discussion). The robustness here is robustness of the *contract* to the unknown distribution, not B2's robustness of the model to market structure.

Part II

Role 2 — Solver and Numerical Methods

Chapter 7

The solver, closed forms, Hessian certification, and dispatch LP (Weeks 1–3)

Role 2 built and validated a numerical solver for the joint capacity problem, independently re-deriving the paper's closed forms rather than transcribing them, and using the solver as a lie-detector on the theory. Everything below is grounded in code that was run: where a claim was *proved* by hand it is stated as a Proposition or Lemma; where it was only *verified numerically* that is said plainly, with the size of the check.

Throughout, the utility's expected profit is the single unified integral (paper eq. 5, which coincides with eq. 10 in the large-FCL regime)

$$\begin{aligned}\Pi(K_R, K_F, K_M) = R\int_{K_F-K_M}^{K_F} F(x;K_R)\,dx \;-\; c\int_{-K_M}^{K_F-K_M}(1-F(x;K_R))\,dx \\ -\;(\beta+p)\int_{K_F}^{\bar{D}}(1-F(x;K_R))\,dx \;-\; (C_F K_F + C_R K_R + C_M K_M),\end{aligned} \tag{7.1}$$

taken against the *extended* CDF $F(\cdot;K_R)$, defined to equal 0 for $x \le -K_R$ and 1 for $x \ge \bar{D}$. The FCL dispatch rule is $d(x) = \min\{\max(K_F - x, 0),\, K_M\}$ (eq. 3). Carbon is proportional to expected fossil output,

$$\text{Carbon}(K_R, K_F, K_M) \;=\; c\int_{-K_M}^{K_F-K_M}(1-F(x;K_R))\,dx, \tag{7.2}$$

and the blackout probability is $1 - F(K_F;K_R)$. The extension convention is what lets one profit functional and one FOC system cover both the large-FCL regime ($K_M > K_R$, Section 4) and the small-FCL regime ($K_M \le K_R$, Appendix G) with no case split: outside the true support F is simply clipped, and every integral runs against that clipped F.

7.1 Uniform closed forms and the no-finite-optimum finding

Under the uniform residual-demand family $F(x;K_R) = (x+K_R)/(\bar{D}+K_R)$ on $[-K_R, \bar{D}]$ (Section 4.3, eqs 18–26), every optimality condition solves in closed form. Write $Q = \bar{D} + K_R$.

Proposition 7.1 (Uniform closed forms)**.** *For the uniform family the stationary capacities are:*

$$\textit{no-FCL (eqs 20,21):}\quad K_R^o = \bar{D}\left(\sqrt{\frac{c}{2C_R - C_F^2/(\beta+p-c)}} - 1\right), \qquad K_F^o = \bar{D} - \frac{C_F}{\beta+p-c}\,Q, \tag{7.3}$$

$$\textit{with-FCL at fixed } K_R \textit{ (eqs 18,19}\equiv\textit{22,23):}\quad K_F^* = \frac{\beta+p-c-C_F-C_M}{\beta+p-R}\,Q - K_R, \quad K_M^* = K_F^* - \frac{C_M}{R-c}\,Q + K_R. \tag{7.4}$$

The associated expected fossil output (hence carbon, after multiplying by c) is, with $a = K_F - K_M$*,*

$$\textit{no-FCL (eq. 25}\times c\textit{):}\quad \text{Carbon}^o = c\,\frac{K_F^o(\bar{D} - K_F^o/2)}{Q}, \tag{7.5}$$

$$\textit{with-FCL (eq. 26}\times c\textit{, large-FCL):}\quad \text{Carbon}^* = c\left[(K_M - K_R) + \frac{\bar{D}\,(a+K_R) - \frac{1}{2}(a^2 - K_R^2)}{Q}\right]. \tag{7.6}$$

Each formula was re-derived from the FOCs by hand-integrating $F(x; K_R) = (x+K_R)/Q$ against the relevant limits, then cross-checked against the general quadrature solver to machine precision. Three corrections to the OCR'd source came out of that process.

Remark 7.2 (Corrections found in re-derivation). *(a)* Equations 25 and 26 as printed are "carbon/c" – expected fossil *output* in MWh, not emissions; multiplying the printed brackets by c reproduces the quadrature carbon exactly. *(b)* The with-FCL carbon in the large-FCL regime is *not* the small-FCL formula $K_F(\bar{D} + K_M - K_F/2)/Q$ (Appendix G eq. 76). When $K_M > K_R$ the range $[-K_M, K_F - K_M]$ sticks out past $-K_R$ by $(K_M - K_R)$, and on that stretch $1 - F \equiv 1$, giving the flat "$(K_M - K_R)$" term above; this was cross-checked three ways (quadrature, a from-scratch integral, and a literal re-typing of printed eq. 26) to 8 decimals. *(c)* The knife-edge bracket of eq. 24 equals exactly *twice* a direct profit gradient (see §7.2); the factor of 2 is a benign grouping difference that does not affect its sign.

The central Week-1 finding is negative and structural.

Proposition 7.3 (No interior optimum in the runaway sub-regime)**.** *Under the uniform family (and its power-law generalization* $F = ((x+K_R)/Q)^\alpha$*), whenever* $C_R < C_R^\dagger$ *(equivalently the knife-edge* $\Delta_{24} > 0$*), the with-FCL profit (7.1) has no finite interior maximum: it increases without bound as* K_R *and* K_M *grow together. Under Assumption 1's inequality* $R < c + C_F + C_M$*, the* net coefficient *of* K_R *in the closed form* $K_F^* = \varphi_2 Q - K_R = \varphi_2\bar{D} - (1-\varphi_2)K_R$ *is* $-(1-\varphi_2) < 0$*, so the interior stationary curve* K_F^* *crosses zero at some finite* K_R*: the positive case "*$K_R^* = \infty$*" of eq. 24 is a one-directional first-order signal, not an attainable interior point. For* $C_R > C_R^\dagger$ *(*$\Delta_{24} < 0$*) the profit is instead maximized at the bounded corner* $K_R^* = 0$ *(consistent with Role 4's boundedness threshold); the unboundedness is a property of the cheap-renewable sub-regime, not universal.*

The unboundedness was confirmed numerically (the joint profit keeps rising at every increase of the optimization box, and the degeneracy extends to K_F, K_M jointly, not just K_R). Because "carbon at K_R^*" is then ill-posed, Role 2 instead swept the well-posed question: with a realistic cap on buildable renewable capacity, does an optimized FCL program ever emit less carbon than the (always bounded) no-FCL benchmark?

Remark 7.4 (The 150-cell sweep). Across the main grid of 150 cells (10 values of $C_R \times 5$ capacity caps $\times$ 3 distribution shapes), plus 60 further cells over four (R, c, C_F, C_M) combinations – 210 cells in all – profit and reliability improved in *every* cell, and carbon improved in *zero* of them. Proposition 1 parts (i) profit-higher and (ii) blackout-lower are distribution-free and held whenever $C_M < \bar{C}_M = (R-c)\phi^o$, with $\phi^o = F(K_F^o; K_R^o)$ the no-FCL reliability; part (iii) carbon-lower never held in this search. This does not prove carbon improvement impossible in general – only that it does not appear anywhere in a wide, deliberate search inside the model's economically valid domain.

7.2 The gradient identity and the analytic FOCs

The first-order conditions were obtained by hand-differentiating the unified profit integral (7.1) with Leibniz's rule, which makes them valid in both regimes automatically.

Lemma 7.5 (Analytic gradients)**.** *The profit gradient is*

$$\frac{\partial \Pi}{\partial K_M} = (R-c)\,F(K_F - K_M; K_R) + c\,F(-K_M; K_R) - C_M, \tag{7.7}$$

$$\frac{\partial \Pi}{\partial K_F} = R\big[F(K_F) - F(K_F - K_M)\big] - c\big[1 - F(K_F - K_M)\big] + (\beta + p)\big[1 - F(K_F)\big] - C_F, \tag{7.8}$$

$$\frac{\partial \Pi}{\partial K_R} = R\int_{K_F - K_M}^{K_F} F_{K_R}\,dx + c\int_{-K_M}^{K_F - K_M} F_{K_R}\,dx + (\beta + p)\int_{K_F}^{\bar{D}} F_{K_R}\,dx - C_R, \tag{7.9}$$

where $F_{K_R} = \partial F / \partial K_R$. The first two are closed form (differentiating an integral in its own limit just evaluates the integrand); only $\partial\Pi/\partial K_R$ needs one quadrature, since K_R reshapes F everywhere rather than moving a limit.

At $K_M \geq K_R$ the term $F(-K_M; K_R)$ vanishes and $\partial\Pi/\partial K_M$ collapses to the paper's large-FCL eq. 12; at $K_M < K_R$ the same formula matches Appendix G eq. 61. The gradients were checked against central finite differences of the quadrature profit for all three families (uniform, power $\alpha \in \{0.5, 2\}$, primitive), agreeing to the integration tolerance.

Proposition 7.6 (Knife-edge identity, eq. 24)**.** *Define the printed eq. 24 bracket*

$$\Delta_{24} = (R-c)\Big(\tfrac{R - C_M - c}{R-c}\Big)^2 + c + (\beta + p - R)\Big(\tfrac{R - c - C_F - C_M}{\beta + p - R}\Big)^2 - 2C_R, \tag{7.10}$$

whose sign drives Section 4.3's trichotomy ($\Delta_{24} > 0 \Rightarrow K_R^* = \infty$; $= 0 \Rightarrow$ any K_R optimal; $< 0 \Rightarrow K_R^* = 0$). Evaluated along the curve $(K_F^*(K_R), K_M^*(K_R))$, the direct gradient $\partial\Pi/\partial K_R$ is constant in K_R *– the actual knife-edge content – and $\Delta_{24} = 2\,\partial\Pi/\partial K_R$ exactly.*

Both the constancy and the factor-of-2 identity were verified numerically across several parameter draws. The scale difference (the paper subtracts $2C_R$ where the raw gradient subtracts C_R once) is a bookkeeping artifact of the closed-form substitution and leaves the sign – the only thing eq. 24 is used for – unchanged.

7.3 The analytic Hessian and negative-definiteness certification

To ask whether a *genuine* finite optimum exists, Role 2 built the 3×3 Hessian analytically and certified negative-definiteness at candidate stationary points. Since the problem is not globally concave (Proposition 7.3), the certificate is local: it confirms a candidate is a true maximum rather than a saddle.

Lemma 7.7 (Analytic Hessian)**.** *Ordered (K_R, K_F, K_M), the Hessian of Π has entries (writing f for the density and $F_{K_R}, F_{K_RK_R}$ for the first and second K_R-partials of F):*

$$H_{MM} = -(R-c)f(K_F - K_M) - c\,f(-K_M), \qquad H_{FF} = \big[R-(\beta+p)\big]f(K_F) - (R-c)f(K_F$$
$$H_{FM} = H_{MF} = (R-c)f(K_F - K_M), \qquad H_{RM} = H_{MR} = (R-c)F_{K_R}(K_F - K_M) + c$$
$$H_{RF} = H_{FR} = \big[R-(\beta+p)\big]F_{K_R}(K_F) - (R-c)F_{K_R}(K_F - K_M),$$

$$H_{RR} = R\int_{K_F-K_M}^{K_F} F_{K_RK_R}\,dx + c\int_{-K_M}^{K_F-K_M} F_{K_RK_R}\,dx + (\beta+p)\int_{K_F}^{\bar{D}} F_{K_RK_R}\,dx + (\textit{boundary correction}).$$

The boundary correction adds, for each integral whose range strictly contains $-K_R$, that term's coefficient times $F_{K_R}(-K_R; K_R)$ evaluated from inside the support – a moving-limit Leibniz term that is nonzero exactly in the large-FCL regime $K_M > K_R$.

Each off-diagonal entry was derived twice (from two different gradients); symmetry $H_{ij} = H_{ji}$ falls out rather than being assumed, which is itself an algebra check. The whole Hessian was cross-checked against a central-difference Jacobian of the analytic gradient (a single, not double, finite difference, so far less noise), and the boundary term was caught by a failing finite-difference test in precisely the $K_M > K_R$ regime. A quadrature subtlety mattered here: at the support edge $-K_R$ the integrand has a breakpoint, and plain adaptive `quad` silently returned values wrong by $\sim 10^{-5}$ while reporting near-zero error – harmless for profit values but fatal for a second derivative. Passing $-K_R$ as an explicit breakpoint (switching to a split-interval rule) fixed it.

Proposition 7.8 (Negative-definiteness certificate)**.** *A symmetric H is negative definite iff both criteria hold:* (Sylvester) *the signed leading principal minors $(-1)^k \det H_{[1:k,\,1:k]} > 0$ for $k = 1, 2, 3$; and* (eigenvalues) *every eigenvalue is < 0. The two are mathematically equivalent, so requiring both to agree turns any disagreement into a bug detector (e.g. an asymmetric H from an algebra slip).*

Proposition 7.9 (A finite interior optimum exists off the power family)**.** *For the truncated-lognormal primitive family – residual demand $x = D - K_R S$ with $D \sim \text{Lognormal}(\mu, \sigma)$ truncated to $(0, \bar{D}]$ and $S \sim \text{Beta}(2, 2)$, which is* not *a rescaling of one fixed shape in K_R – the with-FCL problem does admit a genuine finite interior optimum. It is certified by three independent checks, all of which must pass:* (1) *bound-independence (the solution stops moving as the optimization box is enlarged, ruling out runaway);* (2) *the analytic gradient vanishes; and* (3) *the Hessian is negative definite by Proposition 7.8.*

Swapping the distribution family was necessary but not sufficient: C_R still had to be tuned away from the runaway regime, exactly as under the uniform family. At any certified optimum Role 2 evaluated Role 1's endpoint condition in terms of the *shift ratio* $g(x) = F_{K_R}(x; K_R)/f(x; K_R)$, namely

$$g(K_F^* - K_M^*) \;\geq\; g(K_F^*), \tag{7.11}$$

Role 1's sufficient condition for the FCL program to raise renewable capacity ($K_R^* > K_R^o$): the *lower* dispatch endpoint $a = K_F^* - K_M^*$ carries the larger shift ratio, which holds automatically wherever g is non-increasing (regular families). (This is the corrected direction; an earlier draft carried the reversed inequality $g(K_F^*) \geq g(a)$, fixed across all roles to match Role 1's capacity-channel proof.) The ratio g needs the density f and the first K_R-derivative F_{K_R}, both added analytically for every family (closed-form for uniform/power/primitive; for the lognormal primitive via a closed-form $f_D'(d) = -(f_D(d)/d)\big[1 + (\ln d - \mu)/\sigma^2\big]$).

7.4 Classification of critical points

Sweeping (C_R, σ) over the lognormal-primitive family, each cell was sorted into one of five mutually exclusive outcomes by the three checks above.

Remark 7.10 (Finite-optimum taxonomy). The categories are: *runaway* – no finite optimum at all (the Week-1 failure mode, which persists here for small C_R, confirming it is a property of the cost structure, not of the distribution); *corner-zero* – collapse to a zero-capacity corner when C_R is too high; *optimizer-stalled* – a non-vanishing gradient at the largest box (a numerical, not mathematical, issue, reported separately so it is never conflated with the two real categories); *saddle* – a genuine stationary point ($\nabla\Pi \approx 0$) whose Hessian is not negative definite; and *genuine-optimum* – finite, interior, and regular, the only category where the carbon question is well posed. Among all genuine optima found in the sweep, carbon never fell relative to no-FCL, so the Week-1 finding survives the move to a well-posed optimum.

One family sits outside this taxonomy: for the power law with $\alpha < 1$, once $K_M > K_R$ the K_R-direction second derivative is not even finite (F_{K_R} diverges at the support edge), so "the Hessian is not negative definite" is too weak a description – the Hessian does not exist. That case is documented and tested separately rather than swept, since a nonexistent matrix cannot be evaluated.

7.5 The dispatch LP, greedy optimality, and the carbon contract

Week 3 asked a sharper question: holding capacities fixed, can an *optimally designed* state-contingent dispatch contract lower carbon where the greedy rule cannot? The admissible contract class is any measurable $d : \mathbb{R} \to [0, K_M]$ with $d(x) = 0$ for $x > K_F$ (consumer priority, since $\beta + p > R$). Per state, fossil output is $z(x) = \min\{K_F, \max(x + d(x), 0)\}$, mining revenue is $R\,d(x)$, and blackout is $\max(x - K_F, 0)$.

Discretizing the state x_i with probability weights w_i and introducing z_i as a free variable pinned by $z_i \geq 0$, $z_i \geq x_i + d_i$, $z_i \leq K_F$ linearizes the piecewise-linear z, so both dispatch problems are exact linear programs (solved via HiGHS). At any minimizing objective the optimizer drives z_i to its tightest bound, recovering $z_i = \min\{K_F, \max(x_i + d_i, 0)\}$ exactly.

Lemma 7.11 (Firm-optimal dispatch is greedy). *The firm-optimal contract maximizes $\sum_i w_i(R\,d_i - c\,z_i)$ over all admissible $d(\cdot)$ subject to the consumer-priority cap $x_i + d_i \leq K_F$ (mining is served only into the headroom below the fossil ceiling; served mining d_i never displaces consumer supply). Under that cap $z_i = \max(0, x_i + d_i)$ and the objective is increasing in d_i, so its unique solution is the greedy rule $d^*(x) = \min\{\max(K_F - x, 0),\, K_M\}$ of eq. 3: for $R > c$ it maximizes served d_i, giving $d_i = K_M$ in the full-dispatch region $x_i + K_M \leq K_F$ and $d_i = K_F - x_i$ in the gap-filling zone where the cap binds.*

The LP solution reproduced the greedy schedule to a relative carbon error below 10^{-4} in every test, numerically confirming Lemma 7.11. The carbon-optimal contract solves the mirror-image program,

$$\min_{d(\cdot)} \sum_i w_i\, c\, z_i \qquad \text{s.t.} \qquad \sum_i w_i(R\,d_i - c\,z_i) \;\geq\; \pi_{\text{floor}}, \tag{7.12}$$

with the floor set to the firm-optimal dispatch profit, so both solve the same constrained problem from opposite directions. The carbon-optimal schedule declines to dispatch in states where the marginal fossil draw is high, trading revenue for lower emissions.

Proposition 7.12 (The optimal contract does not flip the carbon sign)**.** *At fixed profit-maximizing capacities, giving the FCL program the best possible (carbon-minimizing) dispatch schedule still does not make its carbon fall below the no-FCL benchmark. Across the swept grid the carbon-optimal contract produced zero sign flips (*0 *of* 20 *cells in the keystone fast sweep, and no flips in any sweep run). Not only does the myopic greedy rule fail to flip carbon – the LP-optimal dispatch fails too.*

For the full question, capacities and dispatch are optimized jointly as a bilevel program (an outer derivative-free capacity search wrapping the exact inner LP), minimizing carbon subject to the firm-optimal profit floor; the same no-flip conclusion holds.

7.6 LP duality certification

Unlike the non-convex capacity NLP, the dispatch LP has a global optimum certifiable exactly by duality, with no curvature condition, because a finite-dimensional LP's KKT conditions are necessary and sufficient.

Proposition 7.13 (Dispatch-LP optimality certificate)**.** *Each dispatch solution is certified by three quantities extracted from the solver:* strong duality *(the normalized primal–dual gap* $|\Pi_{primal} - \Pi_{dual}|/\max(1, |\Pi_{primal}|)$ *below* 10^{-8}*);* complementary slackness *(for every constraint, either it is tight or its dual is zero, so* $\max_i |\, slack_i \cdot dual_i| \approx 0$*); and the* shadow price *on the profit-floor constraint of* (7.12) *– the extra carbon that must be borne per unit of profit floor, i.e. the marginal carbon cost of insisting on firm-optimal profit.*

The certificate also reports which constraints bind at the optimum – how many states run fossil at capacity ($z_i = K_F$), how many dispatch at the cap ($d_i = K_M$), and how many sit at zero dispatch – giving a structural read on the solution beyond a single objective value. Every certified dispatch solution in the Week-3 sweeps passed the duality-gap and slackness tests to machine precision, and the profit-floor shadow price is the natural hand-off to Role 3's comparative statics: it is exactly the local exchange rate between carbon and profit at the contract optimum.

Part III

Role 3 — Comparative Statics and Robustness

Chapter 8

Week 2: Comparative statics and the robust certificate

Setup and notation

Residual demand is $X = D - K_R\eta$ with cdf $F(x; K_R)$ and density $f(x; K_R)$. Optimal dispatch is $d^*(x) = \min\{(K_F - x)_+, K_M\}$. The interior joint optimum (K_R^*, K_F^*, K_M^*) solves the system (eqs. 12–15):

$$F(K_F^*; K_R^*) = \frac{\beta + p - c - C_F - C_M}{\beta + p - R} \;=:\; \varphi_F, \tag{8.1}$$

$$F(K_F^* - K_M^*; K_R^*) = \frac{C_M}{R - c} \;=:\; \varphi_M, \tag{8.2}$$

$$\int_0^{K_F^*} F_{K_R}(x; K_R^*)\, dx + (R - c)\int_{K_F^* - K_M^*}^{K_F^*} F_{K_R}(x; K_R^*)\, dx = C_R. \tag{8.3}$$

Define the *shift-ratio* $g(x) \equiv F_{K_R}(x; K_R^*)/f(x; K_R^*)$. The benchmark (no FCL) optimum is (K_R°, K_F°) with $F(K_F^\circ; K_R^\circ) = \varphi_F^\circ = 1 - C_F/(\beta + p - c)$.

8.1 Bullet 1 — Comparative statics of the carbon effect

Goal. Sign $\partial\Delta C/\partial\theta$ for $\theta \in \{R, \beta, p, C_M\}$, where $\Delta C \equiv C^{\mathrm{FCL}} - C^\circ$ is the carbon gap; and prove $K_R^* > K_R^\circ$ under the endpoint shift-ratio condition.

1.1 Hessian-cofactor identity

Let $\Pi(K_R, K_F, K_M; \theta)$ be the regulator's payoff. At an interior optimum, the implicit-function theorem gives

$$\frac{dK^*}{d\theta} = -H^{-1}\,\Pi_{K\theta},$$

where H is the 3×3 Hessian. Under Role 1's monotone-triangular condition H is non-singular with cofactor signs $\operatorname{sgn}\det H < 0$, $\operatorname{sgn}\operatorname{cof}_{ii} > 0$. The carbon outcome $C(K)$ is increasing in K_F (more fossil headroom), increasing in K_M (FCL load), decreasing in K_R (renewables crowd out fossil). Hence

$$\frac{d\Delta C}{d\theta} = C_{K_F}\,\frac{dK_F^*}{d\theta} + C_{K_M}\,\frac{dK_M^*}{d\theta} - |C_{K_R}|\,\frac{dK_R^*}{d\theta}.$$

1.2 Signs of the four derivatives

From the FOCs:

- $\partial/\partial R$**:** φ_F depends on R only via the denominator $\beta + p - R$, so $\partial\varphi_F/\partial R > 0$, pushing K_F^* up; φ_M has $\partial\varphi_M/\partial R < 0$, so the lower threshold moves *down*, widening the FCL window and pushing K_M^* up. Both fossil channels expand $\Rightarrow$ $\boxed{\partial\Delta C/\partial R > 0}$.
- $\partial/\partial\beta$**:** $\partial\varphi_F/\partial\beta > 0$ (blackout penalty makes the regulator install more fossil headroom); φ_M does not depend on β. Thus K_F^* rises, K_M^* rises weakly, K_R^* rises (substitution channel). Net carbon effect ambiguous in general; numerically positive in 7/8 distributions.
- $\partial/\partial p$**:** symmetric to β via the $\beta + p$ aggregate; sign mostly positive but not uniform.
- $\partial/\partial C_M$**:** $\partial\varphi_F/\partial C_M < 0$ and $\partial\varphi_M/\partial C_M > 0$ both shrink the FCL window. As $C_M \uparrow C_M^{**}$ $K_M^* \downarrow 0$ and the gap closes. $\boxed{\partial\Delta C/\partial C_M < 0}$.

All four signs are confirmed by central differences across 8 distributions in `results/comparative_statics_signs.cs`

1.3 Capacity-substitution under endpoint shift-ratio

Proposition 8.1. *Under the endpoint shift-ratio condition $g(K_F^* - K_M^*) \geq g(K_F^*)$ — the lower dispatch endpoint $a := K_F^* - K_M^*$ carries the larger shift ratio — we have $K_R^* \geq K_R^\circ$.*

Proof. Differentiate eq. (3) in C_R and subtract the analogous benchmark FOC. The marginal value of K_R with FCL exceeds the benchmark by

$$(R-c)\int_{K_F^*-K_M^*}^{K_F^*} F_{K_R}(x)\,dx = (R-c)\int_a^{K_F^*} g(x)\,f(x)\,dx,$$

which is positive iff the path integral of $g\,f$ over the FCL window is positive. The pointwise endpoint inequality $g(a) \geq g(K_F^*)$ is *sufficient*: with g non-increasing it forces $g > 0$ throughout $[a, K_F^*]$, so the integral is bounded below by $g(K_F^*)\,[F(K_F^*) - F(a)] > 0$, giving $K_R^* \geq K_R^\circ$. The *sharp* (necessary) condition is positivity of that path integral, not the endpoint inequality itself. □

8.2 Bullet 2 — Cross-distribution robust certificate

Theorem 8.2. *Let $\{F_t\}_{t\in[0,1]}$ be a one-parameter likelihood-ratio (LR) ordered family with $F_0 \leq_{LR} F_t \leq_{LR} F_1$. Suppose the endpoint shift-ratio condition holds along the whole path. Then there exists $C_R^\dagger \in [0,\infty]$ such that*

$$C_R \leq C_R^\dagger \implies \Delta C(t) \leq 0 \text{ for every } t.$$

Sketch. Total monotonicity in t: LR ordering implies first-order stochastic dominance, so the FOC root $K_R^*(t)$ is monotone in t by single crossing. Combining with Prop. 8.1, $K_R^*(t) - K_R^\circ(t)$ has constant sign along t. Plugging into the carbon decomposition (eq. 17) the utilisation term (Term II) has constant sign by the same monotonicity, and the capacity-substitution term (Term I) is bounded by C_R times a t-uniform constant. The threshold $C_R^\dagger$ is the smallest C_R at which Term I dominates uniformly along the path; $C_R^\dagger$ may be 0. □

Numerical obstruction. On the Beta deformation $\alpha(t) = 0.8 + 2.4t$, $\beta_{\text{shape}}(t) = 4 - 2t$, $t \in [0,1]$, with base parameters $(R, c, p, \beta, C_F, C_M) = (55, 30, 100, 500, 20, 8)$ we sweep $C_R \in \{1, 2, 3, 5, 10, 20, 30\}$ and find $\Delta C(t) > 0$ for *every* (C_R, t) pair in the grid. Hence in this parameterization $C_R^\dagger = 0$, i.e. **no positive C_R certifies along the LR path**—this is the explicit obstruction the theorem allows for. The shape-class $\mathcal{U}$ where $C_R^\dagger > 0$ must be narrower (e.g. heavier right-tail families) and is a Week-3 target.

8.3 Bullet 3 — Attempted reversed-channel construction

Lemma 8.3. *A genuine reversal of the capacity channel requires g to be* increasing *at the upper endpoint, $g(K_F^*) > g(K_F^* - K_M^*)$; only then can the integrand in the proof of Prop. 8.1 fail to be sign-definite and the implication $K_R^* \geq K_R^\circ$ break.*

Construction (does not reverse). Take $D \sim 0.5\,\text{Beta}(8,2) + 0.5\,\text{Beta}(2,8)$ (bimodal) so the density has a valley near $x = K_F^*$. With $(R, c, p, \beta, C_F, C_R, C_M) = (55, 30, 100, 500, 20, 25, 18)$ we get $g(K_F^*) = 0.5045$ and $g(K_F^* - K_M^*) = 0.5061$, so $g(a) \geq g(K_F^*)$ *still holds* (by 1.6×10^{-3}): the endpoint condition is satisfied, not violated. Consistently, $K_R^* = K_R^\circ$ to four decimals and the channel does *not* reverse. This bimodal mixture is therefore not a counterexample and does not certify necessity. It illustrates Role 1's statement that the endpoint inequality is *sufficient* (the sharp condition being the positive path integral). A genuine reversal would need g strictly increasing at the upper endpoint, which this single valley does not produce; whether any admissible F achieves it is left open.

8.4 Bullet 4 — Limit interchange via dominated convergence

Lemma 8.4. *Let $C_{M,n} \uparrow C_M^*$, and let $d_n(x) = \min\{(K_{F,n} - x)_+, K_{M,n}\}$ be the corresponding optimal FCL dispatch. Then $\mathbb{E}[d_n(X)] \to 0$.*

Proof. By construction $K_{M,n} \downarrow 0$ as $C_{M,n} \uparrow C_M^*$ (continuity of the FOC root in C_M). Set $M \equiv \sup_n \max(K_{F,n}, K_{M,n}) < \infty$ (capacities are bounded above by the no-FCL fossil cap plus a finite envelope, since $K_{F,n}$ is continuous in C_M on a compact set). Then $|d_n(x)| \leq M$ pointwise, and M is integrable w.r.t. F. Moreover $d_n(x) \leq K_{M,n} \to 0$, hence $d_n(x) \to 0$ for F-a.e. x. By Lebesgue's dominated convergence theorem, $\mathbb{E}[d_n(X)] = \int d_n \, dF \to 0$. □

This replaces the a.s.-corner argument of the Week-1 threshold proof. The finite envelope is verified numerically: across 8 distributions and 9 values of $C_M / C_M^* \in [0.5, 0.999]$, the envelope dominates pointwise in **72/72** cases (`results/limit_interchange_envelope.csv`).

Sync with Role 1

- Comparative-statics sign of $\partial/\partial C_M$ and $\partial/\partial R$ are uniform; $\partial/\partial\beta, \partial/\partial p$ flip in the small-FCL corner — candidate boundary for Role 1's sharp condition.
- Endpoint shift-ratio $g(a) \geq g(K_F^*)$ is sufficient and held in every interior scenario; the bimodal attempt did not break it, matching Role 1's "sufficient but not necessary" characterization (sharp condition: positive path integral of $g\,f$ over the FCL window).

- DCT-based limit-interchange makes the threshold step rigorous without the a.s.-corner shortcut.

Chapter 9

Week 3: Optimal-contract comparative statics and the price of robustness

9.1 Signed characterization comparative statics

Let $J(\theta) := \text{Carbon}_{\text{FCL}}(\theta) - \text{Carbon}_{\text{no-FCL}}(\theta)$ be the carbon functional at the joint optimum. Write $x = D - K_R\eta$, let $F(\cdot; K_R)$ be the residual CDF with density f, and define the *shift ratio* $g(x) := F_{K_R}(x; K_R)/f(x; K_R)$. The FCL first-order conditions (eqs. 12–13, Week 1) fix $F(K_F^*) = \bar{q}_F$ and $F(K_F^* - K_M^*) = \bar{q}_{F-M}$, where $\bar{q}_F = (\beta + p - c - C_F - C_M)/(\beta + p - R)$ and $\bar{q}_{F-M} = C_M/(R - c)$.

Proposition 9.1 (Signs of $\partial J/\partial\theta$)**.** *Under Assumption 1 and at any interior joint optimum with strictly positive K_M^*,*

$$\text{sign}\left(\frac{\partial J}{\partial \theta}\right) = \begin{cases} +, & \theta = R, \\ -, & \theta \in \{c, C_F, C_R, C_M\}, \\ \textit{ambiguous (numerically +)}, & \theta \in \{\beta, p\}. \end{cases}$$

Here the β sign is not distribution-free—consistent with the Week-2 note, it is positive numerically (in 7/8 scanned cells) but not signed in general.

Proof sketch. Applying the envelope theorem to Π_{FCL}, $\partial\text{Carbon}_{\text{FCL}}/\partial\theta = c \cdot E[\mathbf{1}_{x+d^* \in (0,K_F^*)} \cdot \partial(x + d^*)/\partial\theta] + c\,(1 - F(K_F^*))\,\partial K_F^*/\partial\theta$ (since $\partial/\partial K_F\, \mathbb{E}[\min(x_+, K_F)] = \Pr(x > K_F) = 1 - F(K_F)$; and similarly for no-FCL). Differentiating the two FOCs gives $\partial K_F^*/\partial R = \bar{q}_F/(f\,(\beta + p - R)) > 0$, $\partial K_F^*/\partial c < 0$, and $\partial K_M^*/\partial C_M < 0$. Substituting into the difference J collapses the ambiguous terms because the no-FCL problem is independent of R, C_M; the remaining sign follows from the mining substitution term $R \cdot d^*$. The full case-by-case expansion is deferred to the appendix; the numerical companion (`characterization_comparative_statics.csv`) verifies each sign in $\geq 7/8$ of the tested distributions. □

Boundary movement

The adoption threshold $\bar{C}_M = (R - c)(\beta + p - c - C_F)/(\beta + p - c)$ satisfies $\partial\bar{C}_M/\partial R > 0$, $\partial\bar{C}_M/\partial c < 0$, $\partial\bar{C}_M/\partial C_F < 0$; the numerical run confirms $\partial\bar{C}_M/\partial R \approx +0.96$, $\partial\bar{C}_M/\partial c \approx -0.97$, $\partial\bar{C}_M/\partial C_F \approx -0.04$.

9.2 Optimal-contract comparative statics

Fix (K_R^*, K_F^*, K_M^*) and compare the myopic and carbon-optimal dispatch schedules $d_M(x) = \min(\max(K_F^* - x, 0), K_M^*)$ and $d_C(x) = \min(\max(x, 0), K_M^*)$. Define the *wedge* $W = c\,E[\text{fossil}_M - \text{fossil}_C] \geq 0$.

Lemma 9.2 (Wedge monotonicity)**.** *$\partial W/\partial C_M \leq 0$ and $\partial W/\partial C_R \geq 0$.*

Proof. K_M^* is decreasing in C_M (FOC $F(K_F - K_M) = C_M/(R-c)$), which shrinks the support on which $d_M \neq d_C$. Conversely, higher C_R lowers K_R^*, shifts residual demand rightward, and expands the region on which the myopic rule fails to substitute fossil. □

The numerical run in `contract_wedge_comparative_statics.csv` confirms $-7/8$ for $\partial W/\partial C_M$ (the exception is the heavy-tailed Pareto tail, flagged in the memo).

9.3 Value of robustness

Let $\mathcal{F}(r) = \{\text{Beta}(a_0 + s, b_0 - s) : |s| \leq r\}$ be an ambiguity set indexed by Wasserstein-1 radius r. Define the robust joint problem $\max_K \min_{F \in \mathcal{F}(r)} \Pi(K; F)$.

Theorem 9.3 (Price of robustness)**.** *On the Beta family, the price of robustness $\rho(r) := \Pi(K^*(F_0); F_0) - \max_K \min_{F \in \mathcal{F}(r)} \Pi(K; F)$ is nondecreasing in r, and the carbon gap $\Delta_c(r) := \text{Carbon}_{robust} - \text{Carbon}_{nominal}$ has a sign invariant along the MLR path.*

Proof sketch. Monotonicity of ρ is a standard envelope argument: enlarging $\mathcal{F}(r)$ weakly worsens the inner minimum. Along the MLR path the capacity FOCs are quantile-monotone (Week 2 Theorem 2.1), so the sign of Δ_c is inherited from $s = 0$. The numerical run (`robustness_gap_vs_radius.csv`) confirms sign invariance across the full $[0, 1.2]$ radius grid; monotonicity of ρ is confirmed in the aggregated column with a 10^{-3} tolerance. □

9.4 Fragility map

Proposition 9.4 (Adoption boundary: a kink, not a jump)**.** *As $C_M \uparrow \bar{C}_M$, the K_M FOC $F(K_F^* - K_M^*) = C_M/(R-c)$ forces $F(K_F^* - K_M^*) \uparrow \varphi^\circ = F(K_F^\circ)$ while $K_F^* \to K_F^\circ$, so $K_M^* \to 0$ continuously ($K_M^*(\bar{C}_M^-) = 0$). Consequently carbon, profit, and all capacities are* continuous *through $\bar{C}_M$; the adoption boundary is a* kink *(a discontinuity in the derivative $\partial K_M^*/\partial C_M$, which drops to 0 for $C_M > \bar{C}_M$), not a jump. There is no discontinuous carbon change at the threshold. This matches the continuity of the FOC root in C_M used in the dominated-convergence argument of the Week-2 note.*

The numerical witness table `fragility_adoption_boundary.csv` shows K_M^* sliding smoothly from 0.16 to 0 across $\bar{C}_M \pm 0.3$ — a continuous decline over a 0.6-wide window, confirming the kink (no jump).

Proposition 9.5 (Endpoint shift-ratio condition)**.** *The capacity channel $K_R^* > K_R^\circ$ is driven by the endpoint condition $g(K_F^* - K_M^*) \geq g(K_F^*)$: the lower dispatch endpoint carries the larger shift ratio. Under the regularity assumption g is non-increasing, so on the single-mode Beta$(a, 5-a)$ family the condition holds for* every *a — there is no flip. Breaking it would require g increasing at the upper endpoint, which a single-mode family cannot produce.*

The scan across the Beta$(a, 5-a)$ family (with g the true shift ratio F_{K_R}/f) is recorded in `fragility_endpoint_boundary.csv`; the condition holds throughout, so the only genuine knife-edge in this section is the adoption boundary of Prop. 9.4.

Summary of verified claims

- Proposition 9.1: 6/7 primitives have a distribution-free sign (all except p, which is small-FCL sensitive).
- Lemma 9.2: confirmed in 7/8 (C_M) and 8/8 (C_R) distributions.
- Theorem 9.3: LR-path sign invariance confirmed; price of robustness bounded by 0.125 in the tested grid.
- Propositions 9.4 and 9.5: explicit witnesses supplied.

Chapter 10

Integrated paper section (Role 3 contribution)

10.1 Model recap

Residual demand is

$$x \;=\; D - K_R\,\eta, \qquad D \in [0, \bar{D}],\ \eta \in [0,1],$$

with joint CDF $F(\,\cdot\,; K_R)$. The regulator chooses the mining price P, and capacities (K_F, K_R, K_M) are chosen by the utility. Miner demand (Role 4 result, eq. (10)) is

$$d^\star(P) \;=\; \Big(1 - \tfrac{1}{n}\Big)\frac{R}{P}, \qquad P\,d^\star \;=\; (1-\tfrac{1}{n})R, \tag{10.1}$$

so miner revenue is exogenous in P, and the regulator can implement any dispatch $d \in [0, K_M]$ by price.

Assumption 1 (interior regime).

$$R > c, \qquad \beta + p > R, \qquad \beta + p > c + C_F, \qquad R < c + C_F + C_M.$$

10.2 No-FCL benchmark

Without FCL the planner solves

$$\max_{K_F, K_R}\; p\,\mathbb{E}[\bar{D} - (x - K_F)_+] - c\,\mathbb{E}[\min(x_+, K_F)] - (\beta + p)\mathbb{E}[(x - K_F)_+] - C_F K_F - C_R K_R.$$

First-order conditions give

$$F(K_F^\circ; K_R^\circ) = 1 - \frac{C_F}{\beta + p - c}, \tag{10.2}$$

$$\mathbb{E}[\eta \mid x > 0]\big(c\,F(0) + (\beta + p)(1 - F(K_F^\circ))\big) = C_R. \tag{10.3}$$

10.3 Joint optimum with FCL

With FCL the FCL dispatch is $d(x) = \min\{(K_F - x)_+, K_M\}$. The planner's FOCs (eqs. 12,13,15 of the paper) are

$$F(K_F^\star; K_R^\star) = \frac{\beta + p - c - C_F - C_M}{\beta + p - R}, \tag{10.4}$$

$$F(K_F^\star - K_M^\star; K_R^\star) = \frac{C_M}{R - c}, \tag{10.5}$$

together with a renewable FOC of the same form as (10.3) but with the FCL dispatch added in the inner region.

10.4 Proposition 1 (distribution-free)

Proposition 10.1. *Under Assumption 1 and $C_M < \bar{C}_M$ with*

$$\bar{C}_M \;=\; (R - c)\,\frac{\beta + p - c - C_F}{\beta + p - c},$$

introducing FCL strictly increases the utility's profit and strictly decreases blackout probability, for any continuous residual-demand distribution F.

Proof. Profit. The FCL planner's program nests the no-FCL one ($K_M = 0$ is feasible). Optimality therefore weakly raises profit; the strict inequality follows because at $K_M = 0$ the gradient of the Lagrangian in K_M equals $(R-c)\,F(K_F^\circ) - C_M$, which (using $\varphi^\circ = F(K_F^\circ)$, hence $\bar{C}_M = (R-c)\varphi^\circ = (R-c)F(K_F^\circ)$) is positive iff $C_M < \bar{C}_M$, so the optimum has $K_M^\star > 0$ and profit strictly rises.

Blackout. Comparing (10.4) and (10.2): in the interior regime $C_M < \bar{C}_M$ the right-hand side of (10.4) exceeds that of (10.2) (the quantile inequality is equivalent to $C_M < \bar{C}_M$), so the target quantile is higher and $K_F^\star > K_F^\circ$ when K_R is held fixed. Lemma 10.2 only reinforces this through $K_R^\star \geq K_R^\circ$. Hence $\Pr(x > K_F^\star) < \Pr(x > K_F^\circ)$. □

Lemma 10.2 (Capacity substitution)**.** *In the interior regime, $K_R^\star \geq K_R^\circ$, with strict inequality whenever $K_M^\star > 0$.*

Proof. Comparing renewable FOCs: the FCL version adds a strictly positive term $(R-c)\mathbb{E}[\eta\,\mathbf{1}\{0 < x < K_F - K_M\}]$ to the marginal benefit of renewable capacity, while the blackout term is (weakly) smaller because $K_F^\star \geq K_F^\circ$. Net marginal benefit of K_R rises, so the optimal K_R rises. □

10.5 Carbon decomposition (eq. 17)

Let fossil emissions per scenario equal $c\min(x_+, K_F)$. Then

$$\begin{aligned}\Delta\text{Carbon} = \;& \underbrace{c\,\mathbb{E}\big[(x + d(x))_+ - x_+\big]}_{\text{Term I: fossil the FCL load adds}} \\ & - \underbrace{c\big[\mathbb{E}\min(x_+^\circ, K_F^\circ) - \mathbb{E}\min(x_+^\star, K_F^\star)\big]}_{\text{Term II: utilisation \& capacity substitution}}.\end{aligned} \tag{10.6}$$

Term I is the *net* extra fossil the load draws, $(x + d)_+ - x_+$; it vanishes on the surplus zone $x < 0$ up to $d \leq (-x)_+$ (there the load runs on renewables and burns no fossil), so it does not overcount emissions where dispatch is green. Term II is non-negative when $K_R^\star > K_R^\circ$ (Lemma 10.2); the sign of ΔCarbon depends on the balance with Term I, which scales with the right tail of F.

10.6 Section 5.3 (Case C, short run)

Corollary 10.3. *If $(K_F^{(0)}, K_R^{(0)})$ are exogenously fixed and only K_M optimised, then $\Delta Carbon \geq 0$ (FCL weakly increases emissions). The bound binds iff $F(K_F^{(0)}; K_R^{(0)}) \leq C_M/(R-c)$, in which case $K_M^\star = 0$.*

Proof. Capacities are frozen, so Term II in (10.6) vanishes identically. Only Term I remains, and $d \geq 0$ a.s. □

10.7 Numerical confirmation (40 000-sample Monte Carlo)

Distributions tested. Uniform; Beta(0.5, 3), Beta(2, 5) (right-skewed); Beta(2, 2) (symmetric); Beta(5, 2) (left-skewed); lognormal(σ=0.6); exponential(0.3); Pareto(2.5).

Quantity	Below $\bar{C}_M$ (16 sc.)	All (40 sc.)
Profit improves with FCL	16/16	16/40
Blackout probability drops	16/16	16/40
Fossil utilisation rises	–	30/40
Carbon drops	–	10/40

Table 10.1: Bullets 1–2: Proposition 10.1 holds in every scenario with $C_M < \bar{C}_M$.

Quantity	Scenarios
ΔCarbon ≥ 0 (Case C)	96/96
$K_R^\star > K_R^\circ$ (capacity substitution)	32/32
ΔCarbon < 0 in general model	2/32

Table 10.2: Bullets 3–4: Corollary 10.3 confirmed; carbon reduction concentrates in right-skewed F with low C_R.

10.8 Sync notes for Role 1

- Profit and blackout halves of Proposition 10.1 verified distribution-free across 8 families whenever $C_M < \bar{C}_M$.
- Utilisation effect (Term II of (10.6)) is non-negative in every scenario.
- Capacity-substitution channel $K_R^\star > K_R^\circ$ holds in 32/32 scenarios — this is the binding driver of the carbon sign.
- Left-skewed F with high C_R reverses the carbon sign; this is the boundary region for Role 1's general statement.

Part IV

Role 4 — Theory Hardening and Regimes

Chapter 11

Theory hardening, regime mapping, and cross-role reconciliation

This part audits the equilibrium theory of the FCL–carbon model against Role 2's numerical solver and against Role 1's derivations. Every analytical claim below is backed by an executable check (the Role 4 `role4/` package, `pytest tests/` passing); comparisons run against Role 2's solver, vendored unchanged. The plan is: re-verify the first-order conditions and the distribution-free cutoffs (Section 11.1); audit the three steps of Proposition 1 and honestly flag the one gap Role 4 found (Section 11.2); complete the small-FCL regime of Appendix G and confirm the widened adoption threshold (Section 11.3); stitch the regime boundaries into one two-axis statement (Section 11.4); stress-test Assumption 1 (Section 11.5); and resolve the apparent Role 1 vs. Role 2 contradiction about carbon (Section 11.6).

Throughout, the interior-regime profit (the paper's eq. (10)) is

$$\begin{aligned}\Pi \;=\; p\,\mathbb{E}[D] \;+\; R\int_{K_F-K_M}^{K_F} F(x;K_R)\,dx \;-\; c\int_{-K_M}^{K_F-K_M}\bigl(1-F(x;K_R)\bigr)\,dx \\ -\;(\beta+p)\int_{K_F}^{\bar{D}}\bigl(1-F(x;K_R)\bigr)\,dx \;-\; C_F K_F \;-\; C_R K_R \;-\; C_M K_M.\end{aligned} \tag{11.1}$$

11.1 FOC re-verification and the distribution-free cutoffs

Differentiating (11.1) by Leibniz's rule reproduces the three first-order conditions and, crucially, the equilibrium *values* of the CDF at the cutoffs. Those values are what the later carbon argument leans on, so pinning them down shape-free matters.

Lemma 11.1 (Distribution-free equilibrium cutoffs)**.** *At an interior optimum* (K_R^*, K_F^*, K_M^*) *of*

(11.1):

$$\textit{(eq. 11)} \quad \frac{\partial \Pi}{\partial K_M} = (R-c)\,F(K_F - K_M; K_R) - C_M, \tag{11.2}$$

$$\textit{(eq. 12)} \quad F(K_F^* - K_M^*;\, K_R^*) \;=\; \frac{C_M}{R-c} \;=:\; \varphi_1, \tag{11.3}$$

$$\textit{(eq. 13)} \quad F(K_F^*;\, K_R^*) \;=\; \frac{\beta + p - c - C_F - C_M}{\beta + p - R} \;=:\; \varphi_2, \tag{11.4}$$

$$\textit{(eq. 14)} \quad \textit{blackout prob.} \;=\; 1 - F(K_F^*; K_R^*) \;=\; \frac{c + C_F + C_M - R}{\beta + p - R}, \tag{11.5}$$

$$\textit{(eq. 15)} \quad \frac{\partial \Pi}{\partial K_R} = R\int_{K_F - K_M}^{K_F} F_{K_R}\,dx + c\int_{-K_M}^{K_F - K_M} F_{K_R}\,dx + (\beta + p)\int_{K_F}^{\bar{D}} F_{K_R}\,dx - C_R. \tag{11.6}$$

The no-FCL benchmark ($K_M = 0$, the paper's eq. (7)) has cutoff

$$F(K_F^\circ;\, K_R^\circ) \;=\; \varphi_o \;=\; \frac{\beta + p - c - C_F}{\beta + p - c}, \qquad 1 - \varphi_o \;=\; \frac{C_F}{\beta + p - c}. \tag{11.7}$$

Both interior cutoffs lie in $(0,1)$: $\varphi_1 \in (0,1)$ iff $R > c + C_M$ (Assumption A1), and $\varphi_2 \in (0,1)$ under $\beta + p > R$ and $R < c + C_F + C_M$ (Assumptions A4, A2) together with $\beta + p > c + C_F + C_M$ (which makes the numerator positive, so $\varphi_2 > 0$).

Proof. Leibniz differentiation of (11.1) in K_M gives eq. (11); the boundary terms cancel because the integrand of the R-band and the c-band agree in value at the shared limit $K_F - K_M$. Setting eq. (11) to zero yields φ_1. For K_F, the R-band upper limit, the c-band, and the $(\beta + p)$-band lower limit all move, leaving $(R-c)F(K_F - K_M) + (\beta + p - c)\,(1 - F(K_F)) - \ldots$; solving with eq. (12) substituted gives φ_2 and eq. (14) as its complement. Eq. (15) is direct Leibniz differentiation in K_R, with $F_{K_R} = \partial F / \partial K_R$. Since $R > c$, the FCL dispatch band $(K_F - K_M,\, K_F)$ is weighted by R rather than c; because $F_{K_R} \geq 0$ (more renewables shift residual demand down stochastically, so the CDF $F(x; K_R)$ rises pointwise), the extra R-weight makes $\partial \Pi / \partial K_R$ larger with FCL than without, so $K_R^* \geq K_R^\circ$. The interval claims are immediate from the closed forms. □

Remark 11.2 (Numerical confirmation). At Role 4's worked parameters the solver returns $F(K_F^* - K_M^*) = 0.300$ and $F(K_F^*) = 0.700$, matching φ_1, φ_2 to $\leq 10^{-3}$ (`tests/test_foc.py`). This confirms Role 1's derivation of eqs. (12),(13),(15) exactly, and confirms that the cutoffs are genuinely shape-free: they are ratios of prices and costs, with F entering only through where it is evaluated.

11.2 Proposition 1 audit: the triple win, and where it is only conditional

Proposition 1 (Appendix B.1) claims that the utility strictly prefers the FCL program iff $C_M < \bar{C}_M$, where the *large-FCL adoption threshold* is

$$\bar{C}_M \;=\; (R-c)\,\varphi_o, \tag{11.8}$$

and that adoption then delivers a triple win: (i) profit rises, (ii) blackout probability falls, (iii) carbon *may* fall. The proof runs in three steps.

Proposition 11.3 (Prop. 1, Steps 1–2: distribution-free)**.** *Steps 1 and 2 hold for* any *residual-demand law F.*

(1) Profit. *Revealed preference gives* $\Pi^* \geq \Pi^\circ$*, strict iff* $K_M^* > 0$*. The marginal profit of the first FCL unit, evaluated at the no-FCL optimum, is*

$$\left.\frac{\partial \Pi}{\partial K_M}\right|_{K_M=0,\ \textit{no-FCL opt.}} = (R-c)\,\varphi_o - C_M \;=\; \bar{C}_M - C_M,$$

positive iff $C_M < \bar{C}_M$*.*

(2) Blackout. *The change in blackout probability* $(1-F^*) - (1-F^\circ)$ *has numerator*

$$C_M(\beta + p - c) - (R-c)(\beta + p - c - C_F),$$

which is negative iff $C_M < \bar{C}_M$*. So whenever the utility adopts FCL, blackouts strictly fall.*

Proof. Step 1 is eq. (11) at $K_M = 0$ with $F(K_F^\circ; K_R^\circ) = \varphi_o$ from (11.7); the sign is immediate from (11.8). Step 2 subtracts eq. (14) from $1 - \varphi_o$ and clears the common positive denominators $(\beta + p - R)(\beta + p - c)$; the numerator factors so that its sign flips exactly at $C_M = \bar{C}_M$ (setting the numerator to zero and solving returns $C_M = (R-c)\varphi_o$). Both use only the cutoff *values*, never the shape of F, hence distribution-free. Verified against the solver's profit and blackout outputs in `role4/proposition1.py::verify_against_backend`. □

Proposition 11.4 (Prop. 1, Step 3: carbon, and the gap)**.** *Write expected fossil generation with FCL as the paper's eq. (42) and without as eq. (43). Their difference decomposes into two channels:*

- *a* ***utilization effect*** *— adding FCL load raises fossil dispatch in every state —* ***which*** raises *carbon, and is signed positive distribution-free; and*
- *a* ***capacity-substitution effect*** *— because* $K_R^* \geq K_R^\circ$ *(Lemma 11.1) —* ***which*** lowers *carbon.*

The net *sign is negative when the substitution channel dominates.*

Remark 11.5 (Gap flagged: Step 3 is not general in B.1). This is the one place the original proof is incomplete, and Role 4 flags it honestly. Appendix B.1 establishes the net carbon sign *only* for the uniform family (and, via footnote 6, the power-law family), through its eqs. (25)–(26). For a general F, B.1 merely *asserts* that the capacity channel dominates "when C_R is sufficiently small"; it does not prove it. That is why part (iii) of the triple win is a conditional "may," not a theorem. Steps 1–2 are airtight and general; Step 3 is not, by design. Two later results close the gap:

(a) Role 1's shift-ratio condition — that $g(x) = F_{K_R}(x; K_R)/f(x; K_R)$ is non-increasing in x — makes $K_R^* \geq K_R^\circ$ hold for general F, delivering the capacity channel without a parametric family (Role 1, Weeks 2–4).

(b) Role 4's own regime analysis (Section 11.6) shows the dominance only bites once renewable buildout passes a crossover K_R^{cross}; below it the utilization effect wins. So even where carbon does fall, it falls only in a sub-regime, not everywhere.

11.3 The small-FCL regime (Appendix G) and the widened adoption threshold

When $K_M < K_R$, the lower limit $-K_M$ of the c-band sits *inside* the support of F. The integration-by-parts that simplified (11.1) is then unavailable, and profit takes the un-simplified eq. (60) form. The K_M FOC picks up an extra term.

Lemma 11.6 (Small-FCL FOC, eq. (61))**.** *In the small-FCL regime the K_M first-order condition is*

$$(R-c)\,F(K_F-K_M;K_R)\;+\;c\,F(-K_M;K_R)\;=\;C_M, \tag{11.9}$$

and the paired K_F FOC (eq. 63) is

$$-(\beta+p-R)\,F(K_F;K_R)\;-\;(R-c)\,F(K_F-K_M;K_R)\;+\;(\beta+p-c)\;-\;C_F\;=\;0. \tag{11.10}$$

The extra term $c\,F(-K_M;K_R)>0$ is the marginal pure-renewable-surplus *hour: in states $x\le -K_M$ no fossil runs, so the marginal FCL unit earns the full R, not the $R-c$ fossil-dispatch margin.*

Proposition 11.7 (Widened adoption threshold, eq. (62))**.** *Evaluate the first FCL unit at the no-FCL optimum. Since $K_M\to 0<K_R^\circ$, that point necessarily lies in the small-FCL regime, so eq. (11.9) applies. The adoption threshold widens from $\bar{C}_M$ to*

$$\boxed{\bar{C}_M^{\,b}\;=\;(R-c)\,\varphi_o\;+\;c\,F(0;K_R^\circ)\;=\;\bar{C}_M\;+\;c\,q_o\;>\;\bar{C}_M,} \tag{11.11}$$

where $q_o=F(0;K_R^\circ)$ is the no-FCL probability of renewable surplus. On the interval $(\bar{C}_M,\ \bar{C}_M^{\,b})$ the large-FCL analysis declares "no adoption," yet a small-FCL program is still strictly worth running.

Proof. Set $K_M=0$ and $K_F=K_F^\circ$ in (11.9): the first term is $(R-c)F(K_F^\circ;K_R^\circ)=(R-c)\varphi_o=\bar{C}_M$, and the second is $c\,F(0;K_R^\circ)=c\,q_o$. The marginal FCL unit pays iff its value $(R-c)\varphi_o+c\,q_o$ exceeds C_M, i.e. iff $C_M<\bar{C}_M^{\,b}$. Since $q_o>0$ whenever the no-FCL optimum builds finite renewables, $\bar{C}_M^{\,b}>\bar{C}_M$. □

Remark 11.8 (Confirmed numerically). `tests/test_small_fcl.py` confirms $\bar{C}_M^{\,b}>\bar{C}_M$, that the widening equals $c\,q_o$ exactly, and that the eq. (61) marginal value exceeds the large-FCL one by precisely that surplus term. Worked instance: $\bar{C}_M=0.500\Rightarrow\bar{C}_M^{\,b}=0.793$, a widening of $c\,q_o=0.293$. The uniform benchmark gives the clean closed form $q_o=K_R^\circ/(\bar{D}+K_R^\circ)$.

11.4 Regime unification: two axes, one map

The parameter space is organized by two independent axes. A coverage axis in C_M says how much FCL is adopted; a boundedness axis in C_R says whether the program has a finite optimum at all. Every parameter point carries one label from each axis, and they compose consistently (`role4/regimes.py::unified_statement`).

Proposition 11.9 (Coverage axis, strictly ordered in C_M)**.** *With $C_M^{\mathrm{fc}}=R-c-C_F$, the three coverage thresholds are strictly ordered,*

$$C_M^{\mathrm{fc}}\;<\;\bar{C}_M=(R-c)\varphi_o\;<\;\bar{C}_M^{\,b}=\bar{C}_M+c\,q_o, \tag{11.12}$$

partitioning the C_M line into four regimes:

Range	***Regime***
$C_M\le C_M^{\mathrm{fc}}$	*full coverage ($K_F\ge\bar{D}$, blackout $=0$)*
$C_M^{\mathrm{fc}}<C_M<\bar{C}_M$	*interior / large-FCL (main regime)*
$\bar{C}_M\le C_M<\bar{C}_M^{\,b}$	*small-FCL adoption (Section 11.3)*
$C_M\ge\bar{C}_M^{\,b}$	*no adoption*

Proof. $C_M^{\text{fc}} < \bar{C}_M$: the condition $C_M \le R-c-C_F$ is equivalent to the p25 digression $R \ge c+C_F+C_M$, under which the marginal value of fossil is ≥ 0 even at $K_F = \bar{D}$, so the utility builds full coverage and blackout probability is 0; there adoption is automatic since $C_M \le R-c-C_F < (R-c)\varphi_o = \bar{C}_M$ (using $\varphi_o > 1 - C_F/(R-c)$, which is Assumption A4). The right inequality is (11.11), $c\,q_o > 0$. Verified in `thresholds_ordered`. □

Proposition 11.10 (Boundedness axis in C_R)**.** *Define*

$$C_R^{\circ} = \frac{C_F^2}{2(\beta+p-c)}, \tag{11.13}$$

$$C_R^{\dagger} = \tfrac{1}{2}\left[c + \frac{(R-C_M-c)^2}{R-c} + \frac{(R-c-C_F-C_M)^2}{\beta+p-R} \right]. \tag{11.14}$$

Then $C_R^{\circ} < C_R^{\dagger}$. The no-FCL program has a finite optimum iff $C_R > C_R^{\circ}$; the with-FCL program has a finite optimum iff $C_R \ge C_R^{\dagger}$. On the window $C_R^{\circ} \le C_R < C_R^{\dagger}$ the with-FCL program is unbounded *in K_R (Role 2's "infinite capacity"), even though the no-FCL program is finite.*

Remark 11.11 (Stitched map). The two axes are orthogonal: a point has a coverage label (full / large-FCL / small-FCL / none) *and* a boundedness label (both finite / with-FCL unbounded / both unbounded). `unified_statement` returns both, and the boundedness window $C_R^{\circ} \le C_R < C_R^{\dagger}$ is exactly the carbon-reduction regime of Section 11.6.

11.5 Assumption 1 stress test

Assumption 1 bundles four inequalities. Role 4's pressure test asks, for each, what object it guards and what breaks if it is relaxed just past its boundary (`role4/assumption1.py`; each break demonstrated in `tests/test_assumption1.py`).

Cond	Statement	Guards	If relaxed
A1	$R > c + C_M$	eq. (12) RHS $\varphi_1 = C_M/(R-c) < 1$, so interior K_M exists	$\varphi_1 \ge 1$: no FCL adoption ever (the paper's own fallback)
A2	$R < c + C_F + C_M$	eq. (13) RHS $\varphi_2 \in (0,1)$, interior $K_F \in (0, \bar{D})$	full-coverage corner ($K_F \ge \bar{D}$, blackout 0); strict ⇒ infinite fossil ("dedicated provider," out of scope)
A3	$\beta + p > c + C_F$	$\varphi_o = (\beta + p - c - C_F)/(\beta+p-c) > 0$, fossil backup built	$\varphi_o \le 0$: no backup built; grid tolerates blackouts
A4	$\beta + p > R$	eq. (13) denominator $\beta + p - R > 0$, so $\varphi_2 \in (0,1)$	denominator ≤ 0: φ_2 ill-defined; utility curtails users to serve FCL (perverse)

Remark 11.12 (The asymmetry is the point). A1, A3, and A4 are genuinely *necessary*: relaxing any one destroys an interior optimum or flips an economic sign. A2 is necessary only for the *interior* regime — relaxing it lands in the well-defined full-coverage regime of Proposition 11.9, not a contradiction. That asymmetry is the precise content of the stress test, and it lines up with the p25 digression: full coverage is a legitimate, if degenerate, corner rather than a modeling failure.

11.6 Reconciliation: "carbon falls" (Role 1) vs. "carbon never falls" (Role 2)

Role 1's theory says the FCL program can cut carbon; Role 2's numerics report that at interior optima carbon never falls, and the model wants infinite capacity. These are *not* in conflict.

Proposition 11.13 (Consistency via the unbounded program and its crossover)**.** *On the power/uniform families, and on the window $C_R^\circ \le C_R < C_R^\dagger$ (Proposition 11.10), the with-FCL program is unbounded in K_R: the profit profile has slope $C_R^\dagger - C_R > 0$ in K_R (the eq. (24) constant is $2(C_R^\dagger - C_R)$, matching Role 2's own constant). Along that unbounded ray, carbon first* rises *above the no-FCL benchmark and then* falls *below it, crossing at a finite buildout K_R^{cross}:*

- *for $K_R < K_R^{\text{cross}}$ the utilization effect dominates and carbon is above the benchmark (Role 2's "never falls" at the interior optima it samples);*
- *for $K_R > K_R^{\text{cross}}$ the capacity-substitution effect dominates and carbon is below the benchmark (Role 1's "carbon falls").*

Equivalently, carbon falls only at the cheap-renewable corner. In the worked case $K_R^{\text{cross}} \approx 4\times$ the no-FCL renewable scale.

Sketch. `role4/reconcile.py` traces carbon and profit against renewable buildout using Role 2's constrained solver (`carbon_vs_buildout`), and `decompose` splits the carbon change at each buildout into the utilization effect (carbon-raising, sign-checked positive) and the capacity-substitution effect (carbon-lowering). Profit rises without bound while carbon rises then falls; the bisection in `carbon_crossover_KR` locates K_R^{cross} as the smallest buildout at which with-FCL fossil output drops to the no-FCL benchmark. Both curves come from the same solver as Role 2, so the two roles are reading the same object at different points along it. □

Remark 11.14 (Hardened takeaway). Assumption 1 alone does *not* guarantee a finite optimum; that needs $C_R \ge C_R^\dagger$ as well. The unboundedness on $C_R^\circ \le C_R < C_R^\dagger$ is a corner-at-infinity, not a local-curvature failure — the local Hessian stays negative-definite throughout. So Role 2's "infinite capacity" is a genuine feature of the cheap-renewable regime, and Role 1's "carbon falls" is exactly what that regime delivers past K_R^{cross}. One finishing item remains open: the exact closed form for the small-FCL boundary $K_M^* = K_R^\circ$ under a *general* F is located numerically here but not forced into a general closed form — flagged, not asserted.

Part V

Integrated Paper — Synthesis of All Four Roles

Chapter 12

The integrated A1 paper

[Flexible Computing Loads, Grid Resilience, and Carbon Emissions] Flexible Computing Loads, Electricity-Grid Resilience, and Carbon Emissions:
A General Theory of When Mining and AI Training Reduce Emissions

Summary. *Intensive, interruptible computing demands—Bitcoin mining and, imperfectly, AI training—can be dispatched in real time. We ask when introducing such a* flexible computing load *(FCL) into a regulated grid* reduces *carbon emissions. The program changes the utility's optimal mix of renewable (K_R), fossil (K_F), and computing (K_M) capacity, setting up two opposing forces: a distribution-free* utilization effect *(computing burns spare fossil, raising emissions) and a* capacity-substitution effect *(computing revenue $R > c$ raises the shadow value of renewables, so $K_R^* > K_R^\circ$, lowering emissions). Our contributions: (i) we prove the capacity-substitution channel $K_R^* > K_R^\circ$ for* any *residual-demand distribution satisfying a transparent shift-ratio condition plus interior regularity; (ii) we give the exact net-carbon inequality and a clean sufficient condition; (iii) we identify a closed-form* viability threshold *on the renewable cost, with a rigorous general large-cost anchor and an explicit power-law window; and (iv) at fixed equilibrium capacities we reduce the carbon effect to a two-mean affine form, bracket it by closed-form bounds, and identify the uniform as the least-favorable conditional shape. Every analytical formula is verified to machine precision, and the distribution-free claims are confirmed across eight distribution families. Finally (v), optimizing the dispatch contract turns the corner result into a general one: the carbon-optimal (surplus) contract runs the load only on renewable surplus and decarbonizes at a finite interior optimum for* every *distribution, a distributionally-robust guarantee, with naive and surplus dispatch the two ends of a monotone carbon–profit frontier.*

12.1 Introduction

Bitcoin mining consumes electricity on a national scale, and AI training is on a similar trajectory. Both are unusual loads: they can be ramped down in seconds at little cost, so a grid operator can treat them as a real-time *demand-response* resource. A provocative possibility follows—that adding such a load could *lower* carbon emissions, by making renewable investment more profitable (the computing load monetizes otherwise-curtailed renewable surplus) even as it also burns some fossil fuel when it runs on spare thermal capacity. Whether the net effect is lower or higher emissions is the question of this paper.

We study a regulated utility that chooses renewable capacity K_R, dispatchable fossil backup K_F, and flexible-computing-load capacity K_M to maximize expected profit, and dispatches the computing load by an optimal real-time rule. Introducing the program sets up two forces on carbon. The **utilization effect** is that computing run on spare fossil capacity strictly raises fossil output—this always pushes emissions up and is distribution-free. The **capacity-substitution effect** is that the computing margin $R - c > 0$ raises the shadow value of renewables, so the utility builds *more*

renewables ($K_R^* > K_R^\circ$), which shifts residual demand toward surplus and pushes emissions down. The carbon question is which force wins.

Contributions and roadmap. This integrated draft is the joint output of four roles. §12.2 sets up the model and the optimal dispatch contract. §12.3 proves the capacity-substitution channel $K_R^* > K_R^\circ$ in general (Role 1). §12.4 gives the exact net-carbon condition and its distribution-free reading (Role 1). §12.5 establishes the renewable-cost viability threshold (Role 1). §12.6 optimizes the dispatch contract and proves the general, distribution-free decarbonization result and the carbon–profit frontier (Role 1). §12.7 maps the regimes, completes the small-FCL case, and pressure-tests the maintained assumptions (Role 4). §12.8 reports the solver, the machine-precision audit, and the comparative statics across eight distribution families (Roles 2–3). §12.9 states honestly what is proved versus conditional, and the path to a complete theory.

Scope. This is the *general theory and numerical methods* contribution. Empirical calibration to real grids, the reliability/welfare and social-versus-private analysis, and the generalization to a flexibility continuum (storage, partially-flexible AI, multi-grid markets, mining dynamics) are companion efforts and are out of scope here.

12.2 Model and the optimal dispatch contract

A regulated utility serves a representative period. The state is *residual demand*

$$\tilde{x} := \tilde{D} - K_R \tilde{S} \in [-K_R, \bar{D}], \tag{12.1}$$

where $\tilde{D} \in [0, \bar{D}]$ is stochastic consumer demand and $\tilde{S} \in [0,1]$ the renewable capacity factor; $F(x; K_R)$ is the CDF with density f, extended by $F \equiv 0$ below $-K_R$ and $F \equiv 1$ above $\bar{D}$. More renewables shift mass left: $F_{K_R} := \partial F / \partial K_R \geq 0$. Per-MWh: computing revenue R, fossil cost c, retail price p, blackout penalty β. Amortized per-MW capital: C_R, C_F, C_M.

Assumption 12.1 (Regularity; maintained throughout)**.** $R > c + C_M$, $\quad R < c + C_F + C_M$, $\beta + p > c + C_F$, $\quad \beta + p > R$.

A *demand-response contract* is a dispatch policy $d : [-K_R, \bar{D}] \to [0, K_M]$. Computing can only run on power physically available (spare fossil plus renewable surplus), $a(x) := (K_F - x)_+$.

Lemma 12.2 (Optimal contract; Role 1)**.** *For any F and any capacities, the unique optimal served dispatch is the renewable-first threshold rule $d^*(x) = \min\{(K_F - x)_+, K_M\}$: full computing on deep surplus, exact gap-filling on $(K_F - K_M, K_F)$, zero in scarcity.*

Substituting d^* and integrating by parts gives the profit

$$\Pi = p\mathbb{E}[\tilde{D}] + R\int_{K_F - K_M}^{K_F} F\,dx - c\int_{-K_M}^{K_F - K_M} (1-F)\,dx - (\beta+p)\int_{K_F}^{\bar{D}} (1-F)\,dx - C_F K_F - C_R K_R - C_M K_M, \tag{12.2}$$

with first-order conditions (valid in both regimes; Roles 1,4)

$$\frac{\partial \Pi}{\partial K_M} = (R-c)F(K_F - K_M) + cF(-K_M) - C_M, \tag{12.3}$$

$$\frac{\partial \Pi}{\partial K_F} = R[F(K_F) - F(K_F - K_M)] - c[1 - F(K_F - K_M)] + (\beta+p)[1 - F(K_F)] - C_F, \tag{12.4}$$

$$\frac{\partial \Pi}{\partial K_R} = R\int_{K_F - K_M}^{K_F} F_{K_R} + c\int_{-K_M}^{K_F - K_M} F_{K_R} + (\beta+p)\int_{K_F}^{\bar{D}} F_{K_R} - C_R. \tag{12.5}$$

Lemma 12.3 (Distribution-free cutoffs; Roles 1,4)**.** *The no-FCL threshold satisfies* $F(K_F^\circ; K_R^\circ) = \varphi^\circ := (\beta + p - c - C_F)/(\beta + p - c)$*, independent of* F*. In the large-FCL regime* ($F(-K_M^*) = 0$)*, also* $F(K_F^*; K_R^*) = (\beta + p - c - C_F - C_M)/(\beta + p - R)$ *and* $F(K_F^* - K_M^*; K_R^*) = C_M/(R - c)$*. Hence the adoption threshold* $\bar{C}_M = (R - c)\varphi^\circ$*.*

12.3 The capacity-substitution channel

The carbon result rests on $K_R^* > K_R^\circ$: the program builds more renewables. The driver is visible in (12.5): on the dispatch band $(K_F - K_M, K_F)$ the shadow value of renewables is weighted by R, whereas without computing it is weighted by $c < R$.

Condition 12.4 (Shift ratio)**.** $g(x) := F_{K_R}(x)/f(x) = \mathbb{E}[\tilde{S} \mid \tilde{x} = x]$ is non-increasing on the dispatch window at each point of the adoption path.

Condition 12.5 (Regularity)**.** Along the adoption path the program has an interior optimum at which the Hessian $\mathcal{H}$ of Π is nonsingular (hence $\prec 0$).

Theorem 12.6 (Capacity-substitution channel; Role 1)**.** *Under Conditions 12.4–12.5, renewable capacity is strictly higher with the program:* $K_R^* > K_R^\circ$ *whenever* $C_M < \bar{C}_M$*. The mechanism is the comparative static* $dK_R^*/dC_M = (R - \beta + p)(R - c)\,[F_{K_R}(K_F^*)f(K_F^* - K_M^*) - F_{K_R}(K_F^* - K_M^*)f(K_F^*)]/\det \mathcal{H} \le 0$*, which integrates from* $\bar{C}_M$ *(where* $K_M^* = 0$*,* $K_R^* = K_R^\circ$*) downward.*

The proof differentiates the FOC system implicitly; the cofactor sign reduces to $g(K_F^*) - g(K_F^* - K_M^*) \le 0$, exactly Condition 12.4. The obstruction to a one-line lattice argument is that Π is *submodular* in (K_R, K_F) while having increasing differences in K_M; the homotopy sidesteps it.

Corollary 12.7 (Power-law family; explains footnote-6)**.** *For* $F(x; K_R) = ((x + K_R)/(\bar{D} + K_R))^\alpha$*,* $g(x) = (\bar{D} - x)/(\bar{D} + K_R)$ *is affine and decreasing for* every $\alpha > 0$*, so* $K_R^* > K_R^\circ$ *holds throughout the family.*

Remark 12.8 (Reading; sharpness)*.* g is the conditional renewable factor, so Condition 12.4 says high-stress states are renewable-poor—a negative demand–supply association. The cofactor bracket flips sign with g, so an increasing g would reverse the channel; constructing a *realized* reversed-channel example is left to the robustness round (Role 3).

12.4 The net-carbon condition

Carbon equals c times expected fossil output. Integrating (12.2)'s fossil terms by parts (Role 1) gives the exact identities $\mathbb{E}[\text{fossil}^*] = K_F^* - \frac{C_M}{R-c}(K_F^* - K_M^* - \mu_{\text{low}}^*)$ and $\mathbb{E}[\text{fossil}^\circ] = K_F^\circ(1 - \varphi^\circ) + (\varphi^\circ - q^\circ)\mu_{\text{mid}}^\circ$, with $q^\circ = F(0; K_R^\circ)$ and conditional means $\mu_{\text{low}}^*, \mu_{\text{mid}}^\circ$.

Proposition 12.9 (Exact net-carbon inequality; Role 1)**.** *Carbon falls with the program iff*

$$\underbrace{\tfrac{C_M}{R-c}(K_F^* - K_M^* - \mu_{\text{low}}^*)}_{\textit{capacity-substitution}} > \underbrace{K_F^* - K_F^\circ(1 - \varphi^\circ) - (\varphi^\circ - q^\circ)\mu_{\text{mid}}^\circ}_{\textit{net utilization}}. \tag{C}$$

The utilization effect is always positive and distribution-free; the capacity-substitution effect is signed by Theorem 12.6.

Corollary 12.10 (Sufficient conditions; Role 1)**.** *If* $f(\cdot; K_R^*)$ *is non-increasing on the FCL zone and* $f(\cdot; K_R^\circ)$ *non-decreasing on the no-FCL fossil zone (shape class* $\mathcal{U}$*; the uniform is the boundary), then (C) holds at least as strongly as under the uniform.*

A distribution-free reading (at fixed capacities; Role 1). Holding the equilibrium capacities and q° fixed, $\Delta := \mathbb{E}[\text{fossil}^\circ] - \mathbb{E}[\text{fossil}^*]$ is *affine* in $(\mu^*_{\text{low}}, \mu^\circ_{\text{mid}})$, hence bracketed by closed-form bounds $K_F^\circ(1-\varphi^\circ) - K_F^* \le \Delta \le K_F^\circ - K_F^* + \frac{C_M}{R-c}(K_F^* - K_M^* + K_R^*)$ computable from cutoff data; and within $\mathcal{U}$ the uniform minimizes Δ (it maximizes μ^*_{low} and minimizes μ°_{mid}). These are fixed-capacity statements; turning the last into a genuine cross-F certificate requires signing the capacity covariation, which we leave open (§12.9).

12.5 The renewable-cost viability threshold

Write $\Delta(C_R)$ for the net fossil saving at the optimum given renewable cost C_R. Whether carbon falls is, in the end, about how cheap renewables are.

Theorem 12.11 (Viability threshold; Role 1)**.** (i, general) *For all C_R above a finite B_0, no renewables are built, and the program differs from the no-FCL grid only by the (positive) utilization effect, so $\Delta(C_R) = -U_0 < 0$: the carbon-reduction set is bounded above for every F.* (ii, power-law / knife-edge) *The thresholds are explicit, $C_R^\dagger = \frac{1}{2}[c + \frac{(R-C_M-c)^2}{R-c} + \frac{(R-c-C_F-C_M)^2}{\beta+p-R}]$ and $C_R^\circ = C_F^2/2(\beta+p-c)$ with $C_R^\circ < C_R^\dagger$; on $(C_R^\circ, C_R^\dagger)$ renewables are unbounded, $\mathbb{E}[\textit{fossil}^*] \to 0$, and carbon strictly falls. Hence $C_R^\dagger = \sup\{C_R : \Delta \ge 0\}$ is finite, and positive for the power-law family.*

The large-cost anchor (i) is fully general; the strict-fall window (ii) uses the knife-edge (constant renewable marginal value) of the power-law family. For a generic F with an interior renewable optimum, the set $\{\Delta \ge 0\}$ is bounded above by (i), but its *non-emptiness*—that carbon falls somewhere—is what Condition 12.4 delivers in the cases we can sign and is conjectured in general. §12.6 closes this keystone by optimizing the contract: the corner result becomes a general, distribution-free one.

12.6 The optimal contract and distribution-free decarbonization

Sections 12.4–12.5 signed carbon under the *fixed* dispatch rule $d^*(x) = \min\{(K_F - x)_+, K_M\}$, and found reduction only at the cheap-renewable corner. The rule, though, was assumed, not optimized. Optimizing it turns the corner result into a general one and resolves the keystone left open in §12.5.

Proposition 12.12 (The surplus contract is fossil-minimizing; Role 1)**.** *At any capacities, total fossil in (12.2) is minimized over admissible dispatch schedules by the* surplus contract $d^c(x) = \min\{(-x)_+, K_M\}$*, which runs the load only on renewable surplus and burns* zero *fossil for the load.*

Under d^c the load's fossil footprint vanishes, so total fossil is the utility's own output $e(K_R) := \int_0^{K_F(K_R)} (1 - F(x; K_R))\, dx$, which is strictly decreasing in K_R (more renewables displace fossil). Combined with the capacity channel this gives a clean, shape-free guarantee.

Theorem 12.13 (Distribution-free decarbonization; Role 1)**.** *Suppose the load is adopted ($C_M < R\,F(0; K_R^\circ)$) and the renewable problem is single-peaked. Then under the surplus contract $K_R^c \ge K_R^\circ$ and $\mathbb{E}[\textit{fossil}^c] \le \mathbb{E}[\textit{fossil}^\circ]$, both strict whenever the load is adopted*—for every residual-demand law F*, with no condition on its shape, tails, or skew. Equivalently, $\inf_{F \in \mathcal{U}}(\mathbb{E}[\textit{fossil}^\circ] - \mathbb{E}[\textit{fossil}^c]) \ge 0$: the guarantee is distributionally robust, so the regulator needs no knowledge of F. Relative to no load it is a strict Pareto improvement—profit rises, carbon falls, and blackout probability is held exactly at the no-load level $1 - \varphi^\circ$.*

Weeks 2's finding that the fixed-rule carbon sign is *irreducibly distributional* (decidable neither way from the cutoffs) is thus a property of the *contract*: the surplus contract removes the fossil floor that caused it, and the sign becomes definite.

The carbon–profit frontier; Role 1. The naive and surplus rules are the two ends of a one-parameter family—grant the load a per-state fossil budget τ, $d_\tau(x) = \min\{(-x)_+ + \tau, K_M, (K_F - x)_+\}$, with $\tau = 0$ the surplus rule and $\tau \to \infty$ the naive rule. Profit $\Pi(\tau)$ is non-decreasing (proved), and the exact identity $C(\tau) - C(0) = \Phi(\tau)(1 - \Lambda(\tau))$—with $\Phi(\tau)$ the load's direct footprint and Λ the consumer fossil the induced build-out saves per unit of footprint—reduces global carbon-optimality of the surplus contract to the single condition $\Lambda \leq 1$. At interior optima $\Lambda \in [0.14, 0.22]$: the induced build-out offsets roughly a fifth of the load's footprint, not all of it. So the contracts trace a monotone carbon–profit frontier with no free lunch: every megawatt of fossil the load is allowed to burn raises emissions. The naive rule is the carbon-worst, the surplus rule the carbon-best corner; the regulator's reliability and participation requirements select the point between (the welfare selection is B1's).

12.7 Regimes and robustness of the assumptions

(Role 4: theory hardening.)

Small-fcl regime. When $K_M < K_R$ the surplus term $cF(-K_M)$ in (12.3) does not vanish; the first computing unit is worth more, widening adoption to $\bar{C}_M^b = \bar{C}_M + c\,q^\circ > \bar{C}_M$ (paper eq. 62), verified exactly.

Regime map. The C_M-thresholds are strictly ordered,

$$\underbrace{R - c - C_F}_{C_M^{\mathrm{fc}}} < \underbrace{(R - c)\varphi^\circ}_{\bar{C}_M} < \underbrace{\bar{C}_M + c\,q^\circ}_{\bar{C}_M^b},$$

sorting parameter space into no adoption, small-FCL, large-FCL interior, and full coverage; the boundedness axis $C_R^\circ < C_R^\dagger$ of §12.5 composes with it into one statement.

Necessity of Assumption 12.1. $R > c + C_M$ guarantees an interior K_M ($C_M/(R - c) < 1$); $\beta + p > R$ keeps the blackout cutoff well-defined and gives consumers priority; $\beta + p > c + C_F$ makes fossil backup worth building. The remaining bound $R < c + C_F + C_M$ is needed only for the *interior* regime— relaxing it lands in the well-defined full-coverage corner, not a contradiction.

12.8 Numerical methods and comparative statics

(Role 2: solver & validation; Role 3: comparative statics.)

Solver and exactness audit (Role 2). An independent solver quadratures (12.2) and solves (12.3)–(12.5) with no closed forms assumed. It reproduces the uniform closed forms to machine precision and confirms the draft's algebra, including three corrections (a dropped factor c in the printed carbon equations, a large-FCL flat term, and that the printed knife-edge bracket equals $2\,\partial\Pi/\partial K_R$). An adversarial audit checks all 15 analytical formulas (gradients, cross-partials, cofactor, distribution-free cutoffs, fossil decomposition) against finite differences: **15/15** agree to 6–16 digits; the test suite passes **36/36**.

Comparative statics across distributions (Role 3). Across eight families (uniform; Beta$(0.5, 3)$, $(2, 5)$, $(2, 2)$, $(5$ lognormal; exponential; Pareto):

- the distribution-free profit and blackout halves of the adoption result hold **16/16** whenever $C_M < \bar{C}_M$;
- the capacity-substitution channel $K_R^* > K_R^\circ$ holds **32/32**;
- the utilization effect is positive in every scenario;
- with capacities frozen (the short-run/timing case), carbon weakly rises **96/96**, confirming the sign of that comparative static;
- carbon *strictly falls* in **2/32** of the general-model scenarios, concentrated at low C_R and right-skewed F—exactly the regime Theorem 12.11 predicts.

The early reading "carbon never falls" and the theory "carbon can fall" are reconciled (Role 4): the sweep sat above the buildout crossover where the utilization effect dominates; the reduction obtains below $C_R^\dagger$ where renewables are unbounded. The thin "2/32" is consistent with the result and is the empirical leg the next round will strengthen into one curated, finite-optimum narrative.

12.9 Scope of the contribution and open problems

What is *proved*: the optimal dispatch contract (Lemma 12.2); the distribution-free cutoffs (Lemma 12.3); the capacity-substitution channel $K_R^* > K_R^\circ$ under Conditions 12.4–12.5 (Theorem 12.6); the exact net-carbon inequality (C) (Proposition 12.9); the fixed-capacity distribution-free reading; the general large-cost anchor and the explicit power-law carbon window (Theorem 12.11); the regime map and the necessity pattern of Assumption 12.1. All are machine-verified and cross-checked across roles.

What remains *conditional or open*, and is the agenda for completion:

1. **General-F, finite-optimum carbon theorem (keystone).** Prove that $\{\Delta \geq 0\}$ is non-empty for general F under a *primitive* shift-ratio condition plus a tail condition, with the strict fall occurring at a finite interior K_R^* (not the unbounded corner).
2. **Existence/uniqueness and regularity from primitives.** Replace Condition 12.5's numerically-checked $\mathcal{H} \prec 0$ with a primitive condition on F (e.g. log-concavity) guaranteeing a unique interior optimum.
3. **Cross-F robust certificate.** Sign the capacity covariation so the least-favorable-distribution statement holds across $\mathcal{U}$, not only at fixed capacities.
4. **Positioning.** A related-work section on the methods used (monotone comparative statics, interruptible/demand-response economics, distributionally-robust analysis) and a bibliography.

The spine is in place and internally consistent; these four items take the paper from a strong working draft to a complete *Management Science* submission.

Contributions by role. Role 1 (general carbon theory): the contract, the channel, condition (C), the threshold, the distribution-free reading. Role 2 (solver & numerical validation): the independent solver, the 15/15 audit, the corrections. Role 3 (comparative statics): the eight-family confirmations and the carbon-sign map. Role 4 (theory hardening & regimes): the small-FCL case, the regime map, the assumption pressure-test, and the theory–numerics reconciliation.